INTERNET OF HAPPINESS [IoH]
THE CURRENCY FOR HAPPINESS

sound

brain entrainment incl music , light & sound

impact

articles: 458

publications: 412

citations: 73,330

evidence



INTERNET OF HAPPINESS [IoH]
THE CURRENCY FOR HAPPINESS

research index

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i. brain entrainment modulates our DNA

i. brain entrainment modulates pain, anxiety and heart rate variability

ii. 528hz modulates gene expression, increasing oxytocin, regenerating brain tissue and treating alcohol addicted cell damage

iii. 7.83 HZ modulates DNA, increasing dopamine, inhibiting cancer cell growth and HIV cell growth

ii. fibonacci sequence modulates cardiovascular, cellular and DNA health

iv. fibonacci sequence modulates cosmological, psychological and biological functions of all dna based life, via first, second and third order harmonics - Fibonacci Mod 9, Pisano Period 24 (Fib M9 P24)

v. The Golden Ratio predicts the center of a black hole, a mathematical constant produced by gravity and time, via an infinite tetrahedral, geodesic fractal array, hypothesized as the unified field known or dark energy

vi. Fibonacci sequence and the Golden Ratio identify cancer cells, modulate gene expression and mitosis via DNA tetrahedrons

iii. music modulates heart rate variability, supporting autonomic balance via baroreflex

vii. music modulates heart rate variability, cognition (memory and learning) and emotional regulation, reducing physical pain and improving mental health

viii. music & sub conscious therapeutic affirmations reduce pain & improve recovery

ix. Baroreflex modulates Heart Rate Variability via 111hz Mayer Waves, modulating oxytocin, and in turn the sympathetic and parasympathetic nervous system

iv. music modulates our DNA

x. music modulates gene expression, improving mood and reducing stress via rhythmic and melodic entrainment

xi. music modulates the NMDA gene receptor, improving learning and memory

xii. music modulates interleukin 6 (IL—6) gene, reducing inflammation and human stress conditions

xiii. music modulates endogenous DMT and the DMT gene, a systems biology hypothesis

v. electromagnetism modulates our DNA

xiv. low frequency (sub 10 HZ) electromagnetic brain entrainment modulates depression (electromagnetic waves via Transcranial Magnetic Stimulation TMS)

xv. light modulates DNA (laser and LED)

xvi. electromagnetic fields EMFs modulates DNA

xvii. sound modulates DNA

xviii. Schumann resonance, solar wind and geomagnetic activity modulate heart rate variability, mental health and human culture

xix. Schumann Resonances are harmonics of the speed of light

vi. brain entrainment & music modulate mental health in at risk communities

xx. brain entrainment modulates pain, anxiety & HR variability in at risk communities

xxi. music modulates mental health with homeless and refugees

xxii. music modulates mental health with indigenous youth

xxiii. music modulates mental health with armed servicemen and women

xxiv. virtual reality modulates emotions and heals PTSD with war veterans

xxv. near infra red laser modulates cells and heals traumatic brain injury with war veterans

xxvi. art therapy modulates emotions with war veterans

xxvii. art therapy modulates emotions with at risk youth

vii. quantum computing empowers bioinformatics

xxviii. Quantum advantage enables transformational bioinformatics with gene expression modelling via cloud based quantum computers

viii. oxytocin receptor gene (OXTR) modulates happiness

xxix. Oxytocin receptor gene (OXTR) modulates dopamine receptor gene (DRD2) and Serotonin receptor gene (SLC6A4), counter acting internet app addiction, substance abuse and compulsive behaviour

xxx. 9 alleles combined of OXTR Single-Nucleotide Polymorphisms (SNPs) rs1042778, rs2254298, rs53576 modulate human evolution, from basic human needs of safety, belonging and esteem, to self actualisation, self transcendence and ascension

xxxi. Oxytocin creates connection between human beings, dissolving perceived separation including in context war and racism via empathy

ix. oxytocin modulates trust, growing businesses, economies and stock markets

xxxii. corporate social responsibility (csr) modulates customer happiness, employee happiness and share value via increases in trust, measured via oxytocin

xxxiii. the macroeconomics of happiness, trust and oxytocin modulate stock market growth

xxxiv. gamification and behavioural economics modulate usage of digital health interventions

x. EMDR modulates mental health

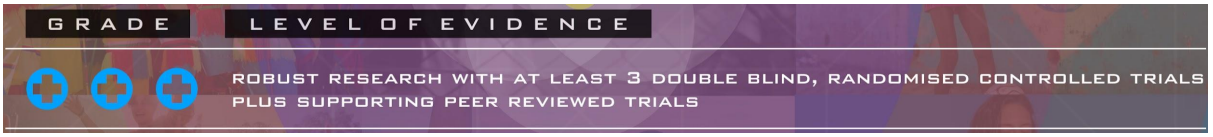
xxxv. EMDR modulates mental health, empowering happiness

xi. mantra, mindfulness & yoga modulate mental health

xxxvi. Mantra, mindfulness & yoga modulate mental health, empowering happiness

brain entrainment modulates our DNA

i. brain entrainment modulates pain, anxiety and heart rate variability



"The use of binaural beat technology in the theta brainwave frequency embedded into music decreases physical and psychological indications of stress. Binaural beat technology embedded with beta and delta frequencies may improve cognitive functioning and sleep quality, respectively."

The Effect of Binaural Beat Technology on the Cardiovascular Stress Response in Military Service Members With Postdeployment Stress
Journal of Nursing Scholarship
https://sigmapubs.onlinelibrary.wiley.com/doi/abs/10.1111/jnu.12304

randomised controlled trial, double blind
citations: 11

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"Acute exposure to theta-frequency binaural beats in a young, healthy sample of college students resulted in increased parasympathetic activation, increased sympathetic withdrawal, and increased self-reported relaxation post-exercise."

"Binaural-beat-associated HRV appeared to be more tightly coupled with self-reported relaxation than placebo-associated HRV."

"These findings support the putative clinical effectiveness of binaural beats in their own right, the effects of which may be synergistically enhanced through combination with other therapeutic factors such as verbal guidance and music."

Auditory driving of the autonomic nervous system: Listening to theta-frequency binaural beats post-exercise increases parasympathetic activation and sympathetic withdrawal
Frontiers in Psychology
https://www.frontiersin.org/articles/10.3389/fpsyg.2014.01248/full

randomised controlled trial, double blind
citations: 37

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"Significant change over time in the parasympathetic parameter HRV–HF power derived from ECG was only found under binaural beats"

"Pre– vs. post saliva samples revealed significant increases only under Placebo in stress–related biomarkers (Cortisol, α–Amylase and Salivary Secretory Immunoglobulin A, SIgA), but not under binaural beats"

"The ability of the binaural beat stimulus to decrease arousal levels and induce/boost altered states of consciousness (ASCs) implies applications in several practical–clinical contexts and calls for further field studies."

Psychophysiological reactivity to auditory Binaural Beats stimulation in the alpha and theta EEG brain-wave frequency bands: A randomized, double–blind and placebo–controlled study in human healthy young adult subjects
Independent University of Madrid, Department of Biological and Health Psychology
https://repositorio.uam.es/handle/10486/663948

randomised controlled trial, double blind
citations: 3

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"When placed under an acute stressor after using music embedded with BBT, participants showed a decrease in sympathetic response and an increased in parasympathetic response. Those who used music alone had the opposite effect. In other words, those who did not use the technology exhibited more cardiovascular signs of chronic stress exposure."

The Sound Mind Warrior Study: Using Sound Technology to Combat Stress in Military Service Members
Sigma Global Nursing
https://sigma.nursingrepository.org/handle/10755/616141

randomised controlled trial, double blind
citations:

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"The use of binaural-rhythm stimulation for Parkinson's Disease, as designed in this study, seems to be an effective, portable, inexpensive and noninvasive method to modulate brain activity"

"It resulted in a normalization of EEG power (altered in Parkinson's Disease), normalization of brain FC (also altered in Parkinson's Disease) and working memory improvement (a normalizing effect)"

"We consider that sound, particularly binaural-rhythmic sound, may be a co-assistant tool in the treatment of Parkinson's Disease"

Short-Term Effects of Binaural Beats on EEG Power, Functional Connectivity, Cognition, Gait and Anxiety in Parkinson's Disease
International Journal of Neural Systems
https://www.worldscientific.com/doi/abs/10.1142/S0129065717500551?journalCode=ijns

randomised controlled trial, double blind
citations: 22

"In conclusion, our findings demonstrate that patients using intra-operative Hemi-Sync tapes whilst undergoing a range of different surgical procedures under standardised general anaesthesia with a laryngeal mask require substantially less fentanyl when compared with controls"

"It is possible that the Hemi-Sync tapes, used intra-operatively or with the addition of pre- and postoperative tapes, could have longer-term effects on postoperative recovery, and may even result in cost savings through reduced analgesic requirement and shorter hospital stay."

"In our study, we found that patients exposed to a Hemi-Sync audio tape whilst undergoing surgery under 'light' general anaesthesia required significantly less analgesia with fentanyl when compared with patients listening to a blank tape or to classical music."

Hemispheric-synchronisation during anaesthesia: a double-blind randomised trial using audiotapes for intra-operative nociception control
Wiley Online Library: Anaesthesia
https://onlinelibrary.wiley.com/doi/full/10.1046/j.1365-2044.1999.00958.x

randomised controlled trial, double blind
citations: 66

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"parents and adolescents stated that homework problems due to inattention improved during the 3-week study. Parents and participants stated that the modality was easy to use and helpful. Therefore, this modality should be studied over a longer time frame in a larger sample to further its effectiveness to reduce the symptom of inattention in those diagnosed with attention-deficit/hyperactivity disorder"

Pilot Feasibility Study of Binaural Auditory Beats for Reducing Symptoms of Inattention in Children and Adolescents with Attention-Deficit/Hyperactivity Disorder
Journal of Paediatric Nursing
https://www.sciencedirect.com/science/article/pii/S0882596308003321

randomised controlled trial, double blind
citations: 42

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"offering binaural beat audio before day case procedures might serve to bring about anxiolysis in the majority of patients without impacting adversely on postoperative functioning"

A prospective, randomised, controlled study examining binaural beat audio and pre-operative anxiety in patients undergoing general anaesthesia for day case surgery
Anaesthesia
https://onlinelibrary.wiley.com/doi/full/10.1111/j.1365-2044.2005.04287.x

randomised controlled trial, double blind
citations: 191

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"In moderately anxious emergency department patients, state anxiety was reduced by 10%–15% following exposure to purpose-designed sound interventions."

Original sound compositions reduce anxiety in emergency department patients: a randomised controlled trial
Medical Journal of Australia
https://www.ncbi.nlm.nih.gov/pubmed/22171868

randomised controlled trial
citations: 39

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"Music, both with and without binaural beat, was proven to decrease anxiety level and lower systolic blood pressure."

"Patients who received binaural beat audio showed additional decrease in heart rate."

"Binaural beat embedded musical intervention may have benefit over musical intervention alone in decreasing operative anxiety."

The effect of music with and without binaural beat audio on operative anxiety in patients undergoing cataract surgery: a randomized controlled trial
Nature
https://www.nature.com/articles/eye2016160

randomised controlled trial
citations: 26

———

"complete group had the greatest reduction in both Fibromyalgia Impact Questionnaire and Health Assessment Questionnaire, with a significant improvement in Health Assessment Questionnaire, suggesting that the combined use of music, binaural beats and vibration exerts a greater effect on Fibromyalgia symptoms"

Effects of music in combination with vibration in acupuncture points on the treatment of fibromyalgia
Journal of Alternative and Complimentary Medicine
https://www.ncbi.nlm.nih.gov/m/pubmed/25635925/#fft

randomised controlled trial
citations: 15

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"the findings suggest that binaural-beat exposure before, and before and during the task produces superior results than exposure during the task"

"Time under exposure contributed significantly to the model indicating that longer periods are advisable to ensure maximum effectiveness"

"binaural-beat exposure is an effective way to affect cognition over and above reducing anxiety levels and the perception of pain without prior training, and that the direction and the magnitude of the effect depends upon the frequency used, time under exposure, and the moment in which the exposure takes place"

Efficacy of binaural auditory beats in cognition, anxiety, and pain perception: a meta-analysis
Psychological Research
https://link.springer.com/article/10.1007/s00426-018-1066-8

citations: 14

———

"Binaural beats are capable of entraining the brain."

"7 Hz binaural beats increase relative power of theta band in parietal and temporal lobes."

"6 min of stimulation is enough to significantly change the relative power."

Cumulative effects of theta binaural beats on brain power and functional connectivity
Biomedical Signal Processing and Control
https://www.sciencedirect.com/science/article/abs/pii/S1746809418300296

citations: 12

———

"Alpha rhythm during brain wave entrainment interacts with the brain stem control in the different HRV bands taken in consideration."

"we measured an increase in low frequency power heart rate variability ranging from 16% to 74%"

"and an increase in high frequency heart rate variability power ranging from 38% to 73%. These are very promising results."

Alpha-rhythm stimulation using brain entrainment enhances heart rate variability in subjects with reduced HRV
World Journal of Neuroscience
http://www.saistmp.com/publications/WJNS_20130906172219 78.pdf

citations: 17

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"BBs enhance alpha-band oscillation synchrony between the auditory cortices during auditory stimulation"

"Increased coherence selectively concerns the alpha band whatever the binaural beat frequency is."

Binaural beats increase interhemispheric alpha-band coherence between auditory cortices
Hearing Research
https://www.sciencedirect.com/science/article/pii/S0378595515300149

citations: 18

———

"Our results suggest that beat stimulation offers a non-invasive approach for the modulation of intracranial EEG characteristics."

Intracranial electroencephalography power and phase synchronization changes during monaural and binaural beat stimulation
European Journal of Neuroscience
https://onlinelibrary.wiley.com/doi/abs/10.1111/ejn.12760

citations: 50

———

"Preliminary evidence suggested that brain wave entrainment is an effective therapeutic tool"

A comprehensive review of the psychological effects of brainwave entrainment
Database of Abstracts of Reviews of Effects (DARE): Quality-assessed Reviews
https://www.ncbi.nlm.nih.gov/books/NBK75019/#_NBK75019_pubdet_

citations: 188

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"high-frequency gamma binaural beats did have a significant impact on the global precedence effect: the precedence effect became smaller, suggesting that visual attention became more focused than in the control condition."

"this is the first study to demonstrate that binaural beats induce an individual processing style that modulates visual attention."

"As we have argued, beats in the gamma range achieve this effect by increasing attentional focusing rather than by suppressing currently irrelevant tasks and task-related information."

More attentional focusing through binaural beats: evidence from the global–local task
Psychological Research
https://link.springer.com/article/10.1007/s00426-015-0727-0

citations: 27

"We conclude that binaural beats may be useful in reducing preoperative anxiety in dentistry."

Effectiveness of binaural beats in reducing preoperative dental anxiety
British Journal of Oral and Maxillofacial Surgery
https://www.sciencedirect.com/science/article/abs/pii/S0266435617300657

citations: 17

Exploratory Evidence for Correlations between Entrained Mental Coherence and Random Physical Systems
Journal of Scientific Exploration
http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.686.6679&rep=rep1&type=pdf

citations: 12

"This study demonstrates that guided imagery with theta music is a useful intervention for palliative care of patients with dyspnea."

"theta music alone was demonstrated to be effective, while soothing non-theta music was not effective."

"Guided imagery with theta music was more effective than theta music alone."

Efficacy of Guided Imagery With Theta Music for Advanced Cancer Patients With Dyspnea: A Pilot Study
Biological Research for Nursing
https://journals.sagepub.com/doi/abs/10.1177/1099800409347556

citations: 29

"Binaural beat technology may exhibit positive effect on self-reported psychologic measures, especially anxiety"

Binaural Beat Technology in Humans: A Pilot Study To Assess Psychologic and Physiologic Effects
The Journal of Alternative and Complimentary Medicine
https://www.liebertpub.com/doi/abs/10.1089/acm.2006.6196

citations: 121

"the beta-frequency beats were associated with less negative mood."

"Results suggest that the presentation of binaural auditory beats can affect psychomotor performance and mood."

"This technology may have applications for the control of attention and arousal and the enhancement of human performance."

Binaural Auditory Beats Affect Vigilance Performance and Mood
Physiology and Behavior
https://www.sciencedirect.com/science/article/abs/pii/S0031938497004368

citations: 245

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"alpha brain stimulation induces a constant improvement in the total power spectral density of heart rate variability (and thus induces an increase in total heart rhythm variability) and in particular in the parasympathetic (high frequency) band spectrum of hrv"

A Fast Fourier Transform Analysis of Time Series Data of Heart Rate Variability During Alfa-Rhythm Stimulation in Brain Entrainment
NeuroQuantology
http://www.neuroquantology.com/index.php/journal/article/view/632

citations: 6

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Isochronic Tones in the Schumann Resonance Frequency for the Treatment of Anxiety: A Descriptive Exploratory Study
Faculty of Saybrook University
https://search.proquest.com/openview/4ff918413d437dd05406e7c667256d5c/1?pq-origsite=gscholar&cbl=18750&diss=y

citations:

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"Listening to binaural beat tapes in the delta/theta electroencephalogram range may be beneficial in reducing mild anxiety."

"Future studies should account for music preference among participants and include age as a factor in outcomes, incentives to foster tape listening, and a physiologic measure of anxiety reduction."

Use of binaural beat tapes for treatment of anxiety: a pilot study of tape preference and outcomes
Alternative Therapy Health Medicine
https://www.ncbi.nlm.nih.gov/m/pubmed/11191043/

citations: 108

———

An Alpha-wave-based binaural beat sound control system using fuzzy logic and autoregressive forecasting model
IEEE
https://ieeexplore.ieee.org/document/4654631

citations: 12

Cortical Steady-State Responses to Central and Peripheral Auditory Beats
Cerebral Cortex
https://academic.oup.com/cercor/article/18/5/1193/349492

citations: 86

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ii. 528hz modulates gene expression, increasing oxytocin, regenerating brain tissue and treating alcohol addicted cell damage.



*"**In the 528 Hz condition**, mean levels of **cortisol significantly decreased**, **chromogranin A tended to decrease**, and **oxytocin doubled after only 5 minutes of music exposure**. However, no significant change was observed in any salivary biomarkers in the 440 Hz condition."*

*The influence of music on the autonomic nervous system and endocrine system varies depending on the frequency of the music, and furthermore, that **528 Hz music has an especially strong stress-reducing effect**, even **following only five minutes of exposure**."*

Effect of 528 Hz Music on the Endocrine System and Autonomic Nervous System

Scientific Research Publishing
https://www.scirp.org/pdf/Health_2018090514214117.pdf

citations: 4

———

"sound waves with 528 Hz frequency in 100 dB intensity induce testosterone production in brain by enhancing StAR and SF-1 and reducing P450 aromatase gene expression."

"Frequency of 528 Hz also reduces total concentration of reactive oxidative species in brain tissue."

"Prolonged exposure to this sound wave showed reduction of anxiety related behaviors in rats. The results reveal that reduced anxiety is related to increased concentration of testosterone in brain. This study may lead to ascertain a possible therapy in which sounds may be utilized to reduce anxiety in individual."

Influence of various intensities of 528 Hz sound-wave in production of testosterone in rat's brain and analysis of behavioral changes

Genes and Genomics
https://link.springer.com/article/10.1007/s13258-018-0753-6

citations:

———

"The current results show that in IC50 of ethanol, the frequency of 528 Hz increased cells viability about 20% and the level of ROS production has been reduced up to 100%."

"In this research the complex effects of two agents such as ethanol along with 528 Hz sound wave was studied and results of human astrocyte cell culture exposed with ethanol and 528 Hz sound wave showed that the ethanol damages reduced by the sound wave and therefore it can be useful for alcohol addicted peoples."

"The neural changes induced by alcohol consumption may be related to the disorder of signaling pathways such as protein expression and gene transcription at the cellular level. Therefore, the use of these sound waves can be useful to reduce the toxic effects of ethanol on astrocytes cells culture."

The Effects of 528 Hz Sound Wave to Reduce Cell Death in Human Astrocyte Primary Cell Culture Treated with Ethanol

Journal of Addiction Research & Therapy
https://www.omicsonline.org/open-access/the-effects-of-528-hz-sound-wave-to-reduce-cell-death-in-human-astrocyteprimary-cell-culture-treated-with-ethanol-2155-6105-1000335.php?aid=91771

Citations: 2

———

"Central Nervous System, high frequencies, psychological effects"

''the resulting estimated pitch function (where the minimal discharge interval is assumed to occur at 15khz) is in excellent agreement with that psychoacoustically generated function of reference frequency 528hz which may be expected to possess the least bias''

Neural Coding of High-Frequency Tones
NASA
https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/19760013713.pdf

citations: 2

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528hz, value of 6 in modulus 9, is the central harmonic of the fibonacci infinite fractal sequence, solfeggio infinite fractal sequence and mitosis infinite fractal sequence.

additional research here

iii. 7.83 HZ modulates DNA, increasing dopamine, inhibiting cancer cell growth and HIV cell growth



"exposure to square wave with 7.83±0.3 Hz (sweep step 0.1 Hz) was shown to inhibit the growth of B16F10 melanoma tumor cells."

Inhibition of B16F10 Cancer Cell Growth by Exposure to the Square Wave with 7.83+/-0.3Hz Involves L- and T-Type Calcium Channels
Electromagnetic Biology and Medicine
https://www.tandfonline.com/doi/abs/10.1080/15368378.2020.183949

citations: 3

———

"The results yielded a 17% inhibition rate under 7.83 Hz compared with that of the control group.

Moreover, sweep frequencies in narrow intervals (7.83 ± 0.1 Hz for the step 0.05 Hz) caused an inhibition rate of 26.4%, and inhibitory effects decreased as frequency sweep intervals increased.

These results indicate that a Schumann resonance frequency of 7.83 Hz can inhibit the growth of cancer cells and that using a specific frequency type can lead to more effective growth inhibition."

Effects of extremely low-frequency electromagnetic fields on B16F10 cancer cells
Electromagnetic Biology and Medicine
https://www.tandfonline.com/doi/abs/10.1080/15368378.2019.1591438

citations:

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"The method uses the Schuman frequency, or any ELF, to induce the DNA solution to emit electromagnetic signals in the range 300 - 4000 Hz that are producing spectra that result to be typical for each disease"

New Perspectives of Bioelectromagnetics in Biology and in Medicine
DNA Spectra for Diagnostic Purposes
https://iopscience.iop.org/article/10.1088/1742-6596/329/1/012011/meta

citations: 4

As discovered by Nobel Prize winner Luc Montagnier, who discovered HIV.

"DNA signalling is stimulated by 7 Hz naturally occurring waves on earth. Waves produced by the human brain are also in the range of 7 H z."

"The stimulation by the electromagnetic background of very low frequency is essential. The background is either produced from natural sources (the Schumann resonances [4] which start at 7.83 Hz) or from artificial sources"

"This result shows that, upon 7 H z excitation, the transmission into pure water of the oscillation of the nanostructures initially originated from DNA has been achieved."

DNA waves and water
Journal of Physics
https://www.researchgate.net/publication/230937823_DNA_waves_and_water

citations: 238

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"In particular, it is immediately recognized the observed relevance of extremely low signal (ELS) in the phenomena under study. The stimulation caused by the electromagnetic background of very low frequency is indeed observed to be essential in order for the DNA-water system to emit the EMS. In the experiments, the background ELS is either produced from natural sources (the Schumann resonances which start at 7.83 Hz (Montagnier, A ¨ıssa, Del Giudice, et al., 2011; Nickolaenko and Hayakawa, 2002)) or from artificial sources."

Transduction of DNA information through water and electromagnetic waves
Electromagnetic Biology and Medicine
https://arxiv.org/pdf/1501.01620.pdf

citations: 96

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"Electromagnetic signals of low frequency have been shown to be durably produced in aqueous dilutions of the Human Immunodeficiency Virus DNA.

In vivo, HIV DNA signals are detected only in patients previously treated by antiretroviral therapy and having no detectable viral RNA copies in their blood.

We suggest that the treatment of AIDS patients pushes the virus towards a new mode of replication implying only DNA, thus forming a reservoir insensitive to retroviral inhibitors. Implications for new approaches aimed at eradicating HIV infection are discussed."

Electromagnetic detection of HIV DNA in the blood of AIDS patients treated by antiretroviral therapy
Interdiscipinary Science (by Nobel Prize winner Luc Montagnier, who discovered HIV)
https://www.ncbi.nlm.nih.gov/pubmed/20640802

citations: 65

"The Gateway Process is a training system for obtaining power, concentration, and consistency of the amplitude and frequency of brain waves between the left and right hemispheres to make consciousness change,"

"the entire body, based on its own micromotion, functions as a tuned vibrational system which transfers energy in a range between 6.8 and 7.5 Hertz into the Earth's ionospheric cavity, which itself resonates at about 7-7.5 Hertz."

Analysis and Assessment of Gateway Process
US Central Intelligence Agency (CIA)
https://www.cia.gov/readingroom/docs/CIA-RDP96-00788R001700210016-5.pdf

citations: 6

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*"This is the first study to provide evidence of extrastriatal DA **D2 (DRD2 dopamine receptor)** modulation following **10 Hz** repetitive transcranial magnetic stimulation"*

rTMS of the Left Dorsolateral Prefrontal Cortex Modulates Dopamine Release in the Ipsilateral Anterior Cingulate Cortex and Orbitofrontal Cortex | PLOS ONE | Citations: 336

_____

*"All subjects underwent 3 PHNO-PET scans preceded by rTMS (sham, 1 Hz or **10 Hz**), on 3 separate days. Low frequency rTMS (1 Hz), targeting the insular cortex, significantly decreased **dopamine** levels in the substantia nigra, sensorimotor striatum and associative striatum."*

Deep TMS of the insula using the H-coil modulates dopamine release: a crossover [11C] PHNO-PET pilot trial in healthy humans

Brain Imaging And Behavior | Citations: 27

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*"Five patients with a major depressive episode (DSM-IV) underwent an acute **10 Hz** rTMS challenge with 3000 stimuli over the left dorsolateral prefrontal cortex during an IBZM-SPECT bolus and constant infusion protocol.*

*Data of nine SPECT investigations showed a significant reduction by 9.6 ± 6.2% in IBZM **binding to** striatal **dopamine D2 receptors** after rTMS challenge compared to baseline.*

*The reduction of IBZM binding observed after rTMS challenge is suggestive of a **release** in **endogenous dopamine** induced by prefrontal rTMS."*

Striatal dopamine release after prefrontal repetitive transcranial magnetic stimulation in major depression: Preliminary results of a dynamic [123I] IBZM SPECT study

Journal of Psychiatric Research | Citations: 161

“dopamine modulates addiction”

10-Hz Repetitive Transcranial Magnetic Stimulation of the Left Dorsolateral Prefrontal Cortex Reduces Heroin Cue Craving in Long-Term Addicts

Biological Psychiatry | Randomised Controlled Trial | Citations: 102

———

“10 Hz rTMS applied at 2000 stimuli and 110% intensity may induce significant changes in resting EEG in healthy subjects”

The effects of 10 Hz repetitive transcranial magnetic stimulation on resting EEG power spectrum in healthy subjects

Neuroscience Letters | Citations: 69

———

“Repetitive transcranial magnetic stimulation (rTMS) alters brain neurochemistry (dopamine, serotonin)

Different rTMS parameters had selective effects on neurochemicals (dopamine, serotonin)

method is a precise and sensitive tool to measure neurochemical changes”

Simultaneous quantification of dopamine, serotonin, their metabolites and amino acids by LC-MS/MS in mouse brain following repetitive transcranial magnetic stimulation

Neurochemistry International | Citations: 14

———

“In line with earlier investigations, our results suggest a moderate — potentially clinically relevant — treatment effect of prefrontal 10 Hz rTMS stimulation in chronic patients.”

Effects of 10 Hz repetitive transcranial magnetic stimulation (rTMS) on clinical global impression in chronic schizophrenia

Psychiatry Research | Citations: 48

———

One session of high frequency repetitive transcranial magnetic stimulation (rTMS) to the right prefrontal cortex transiently reduces cocaine craving

Drug and Alcohol Dependence | Citations: 276

"delivered with 10 Hz stimulation frequency"

Dopamine release in human striatum induced by repetitive transcranial magnetic stimulation over dorsolateral prefrontal cortex

The Journal of Nuclear Medicine | Citations 2

———

*"**Fifty veterans with PTSD** received 10 days of sham-controlled iTBS (1,800 pulses/day), followed by 10 unblinded sessions."*

*"**iTBS appears to be a promising new treatment for PTSD**."*

*"intermittent **theta** burst **stimulation** fires a frequency of 50hz at 5 cycles per minute, **5hz**, for a therapeutic duration of 3 minutes"*

Theta-Burst Transcranial Magnetic Stimulation for Posttraumatic Stress Disorder

AM J Psychiatry | Randomized Controlled Trial

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*"We screened **195 articles** identified upon search from PubMed, EBSCO/CINAHL, and CENTRAL; 23 were retrieved for inclusion assessment. Ultimately **10 articles were included in the meta-analyses**. Of these, nine were single-center trials and one was a multicenter Canadian study1. Four studies took place in Germany two in Taiwan and one each in Australia, Belgium, Canada, and Israel"*

*"The finding of **positive effects of TBS on major depressive disorder** confirm prior meta-analyses, expand upon prior findings, and support the growing interest in application of TBS protocols for treating neuropsychiatric disorders such as major depressive disorder."*

Theta burst stimulation for the acute treatment of major depressive disorder: A systematic review and meta-analysis

Nature | Meta Analysis

———

*"In summary, greater accumulated exposure to active **iTBS demonstrated clinically meaningful improvements in the year following stimulation**, and default mode connectivity could be used to predict longer-term outcomes."*

One-year clinical outcomes following theta burst stimulation for post-traumatic stress disorder

Nature

fibonacci sequence modulates cardiovascular, cellular and DNA health

iv. fibonacci sequence modulates cosmological, psychological and biological functions of all dna based life, via first, second and third order harmonics - Fibonacci Mod 9, Pisano Period 24 (Fib M9 P24)



Fibonacci sequence as an infinite repeating cycle of 9 harmonics, within 24 positions

"Modulo was initially introduced into mathematics in the context of modular arithmetic by Carl Friedrich Gauss in 1801."

"In theoretical mathematics, modular arithmetic is one of the foundations of number theory, touching on almost every aspect of its study, and it is also used extensively in group theory, ring theory, knot theory, and abstract algebra.

In applied mathematics, it is used in computer algebra, cryptography, computer science, chemistry and the visual and musical arts."

"The fibonacci sequence is simplified as an infinitely repeating fractal of 24 within modulus 9"

Modular Arithmetic
Collective references including Wolfram Mathematica, summarised on Wikipedia
https://en.wikipedia.org/wiki/Modular_arithmetic

citations:
———

Methods of factoring and modular arithmetic
Google Patent Registration
https://patents.google.com/patent/US6714954B2/en

citations: 205
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Modular arithmetic operation system
Google Patent Registration
https://patents.google.com/patent/US5499299A/en

citations: 54
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*"**Solfeggio** is the musical term given to **frequencies** which **modulate** within ratios of **Fibonacci Sequence Mod 9 Pisano Period 24**.*

9 Seed	1-8 Seed	2-7 Seed	3 Seed	4-5 Seed	5-4 Seed	6 Seed	7-2 Seed	8-1 Seed
in hertz								
123	135	147	159	162	174	186	198	
222	234	246	258	261	273	285	297	219
333	345	357	369	372	384	396	318	321
444	456	468	471	483	495	417	429	432
555	567	579	582	594	516	528	531	543
666	678	681	693	615	627	639	642	654
777	789	792	714	726	738	741	753	765
888	891	813	825	837	849	852	864	876
999	912	924	936	948	951	963	975	987

What makes a scale a solfeggio?

1. All of the tones in sequential order must be 111 hz apart using modulus 9.

2. Each scale must be 12 hz apart from its adjoining scale using modulus 9.

3. Adding any two or more frequencies will always yield another solfeggio frequency.

4. The sum of the scale frequencies will always add up to 999 hz.(modulus 9).

5. The horizontal sum of all frequencies across all scales always adds up to 999 hz.

6. All frequencies sum to one of 3,6,9 in modulus 9.

Solfeggio Sequence
GW Hardin
http://www.soundsofwonder.com/articles.html

citations:
———

Fibonacci Sequence Modulus 9 Pisano Period 24 (Fib M9 P24) is the digital root (modulus 9) of the Fibonacci sequence, as 3 nested fractals within a 24 cycle, an icositetragon (24 sided polygon) with interior angles summing to 3960 degrees.

[I o H] hypothesize these triplet of fibonacci sequences are universal across biological, cosmic and quantum levels, following nucleotide triplets within DNA through to fundamental sub atomic particles of electrons, protons & neutrons. What is revealed are three interlocking harmonic cycles of the one fibonacci sequence. [I o H] hypothesize Fib M9 P24 is a shadow of the 8th dimensional quasicrystal E8, proposed as the unified field by Garret Lisi & Quantum Gravity Research.

Fib M9 P24 fractal α ~ 7.83hz (9) | (7.83hz in modulus 9 =9)

Fib M9 P24 nested fractal sequence of 1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9 forms a spiral

Fib M9 P24 fractal β ~ 111hz (3) | (111hz in modulus 9 =3)

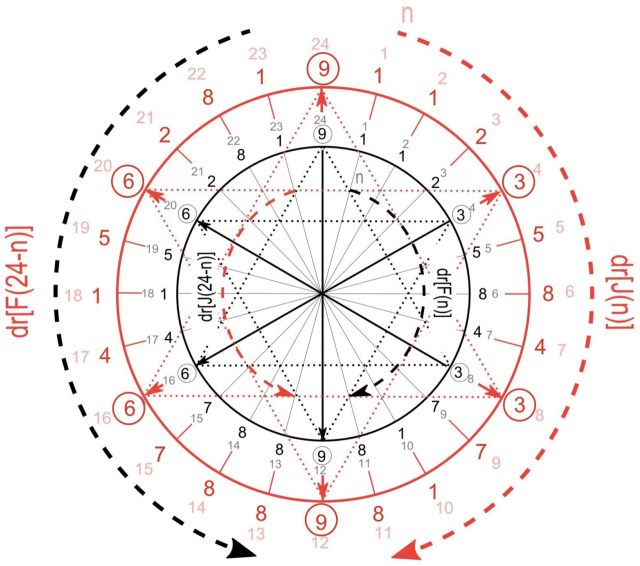
A mitosis nested fractal of 1, 2, 4, 8, 7, 5 modulates within this sequence as a hexagonal toroidal geometry

Fib M9 P24 fractal γ ~ 528hz (6) | (528hz in modulus 9 = 6)

A solfeggio nested fractal of 3, 6, 9, 3, 6, 9 modulates within this sequence in a hexagonal star tetrahedral geometry

position	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
Fibonacci Sequence	0	1	1	2	3	5	8	13	21	34	55	89	144	233	377	610	987	1,597	2,584	4,181	6,765	10,946	17,711	28,657	46,368	75,025	121,393	196,418	317,811	514,229	832,040	1,346,269	2,178,309	3,524,578
Fibonacci Mod 9 Pisano 24 - α	0	1	1	2	3	5	8	4	3	7	1	8	0	8	8	7	6	4	1	5	6	2	8	1	0	1	1	2	3	5	8	4	3	7
fibonacci Mod 9 Pisano 24 - workings	0	1	1	2	3	5	8	4	3	7	1	8	0	8	8	7	6	4	1	5	6	2	8	1	0	1	1	2	3	5	8	4	3	7
fibonacci Mod 9 Pisano 24 - γ	9				3				3				9				6				6				9				3				3	
fibonacci Mod 9 Pisano 24 - workings		4				17				16				23				10				11				4				17				16
fibonacci Mod 9 Pisano 24 - β		4				8				7				5				1				2				4				8				7

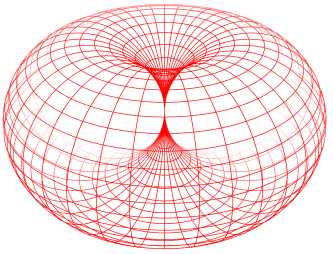
view workings here in **science** tab



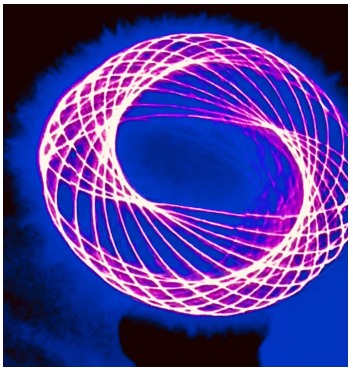
Reverse Fibonacci sequence and its description
Academia | citations: 1



7.83hz x 111hz + 528hz - 2D



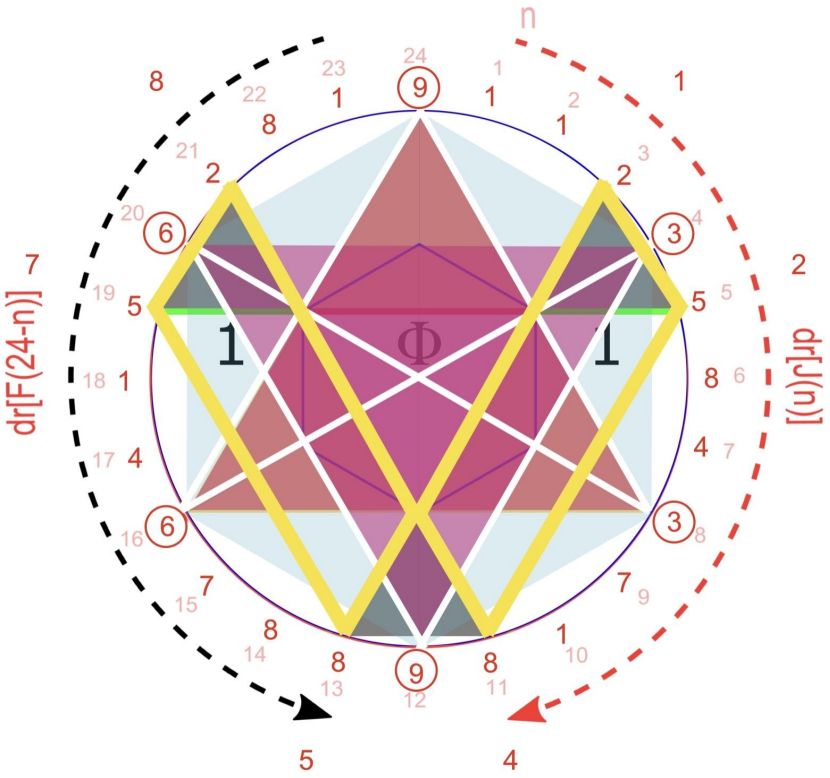
Fib M9 P 24 in 3D is toroidal



7.83hz x 111hz + 528hz - 3D

"sound modulating water via cymascope photography experiments including life viewer patent by David Schiermeyer"

Cymatics for the cloaking of flexural vibrations in a structured plate
Nature
https://www.nature.com/articles/srep23929



Fibonacci Sequence Mod 9, Pisano Period 24

"Results showed a significant correlation between celestial mechanics and Fibonacci numbers measured in 24-hour periods with an average deviation of less than 3%."

"Specifically, the Fibonacci series expressed in 24 hour-based units predicted the period of rotation, precession, and orbit across multiple time scales."

"The basic Fibonacci sine wave pattern has implications that might extend to quantum mechanics and the observable properties of a system. To draw such a conclusion or indication for quantum mechanics, it may be noted that the golden ratio and Fibonacci numbers appear in quantum mechanics"

Modeling Celestial Mechanics Using the Fibonacci Numbers
ResearchGate
https://www.researchgate.net/profile/Robert_Sacco/publication/335313110_Modeling_Celestial_Mechanics_Using_the_Fibonacci_Numbers/links/5d5d9396a6fdcc55e81ed976/Modeling-Celestial-Mechanics-Using-the-Fibonacci-Numbers.pdf?origin=publication_detail

citations: 4

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"synchronicity in the context of a mathematical system in which mind and matter play complementary roles in a reality that has a mathematical structure"

"Jung recognized the conceptual value of postulating that synchronicity operated based on the Fibonacci numbers because of their ubiquity in nature. Jung observed that "...synchronicity... is a secondary effect of the primary coincidence of mental and physical events (as in the Fibonacci series). The bridge seems to be formed by the numbers"

"The central point to the FLCM (Fibonacci Life Chart Method) analysis of human development is that human development depends on the self-organizing properties of the Fibonacci sequence.

When the Fibonacci numbers progressively increase, therefore, the self-organizing and fractal properties result in biological and psychological phase transition.

Phase transitions can result from self-organization processes at multiple levels (e.g., molecules, genes, cell, organ, organ system, organism, behavior, and environment) that can influence each other. The self-organized hierarchy begins at the mathematical and quantum level."

"synchronicity will operate based on Fibonacci numbers controlling the interplay of genes and brain development, which change coherence of brain waves, which cycle around again to influence brain development."

Fibonacci Harmonics: A New Mathematical Model of Synchronicity
Applied Mathematics
https://www.scirp.org/html/9-7403945_85739.htm

citations: 4

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"Fibonacci Life-Chart Method (FLCM) categorizes phase transitions and Phi fractal scaling in human development based on the occurrence of Fibonacci numbers in biological cell division and self-organizing systems."

The Fibonacci Life-Chart Method (FLCM) as a foundation for Carl Jung's theory of synchronicity
Journal of Analytical Psychology
https://onlinelibrary.wiley.com/doi/abs/10.1111/1468-5922.12204

citations: 9

"Fibonacci numbers can be divided based on modulo 9 of the iterative digit sum and the following 9 sequences can be obtained:"

"The iterative digit sum of Fibonacci numbers produced numbers that are recursive, a pattern of numbers that are repeated after every 24th consecutive terms. This can also be used in predicting the iterative digit sum of large Fibonacci numbers."

Digit and Iterative Digit Sum Of Fibonacci Numbers, Their Identities and Powers
International Journal of Applied Engineering Research
http://eprints.covenantuniversity.edu.ng/6530/#.X91ofIE8afC

Table 1: Pattern of the iterative digit sum of Fibonacci numbers.

Iterative digit sum	Terms
1	1, 25, 49, …
1	2, 26, 50, …
2	3, 27, 51, …
3	4, 28, 52, …
5	5, 29, 53, …
8	6, 30, 54, …
4	7, 31, 55, …
3	8, 32, 56, …
7	9, 33, 57, …
1	10, 34, 58, …
8	11, 35, 59, …
9	12, 36, 60, …
8	13, 37, 61, …
8	14, 38, 62, …
7	15, 39, 63, …
6	16, 40, 64, …
4	17, 41, 65, …
1	18, 42, 66, …
5	19, 43, 67, …
6	20, 44, 68, …
2	21, 45, 69, …
8	22, 46, 70, …
1	23, 47, 71, …
9	24, 48, 72, …

citations: 8

"This study found the FLCM produced a developmental pattern characterized by eight recognizable stages.

This finding constitutes a new classification of Erikson's eight developmental stages. The present research provides support for Erikson's epigenetic view of predetermined, sequential stages to human development based on the occurrence of Fibonacci numbers in biological cell division and self-organizing systems.

This method may help identify populations at risk for psychological disorder, which would allow early intervention."

Re-Envisaging the Eight Developmental Stages of Erik Erikson: The Fibonacci Life-Chart Method (FLCM)
Journal of Educational and Developmental Psychology
https://www.researchgate.net/publication/269958479_Re-Envisaging_the_Eight_Developmental_Stages_of_Erik_Erikson_The_Fibonacci_Life-Chart_Method_FLCM

citations: 46

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"This is the first study to demonstrate the appearance of the Fibonacci Cascade within the distribution of coronary artery lesions in the human heart."

Association of the Fibonacci Cascade with the distribution of coronary artery lesions responsible for ST-segment elevation myocardial infarction
The American Journal of Cardiology
https://www.ajconline.org/article/S0002-9149(03)00731-8/fulltext

citations: 26

———

"The new phenotypic, geometrical biomarkers based on the golden ratio (φ) were associated with HRV indices and especially with those computed at the frequency domain. These associations were inde- pendent from age, gender, body mass index, mean arte- rial pressure, pulse pressure and mean 24-hr heart rate"

A novel geometrical analysis of the arterial pulse based on the golden ratio φ (PHI): Association with heart rate variability
Archives of the Balkan Medical Union
https://www.researchgate.net/profile/Dimitrios_Vrachatis/publication/326005385_A_novel_geometrical_analysis_of_the_arterial_pulse_based_on_the_golden_ratio_ph_PHI_Association_with_heart_rate_variability/links/5b546a8e0f7e9b240ffa57b9/A-novel-geometrical-analysis-of-the-arterial-pulse-based-on-the-golden-ratio-ph-PHI-Association-with-heart-rate-variability.pdf

citations: 2

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"overall cardiac and ventricular dimensions in a normal heart are consistent with the golden ratio and angle, representing optimum pump structure and function efficiency, whereas there is significant deviation in the disease state. These findings could have anatomical, functional and prognostic value as markers of early deviation from normality."

The human heart: Application of the golden ratio and angle
International Journal of Cardiology
https://www.sciencedirect.com/science/article/abs/pii/S0167527311005079

citations: 45

———

"Golden ratio and Fibonacci Series ubiquitously exist in every manifestation of life.

'Fibonacci Cascade' associates with branching patterns of coronary arteries.

Left ventricular diameters represent the concept of golden ratio.

Golden Ratio manifests itself in systolic and diastolic intervals of the heart.

Evaluation of golden proportions might serve as a potential therapeutic guide"

Golden Ratio and the heart: A review of divine aesthetics
International Journal of Cardiology
https://www.sciencedirect.com/science/article/abs/pii/S0167527316305885

citations: 28

———

"The need for energy and nutrients are optimized in golden ratio.

Our heart is beating and pumping in golden ratio.

The harmony in our body and cardiovascular system seem to be regulated by subtle forces."

Golden ratio: A subtle regulator in our body and cardiovascular system?
International Journal of Cardiology
https://www.sciencedirect.com/science/article/abs/pii/S016752731631854X

citations: 18

———

"Fibonacci numbers and the golden ratio can be found in nearly all domains of Science, appearing when self-organization processes are at play and/or expressing minimum energy configurations.

Several non-exhaustive examples are given in biology (natural and artificial phyllotaxis, genetic code and DNA), physics (hydrogen bonds, chaos, superconductivity), astrophysics (pulsating stars, black holes), chemistry (quasicrystals, protein AB models), and technology (tribology, resistors, quantum computing, quantum phase transitions, photonics)."

Fibonacci Numbers and the Golden Ratio in Biology, Physics, Astrophysics, Chemistry and Technology: A Non-Exhaustive Review
arXiv Cornwall University
https://arxiv.org/abs/1801.01369

citations: 6

———

"Golden Ratio (GR) has multiple representations in cardio-vascular physiology."

We proposed a synthetic fractal indicator (SFI) for ECG analysis during exercise. The SFI and heart rate showed very strong correlations at rest and postexercise.

Our results showed a fractal dimension of the dynamic of the analysed ECG ratios."

ECG response to submaximal exercise from the perspective of Golden Ratio harmonic rhythm
Biomedical Signal Processing and Control
https://www.sciencedirect.com/science/article/abs/pii/S1746809417302276

citations: 11

———

"Kowalczys et al. [1] found mean daytime systolic blood pressure and diastolic blood pressure of patients with stable congestive heart failure as 114 mmHg and 70 mmHg, respectively. The ratio of SBP to DBP gives 1.62, which is very close to the golden ratio"

Golden ratio in congestive heart failure: A promising proportion for prognosis and decompensation
Cardiology Journal
https://www.researchgate.net/profile/Ertan_Yetkin/publication/346955762_Golden_ratio_in_congestive_heart_failure_A_promising_proportion_for_prognosis_and_decompensation/links/5fd517a692851c13fe80f2fd/Golden-ratio-in-congestive-heart-failure-A-promising-proportion-for-prognosis-and-decompensation.pdf

citations:
———

Golden Ratio is beating in our heart
Trakya University Faculty of Medicine Department of Cardiology, Edirne, Turkey
https://www.researchgate.net/profile/Ertan_Yetkin/publication/253332878_Golden_Ratio_is_beating_in_our_heart/links/5a525d18a6fdcc7690026854/Golden-Ratio-is-beating-in-our-heart.pdf

citations: 27
———

"we develop a matrix model of Fibonacci cell kinetics based on an asymmetric pause between mitosis by daughter cells"

"our result suggests a role for Fibonacci cell kinetics in the genesis of multicellular hollow and tubular structures"

Ring shape Golden Ratio multicellular structures are algebraically afforded by asymmetric mitosis and one to one cell adhesion
bioRxiv
https://www.biorxiv.org/content/10.1101/450528v1.abstract

citations:
———

"The study strongly suggests that esthetics can indeed be made scientific rather than the need to resort to subjective perceptions as in the past."

The biologic significance of the divine proportion and Fibonacci series
American Journal of Orthodontics
https://www.sciencedirect.com/science/article/abs/pii/0002941682900732

citations: 514
———

“without the golden ratio, we would not have the form or function of the proton, cell, athlete, horse, species, planet, solar system or galaxy.”

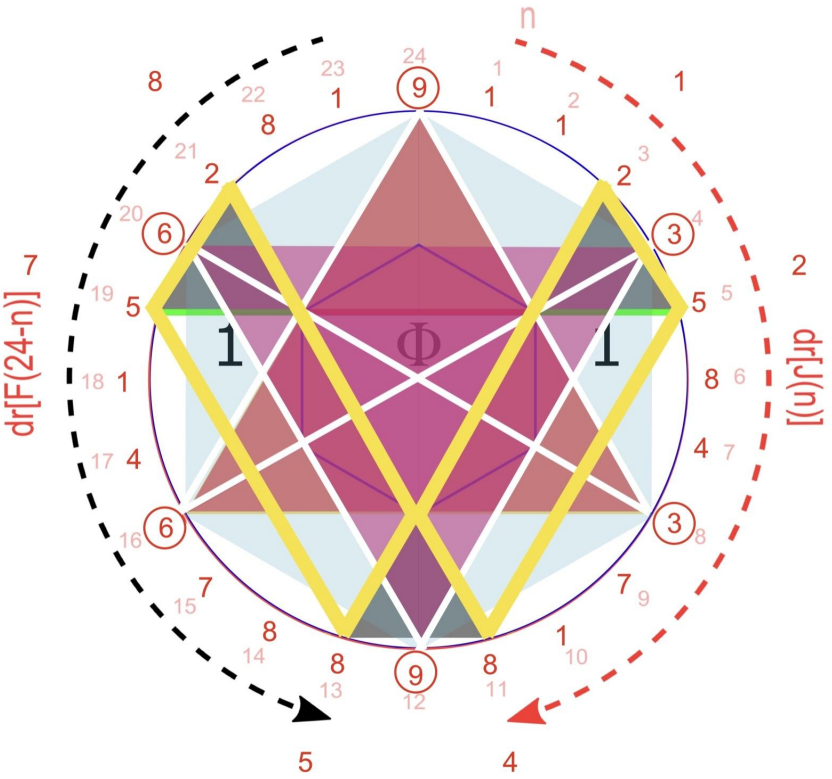
Golden ratio beauty as scientific function
Assessing Well-Being: Aesthetic and Political Atmospheres
https://riviste.unimi.it/index.php/Lebenswelt/article/view/9457

citations: 2

“recent evidence supports a century-old concept that maldistribution of chromosomes (and resultant genomic instability) due to abnormalities in mitosis itself is the primary cause of malignancy rather than a mere byproduct”

A new theory of the origin of cancer: quantum coherent entanglement, centrioles, mitosis, and differentiation
Biosystems
https://www.sciencedirect.com/science/article/abs/pii/S0303264704000619

citations: 56



“we report the crystal structure of human OTR in complex with retosiban, a nonpeptidic antagonist developed as an oral drug for the prevention of preterm labor. Our structure reveals insights into the detailed interactions between the G protein–coupled receptor (GPCR) and an OTR-selective antagonist.

Here, we report the first crystal structure of OTR.

Together, these results further our molecular understanding of the oxytocin/vasopressin receptor family and will facilitate structure-guided development of new therapeutics.”

Crystal structure of the human oxytocin receptor
Structural Biology
https://advances.sciencemag.org/content/6/29/eabb5419

citations: 5

Supplementary Materials for Crystal structure of the human oxytocin receptor
Structural Biology
https://advances.sciencemag.org/content/advances/suppl/2020/07/13/6.29.eabb5419.DC1/abb5419_SM.pdf

citations: 5

"The first crystal structure of the pituitary hormone oxytocin complexed with its carrier protein neurophysin has been determined and refined to 3.0 Å resolution."

Crystal structure of the neurophysin—oxytocin complex
Nature: Nature Structural & Molecular Biology
https://www.nature.com/articles/nsb0296-163

citations: 113

Oxytocin Receptor Signaling in Vascular Function and Stroke
Frontiers in Neuroscience
https://www.frontiersin.org/articles/10.3389/fnins.2020.574499/full

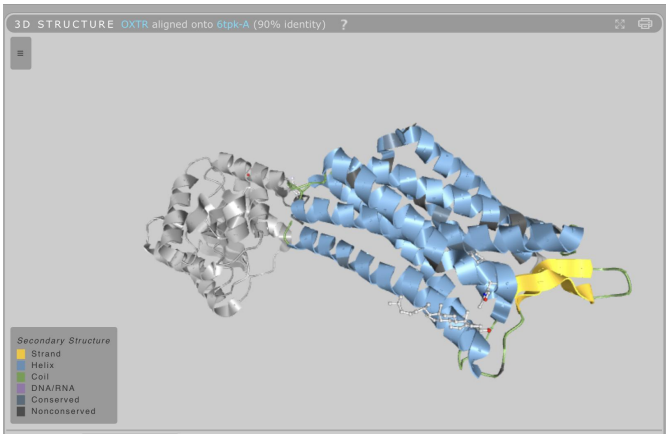
citations:

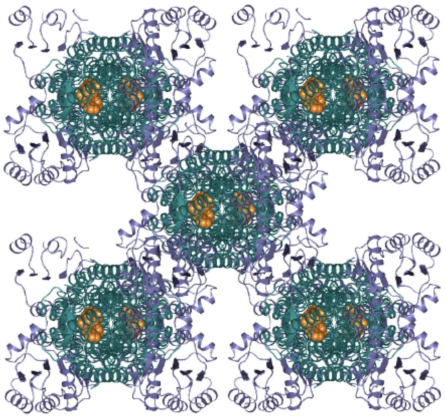
"three models are constructed and refined using known 3D crystal structures of evolutionary related proteins"

Molecular modeling of 3D structure of the oxytocin receptor.
Discovery of novel oxytocin receptor agonists via molecular docking studies.
Uit Munin
https://munin.uit.no/handle/10037/18312

citations:

3D Structure of OXTR
Aquaria
https://aquaria.ws/P30559/6tpk/A





AQUARIA

citations:

"Occurrence of Fibonacci Sequence Numbers in HOX Gene Organization and Expression"

Occurrence of Fibonacci numbers in development and structure of animal forms: Phylogenetic observations and epigenetic significance
Natural Science
https://www.scirp.org/html/4-8301619_18625.htm

citations: 7

"we demonstrate the existence of a palindromic-like mirror structure that exists in real genomes and disappears completely in synthetic genomes. This biomathematic meta-organisation is based on characteristic proportions of Fibonacci numbers between DNA single strand nucleotides proportions TC / AG on the one hand and TG / AC on the other hand."

Epigenetics Theoretical Limits of Synthetic Genomes: The Cases of Artificials
IBM European Research Center on Artificial Intelligence
https://www.preprints.org/manuscript/201907.0120/v1/download

citations: 3

"In the case of the Genetic Code, the codons form the genotypes and amino acids phenotypes. The pattern of synthesis of the different groups in amino acids is observed in terms of Fibonacci sequence, 1 1 2 3 5 8.

The asymmetric form of the Genetic Code table, the Genetic Tableau, has a pattern formed by the six right-hand Fibonacci spirals, 1 1 2. A similar pattern for the six Fibonacci spirals 1 1 2 3 5 8 is observed by matching the elements of the DOGU group of polyhedra classifying proteins, [3].

This precision in pattern formation is intrinsic in the Fibonacci sequence, (Fn= Fn-2+Fn-1)."

Epigenetics of Self-Organization of Biomolecules
Biophysical Journal
https://www.cell.com/biophysj/fulltext/S0006-3495(09)05422-8

citations:

"relevance and the applications possibilities of the Fibonacci sequence, and also its q-deformed or quantum extension, in the study of the genetic code(s).

First, after the presentation of a new formula, an indexed double Fibonacci sequence, comprising the first six Fibonacci numbers, is shown to describe the 20 amino acids multiplets and their degeneracy as well as a characteristic pattern for the 61 meaningful codons.

Next, the twenty amino acids, classified according to their increasing atom-number (carbon, nitrogen, oxygen and sulfur), exhibit several Fibonacci sequence patterns. Several mathematical relations are given, describing various atom-number patterns.

A Mathematical Model for the Genetic Code (S) Based on Fibonacci Numbers and their q-Analogues
NeuroQuantology
https://www.researchgate.net/publication/282428966_A_Mathematical_Model_for_the_Genetic_Code_S_Based_on_Fibonacci_Numbers_and_their_q-Analogues

citations: 13

———

"This constitutes a very high level of constraint that applies to the relative proportions of TCAG bases involved in a "Fibonacci Hyper Constraints Super Resonances".

"Infinite diversity of digital and fractal structures of DNA and genomes: During our Artificial Intelligence research at IBM, based on our model "Fractal Chaos" [30-32], we demonstrated the numerical hyper-sensitivity of an artificial neural network positioned in the vicinity of Phi and subjected to perturbations of Fibonacci number type"

"Finally, from 2010 [26,27] we published various articles demonstrating that the golden ratio controls and finely adjusts the proportions of triplets codons to the scale of the entire human genome. This multi-level diversity of Phi is as astonishing as it is radiant"

Sapiens Mitochondrial DNA Genome Circular Long Range Numerical Meta Structures are Highly Correlated with Cancers and Genetic Diseases mtDNA Mutations
Journal of Cancer Science & Therapy
https://www.hilarispublisher.com/open-access/sapiens-mitochondrial-dna-genome-circular-long-range-numericalmeta-structures-are-highly-correlated-with-cancers-and-genetic-disea-1948-5956-1000469.pdf

citations: 13

———

"Fragment-based drug discovery (FBDD) is complementary to high-throughput screening. The approach has two key stages: identifying the starting fragment hit to be developed and generating the lead compound from the starting fragment hit. Here, we provide an overview of FBDD and introduce two indices originally developed at Astellas Pharma.

Both indices are based on the 'Golden Ratio'."

Two 'Golden Ratio' indices in fragment-based drug discovery
Drug Discovery Today
https://www.sciencedirect.com/science/article/abs/pii/S1359644608003723

citations: 58

———

"Calculating the scattering intensity of an N-atom system is a numerically exhausting O(N2) task. Using an exact expression for the scattering intensity I(q) at a given wavevector q, the rotationally averaged intensity I(q) is computed by evaluating I(q) in several scattering directions. The orientations of the q vectors are taken from a quasi-uniform spherical grid generated by the golden ratio."

"Using various biomolecules as examples, this technique is compared with an established multipole expansion method. For a given level of speed, the technique is more accurate than the multipole expansion for anisotropically shaped molecules, while comparable in accuracy for globular shapes."

Rapid and accurate calculation of small-angle scattering profiles using the golden ratio
Journal of Applied Crystallography
https://scripts.iucr.org/cgi-bin/paper?he5605

citations: 30

———

"The Universal Genetic Code Table not only maps codons to amino acids, but serves as a global checksum matrix.Frequencies of the 64 codons in the whole human genome scale are a self-similar fractal expansion of the universal genetic code."

"The 6 folding steps of codon populations modeled by the binary divisions of the "Dragon fractal paper folding curve" show evidence of 2 attractors. The numerical relationship between the attractors is derived from the Golden Ratio."

"Codon frequencies in the human genome are clustered around 2 fractal-like attractors, strongly linked to the golden ratio 1.618."

Codon populations in single-stranded whole human genome DNA Are fractal and fine-tuned by the Golden Ratio 1.618
Interdisciplinary Sciences: Computational Life Sciences
https://link.springer.com/article/10.1007/s12539-010-0022-0

citations: 81

"Golden Ratio Simulated Annealing (GRSA) for Protein Folding Problem (PFP) is presented. GRSA is similar to Multiquenching Annealing (MQA) and Threshold Temperature Simulated Annealing (TTSA) algorithms. In contrast to MQA and TTSA, GRSA uses several strategies in order to reduce the execution time for finding the best solution of PFP."

"GRSA has a special phase which detects the thermal equilibrium by a least squares method. In addition, a reheat strategy to escape from local optima is applied."

Golden Ratio Simulated Annealing for Protein Folding Problem
International Journal of Computational Methods
https://www.worldscientific.com/doi/abs/10.1142/S0219876215500371

citations: 6

———

"Protein structure prediction (PSP) has important applications in different fields, such as drug design and disease prediction"

3D Protein structure prediction with genetic tabu search algorithm
BMC Systems Biology
https://bmcsystbiol.biomedcentral.com/articles/10.1186/1752-0509-4-S1-S6

citations:

———

"Protein-fragment seqlets typically feature about 10 amino acid residue positions that are fixed to within conservative substitutions but usually separated by a number of prescribed gaps with arbitrary residue content. By quantifying a general amino acid residue sequence in terms of the associated codon number sequence, we have found a precise modular Fibonacci sequence in a continuous gap-free 10-residue seqlet with either 3 or 4 conservative amino acid substitutions.

This modular Fibonacci sequence is genuinely biophysical, for it occurs nine times in the SWISS-Prot/TrEMBL database of natural proteins."

A modular Fibonacci sequence in proteins
Quantitative Biology: Genomics
https://arxiv.org/abs/q-bio/0409038

citations:

———

"The optimization results by conformational space annealing are presented for an off-lattice protein model consisting of hydrophobic and hydrophilic residues in Fibonacci sequences. The ground-state energies found are lower than those reported in the literature."

Structure optimization by conformational space annealing in an off-lattice protein model
Physical Review E: Covering statistical, non linear, biological and soft matter physics
https://journals.aps.org/pre/abstract/10.1103/PhysRevE.72.011916

citations:
———

"A fascinating aspect of both quasicrystals and aperiodic mosaics is that the golden ratio of mathematics and art, the mathematical constant τ (tau), occurs over and over again. For instance, the ratio between the numbers of fat and thin rhombi in Penrose's mosaic is τ. Similarly, the ratio of various distances between atoms in quasicrystals is always related to τ."

Crystals of golden proportions
Nobel Prize in Chemistry
https://www.nobelprize.org/uploads/2018/06/popular-chemistryprize2011.pdf

citations: 3
———

The golden ratio in special relativity
Chaos Solitons and Fractals
https://www.researchgate.net/publication/229317480_The_golden_ratio_in_special_relativity

citations: 40
———

Electronic structure and transport properties of double-stranded Fibonacci DNA
Physical Review
https://journals.aps.org/prb/abstract/10.1103/PhysRevB.74.245105

citations: 27
———

"In this work, we present some results that are indirect evidences of Fibonacci String model fro DNA repetitive sequences growth"

An indirect evidence of the Fibonacci Strings model for DNA Repetitive Sequences growth
Association for Bioinformatics and Computational Biology
https://www.alice.cnptia.embrapa.br/bitstream/doc/1519/1/PLIndirectxmeeting2007.pdf

citations:
———

"This work presents a mathematical model that establishes an interesting connection between nucleotide frequencies in human single-stranded DNA and the famous Fibonacci's numbers.

Under these two hypotheses, it is possible to predict the human nucleotide frequencies with accuracy.

This result may be used as evidence to the Fibonacci string model that was proposed to the sequence growth of DNA repetitive sequences."

Nucleotide Frequencies in Human Genome and Fibonacci Numbers
Bulletin of Mathematical Biology
https://link.springer.com/article/10.1007/s11538-007-9261-6

citations: 63

Fibonacci-Cayley Numbers and Repetition Patterns in Genomic DNA
Annals of Combinatorics
https://link.springer.com/article/10.1007%2Fs00026-003-0185-z

citations: 11

"The golden ratio, ϕ = 1.61803..., has often been found in connection with biological phenomena, ranging from spirals in sunflowers to gene frequency."

Is the golden ratio a universal constant for self-replication?
PLoS ONE
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0200601

citations: 12

"The optimization results by conformational space annealing are presented for an off-lattice protein model consisting of hydrophobic and hydrophilic residues in Fibonacci sequences."

Structure optimization by conformational space annealing in an off-lattice protein model
Physical Review
https://journals.aps.org/pre/abstract/10.1103/PhysRevE.72.011916

citations: 83

"Is Phi the golden ratio a universal constant? We have just demonstrated here how it manifests itself in the codes of the living in forms often different and diversified: in the form HGO (3-Phi) / 2 on the scale of the whole human genome"

Golden ratio and Klein bottle Logophysics: the Keys of the Codes of Life and Cognition (Spurred by Liu & Sumpter, "Is the golden ratio a universal constant for self-replication?"
National University of Quilmes, Quantum Biosystems
https://www.researchgate.net/profile/Diego_Rapoport/publication/329170592_Golden_ratio_and_Klein_bottle_Logophysics_the_Keys_of_the_Codes_of_Life_and_Cognition_Spurred_by_Liu_Sumpter_Is_the_golden_ratio_a_universal_constant_for_self-replication_-comment_final_revised_versio/links/5ca6135c92851c64bd4f6cf5/Golden-ratio-and-Klein-bottle-Logophysics-the-Keys-of-the-Codes-of-Life-and-Cognition-Spurred-by-Liu-Sumpter-Is-the-golden-ratio-a-universal-constant-for-self-replication-comment-final-revised-ver.pdf?origin=publication_detail

citations: 6

"In this paper, we propose a machine learning classification model by integrating the Parameter free Thresholding Adjacency Statistics (PFTAS) with Fibonacci-p patterns for breast cancer detection."

Breast cancer classification using parametric free thresholding adjacency statistics based Fibonacci patterns
Society of Photo-Optical Instrumentation Engineers
https://www.spiedigitallibrary.org/conference-proceedings-of-spie/11399/113990Q/Breast-cancer-classification-using-parametric-free-thresholding-adjacency-statistics-based/10.1117/12.2558613.short?SSO=1

citations:

"we present a new hybrid artificial plant optimization algorithm, in which one new golden section operator is designed to mimic the characteristics in the branch growth process, it enhances the population diversity."

"This algorithm was tested in the two-dimensional AB model for several Fibonacci protein sequences and real protein sequences. The experimental results show it can predicts the protein folding structure precisely."

Protein Folding Structure Prediction with Artificial Plant Optimization Algorithm Based Golden Section and Limited Memory Broyden–Fletcher–Goldfarb–Shanno
Journal of Bionanoscience
https://www.ingentaconnect.com/content/asp/jobn/2013/00000007/00000001/art00019

citations: 14

"Fibonacci AB protein sequences of length between 13 and 55 residues in three dimensions have been extensively studied to understand protein folding.

Here, an efficient global optimization method, conformational space annealing, is applied to the three-dimensional off-lattice AB model protein with the 89-residue Fibonacci sequence whose properties are not known."

Three-dimensional off-lattice AB model protein with the 89-residue Fibonacci sequence
Chaos, Solitons & Fractals
https://www.sciencedirect.com/science/article/abs/pii/S0960077916301345

citations: 1

———

"Experimental results illustrate that the suggested algorithm is effective when the algorithm is applied to the Fibonacci sequences and four real protein sequences in the Protein Data Bank."

The Application of Artificial Bee Colony Algorithm in Protein Structure Prediction
Bio-Inspired Computing - Theories and Applications
https://link.springer.com/chapter/10.1007/978-3-662-45049-9_42

citations: 10

———

"To illustrate the efficiency, we apply APOA with splitting strategy to short sequences, Fibonacci sequences and real protein sequences, simulation results show it is effective."

Using Splitting Artificial Plant Optimization Algorithm to Solve Toy Model of Protein Folding
Journal of Computational and Theoretical Nanoscience
https://www.ingentaconnect.com/content/asp/jctn/2012/00000009/00000012/art00041

citations: 34

———

"Fibonacci sequence, which is defined in (4), is widely used in protein fold prediction"

An Improved Particle Swarm Optimization for Protein Folding Prediction
I.J. Information Engineering and Electronic Business
http://www.mecs-press.net/ijieeb/ijieeb-v3-n1/IJIEEB-V3-N1-1.pdf

citations: 39

———

"The authors studied 100 human skulls and 70 skulls of 6 other mammalian species and calculated 2 ratios: 1) the nasioiniac arc divided by the parieto-occipital arc (between bregma and inion), and 2) the parieto-occipital arc divided by the frontal arc (between nasion and bregma).

*The authors report that in humans these 2 ratios coincide (1.64 ± 0.04 and 1.57 ± 0.10) and approximate **Φ**. In the other 6 mammalian species, these 2 ratios were not only different, but also unique to each species.*

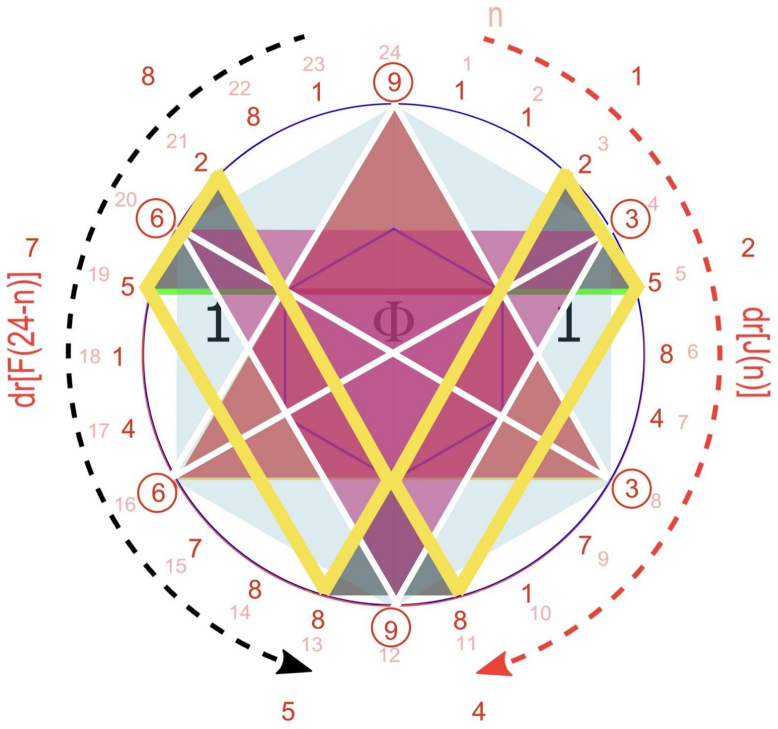
*The authors hypothesize that the Golden Ratio (**Φ**) principle, documented in other biological systems, may be present in the architecture and evolution of the human skull."*

Mammalian Skull Dimensions and the Golden Ratio (Φ)
Journal of Cranofacial Surgery
https://journals.lww.com/jcraniofacialsurgery/Fulltext/2019/09000/Mammalian_Skull_Dimensions_and_the_Golden_Ratio.34.aspx

citations:
———

v. The Golden Ratio predicts the center of a black hole, a mathematical constant produced by gravity & time, via an infinite tetrahedral, geodesic fractal array, hypothesized as the unified field, dark energy.





"For example it appears, notably, in black hole physics: the golden ratio is precisely the point where a black hole's modified heat changes from positive to negative, and it is part of the equation for the lower bound on black hole entropy. M4 / J2 = Φ

The golden ratio even relates to the loop quantum gravity parameter to black hole entropy"

2πγ = Φ

It is part of the equation for the lower bound on black hole entropy: 8πSl2P / ekA = Φ

Quantum Walk on a Spin Network and the Golden Ratio as the Fundamental Constant of Nature
Institute for Gravitation and the Cosmos, Pennsylvania State University
https://www.quantumgravityresearch.org/wp-content/uploads/2017/07/quantum-walk-on-a-spin-network-and-the-golden-ratio-as-the-fundamental-constant-of-nature-7.17-final-reference.pdf

citations: 6

Emergence of an Aperiodic Dirichlet Space from the Tetrahedral Units of an Icosahedral Internal Space
Quantum Gravity Research
https://www.quantumgravityresearch.org/wp-content/uploads/2017/11/Emergence-of-an-Aperiodic-Dirichlet-space-from-the-tetrahedral-02.20.17-final-AS.pdf

citations: 4

———

An Icosahedral Quasicrystal as a Packing of Regular Tetrahedra
Acta Physica Polonica
https://www.quantumgravityresearch.org/wp-content/uploads/2018/02/Icosahedral-Quasicrystal-as-a-packing-of-regular-tetrahedra-7.14.16-Final-Version-Acta-Physica-Polonica-No-Address.pdf

citations: 12

———

Cabinet of curiosities: the interesting geometry of the angle β=arccos((3φ – 1)/4)
Fractal and Fractional
https://res.mdpi.com/d_attachment/fractalfract/fractalfract-03-00048/article_deploy/fractalfract-03-00048.pdf

citations: 10

———

Periodic Modification of the Boerdijk-Coxeter Helix (Tetrahelix)
Mathematics
https://res.mdpi.com/d_attachment/mathematics/mathematics-07-01001/article_deploy/mathematics-07-01001-v2.pdf

citations: 8

———

Golden, Quasicrystalline, Chiral Packings of Tetrahedra
Quantum Gravity Research
https://www.quantumgravityresearch.org/wp-content/uploads/2017/03/Golden-Quasicrystalline-Chiral-Packing-of-Tetrahedra.pdf

citations:

———

An Icosahedral Quasicrystal and E8 Derived Quasicrystals
Quantum Gravity Research
https://www.quantumgravityresearch.org/wp-content/uploads/2019/05/An-Icosohedral-Quasicrystal-and-E8-derived-Quasicrystal-06.08.16-FF.pdf

citations: 7

———

Emergence of an Aperiodic Dirichlet Space from the Tetrahedral Units of an Icosahedral Internal Space
Quantum Gravity Research
https://www.quantumgravityresearch.org/wp-content/uploads/2017/11/Emergence-of-an-Aperiodic-Dirichlet-space-from-the-tetrahedral-02.20.17-final-AS.pdf

citations: 4

———

On the Ramanujan's Fundamental Formula for Obtain a Highly Precise Golden Ratio: Mathematical Connections with Black Holes Entropies and Like-Particle Solutions
Physics Vixra

https://www.semanticscholar.org/paper/On-the-Ramanujan%E2%80%99s-Fundamental-Formula-for-Obtain-a-Nardelli-Nardelli/eeb602211b1d63dbb24e2b8d35c874bd94049a31?p2df

citations:

———

"In this paper we show that the golden ratio is present in the Schwarzschild–Kottler metric."

"In this way, when we find the golden ratio in the geodesic structure of black holes, it gives us the future possibility of studying gravitation with fractal geometry, the geometry of nature."

The golden ratio in Schwarzschild–Kottler black holes
The European Physical Journal
https://link.springer.com/article/10.1140/epjc/s10052-017-4670-7

citations: 7

———

A Link Between Black Holes and the Golden Ratio
arXiv Cornwall University
https://arxiv.org/abs/1106.1600

citations: 8

———

Fibonacci Numbers and the Golden Ratio in Biology, Physics, Astrophysics, Chemistry and Technology: A Non-Exhaustive Review
arXiv Cornwall University
https://arxiv.org/abs/1801.01369

citations: 6

———

The Golden Ratio Time Algorithm
Researchgate
https://www.researchgate.net/publication/332879052_The_Golden_Ratio_Time_Algorithm

citations: 15

Golden Ratio Axioms of Time and Space
Researchgate
https://www.researchgate.net/publication/328738109_Golden_Ratio_Axioms_of_Time_and_Space

citations: 25

Black Holes and the Golden Ratio
Energy Technology Data Exchange, World Energy Base
https://www.osti.gov/etdeweb/biblio/21563419

citations:

Testing general relativity using golden black-hole binaries
Physical Review
https://journals.aps.org/prd/abstract/10.1103/PhysRevD.94.021101

citations: 67

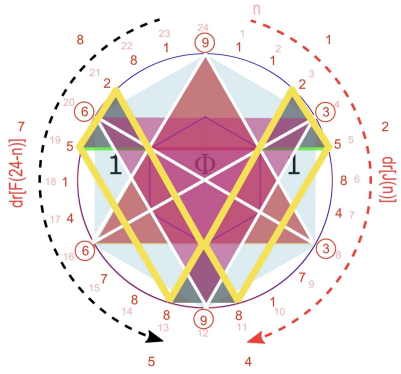
Golden Ratio Gain Enhancement in Coherently Coupled Parametric Processes
Nature
https://www.nature.com/articles/s41598-018-30014-7

citations: 6

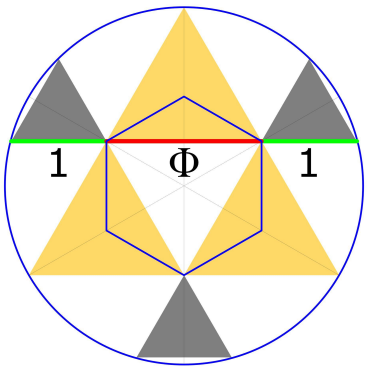
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vi. Fibonacci sequence and the Golden Ratio identify cancer cells, modulate gene expression and mitosis via DNA tetrahedrons





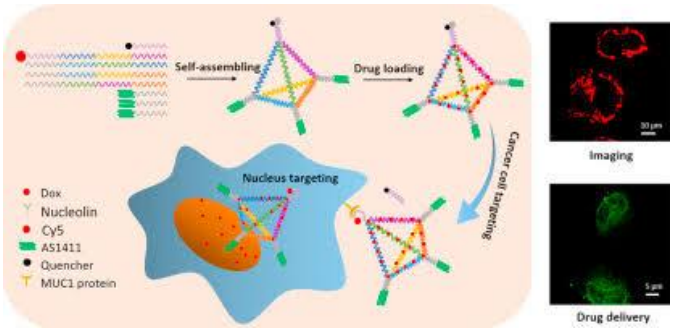
Fib M9 P24



golden ratio φ in a tetrahedron



Fib M9 P24 via cymascope



A dual-targeting DNA tetrahedron nanocarrier for breast cancer cell imaging and drug delivery
Talanta

https://www.sciencedirect.com/science/article/abs/pii/S0039914017311748

citations: 36

Design, fabrication and applications of tetrahedral DNA nanostructure-based multifunctional complexes in drug delivery and biomedical treatment
Nature

https://www.nature.com/articles/s41596-020-0355-z

citations: 14

Tetrahedral DNA Nanomaterial Regulates the Biological Behaviors of Adipose-Derived Stem Cells via DNA Methylation on Dlg3
ACS Applied Materials and Interfaces

https://pubs.acs.org/doi/abs/10.1021/acsami.8b12408#

citations: 19

A DNA Tetrahedron Nanoprobe with Controlled Distance of Dyes for Multiple Detection in Living Cells and in Vivo
Analytical Chemistry

https://pubs.acs.org/doi/abs/10.1021/acs.analchem.7b00889

citations: 42

Multifunctional Double-Bundle DNA Tetrahedron for Efficient Regulation of Gene Expression
ACS Applied Materials and Interfaces

https://pubs.acs.org/doi/abs/10.1021/acsami.0c08886

citations: 2

DNA Tetrahedron Based Biosensor for Argonaute2 Assay in Single Cells and Human Immunodeficiency Virus Type-1 Related Ribonuclease H Detection in Vitro
Analytical Chemistry

https://pubs.acs.org/doi/abs/10.1021/acs.analchem.9b00011

citations: 14

DNA tetrahedron-based nanogels for siRNA delivery and gene silencing
Chemical Communications

https://pubs.rsc.org/no/content/articlelanding/2019/cc/c9cc00175a/unauth#!divAbstract

citations: 30

DNA tetrahedron nanostructures for biological applications: biosensors and drug delivery
RSC: Analyst

https://pubs.rsc.org/am/content/articlelanding/2017/an/c7an01154g/unauth#!divAbstract

citations: 72

Fluorescence Resonance Energy Transfer-Based DNA Tetrahedron Nanotweezer for Highly Reliable Detection of Tumor-Related mRNA in Living Cells
ACS Nano

https://pubs.acs.org/action/doSearch?field1=Contrib&text1=Sitao++Xie

citations: 132

A DNA tetrahedron-based molecular beacon for tumor-related mRNA detection in living cells
Chemical Communications

https://pubs.rsc.org/--/content/articlelanding/2016/cc/c5cc09980c/unauth#!divAbstract

citations: 72

Development of DNA tetrahedron-based drug delivery system
Drug Delivery

https://www.tandfonline.com/doi/full/10.1080/10717544.2017.1373166

citations: 31

Accurate cancer cell identification and microRNA silencing induced therapy using tailored DNA tetrahedron nanostructures
RSC Chemical Science

https://pubs.rsc.org/az/content/articlehtml/2020/sc/c9sc04823e

citations: 27

Co-delivery of DNAzyme and a chemotherapy drug using a DNA tetrahedron for enhanced anticancer therapy through synergistic effects
New Journal of Chemistry

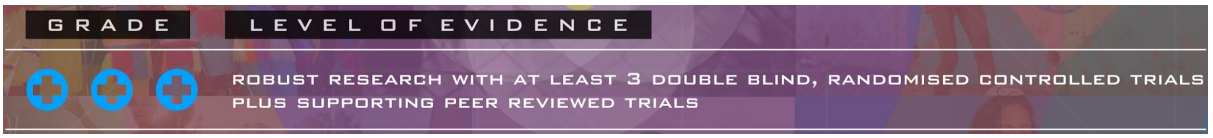
https://pubs.rsc.org/lv/content/articlelanding/2019/nj/c9nj02818h/unauth#!divAbstract

citations: 3

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music modulates heart rate variability, supporting autonomic balance via baroreflex

vii. music modulates heart rate variability, cognition (memory and learning) and emotional regulation, reducing physical pain and improving mental health



"Music or music in combination with therapeutic suggestions during surgery and postoperative recovery can be used as an adjunct in multimodal regimes to improve patient recovery process"

The effect of music and music in combination with therapeutic suggestions on postoperative recovery
Linköping University, Department of Medicine and Care, Nursing Science
http://www.diva-portal.org/smash/record.jsf?pid=diva2%3A248080&dswid=8121

randomised controlled trial, double blind
citations: 23

———

"study results are similar to that of some previous studies and has clearly proven that music and music with therapeutic suggestions have a definite role in the improvement on the postoperative events as this group required less rescue analgesics and antiemetics."

Does Intraoperative Music Therapy or Positive Therapeutic Suggestions during General Anaesthesia affect the postoperative outcome? A double blind randomised controlled trial.
Indian Journal of Anaesthesia
http://medind.nic.in/iad/t06/i4/iadt06i4p258.pdf

randomised controlled trial, double blind
citations: 17

———

"intra-operative music and music in combination with therapeutic suggestions may have some beneficial effects on postoperative recovery after hysterectomy"

Improved recovery after music and therapeutic suggestions during general anaesthesia: a double-blind randomised controlled trial
Acta Anaesthesioligica Scandanavica
https://onlinelibrary.wiley.com/doi/abs/10.1034/j.1399-6576.2001.045007812.x

randomised controlled trial, double blind
citations: 243

———

"Nature-based sounds help in the compatibility with emotional stress and stimulation of relaxing reactions by diverting the listeners' attention from anxiety, pain, and negative experiences to more pleasant one"

"The results of our study showed that using nature based sounds can reduce pain."

"listening to music was effective in reducing pain. It is es- sential to note that neither types of music are superior to others for increasing the relaxation of patients. The most important point is interesting in music (25). Most authors suggested that music should be wordless in order to allow the patient to focus on the music rather than the meaning of words (26)."

Comparative Effect of Nature-Based Sounds Intervention and Headphones Intervention on Pain Severity After Cesarean Section: A Prospective Double-Blind Randomized Trial
Anaesthesiology and Pain Medicine
https://www.researchgate.net/profile/Ali_Abbasijahromi/publication/332592617_Comparative_Effect_of_Nature-Based_Sounds_Intervention_and_Headphones_Intervention_on_Pain_Severity_After_Cesarean_Section_A_Prospective_Double-Blind_Randomized_Trial/links/5cc84bca4585156cd7bc0e1f/Comparative-Effect-of-Nature-Based-Sounds-Intervention-and-Headphones-Intervention-on-Pain-Severity-After-Cesarean-Section-A-Prospective-Double-Blind-Randomized-Trial.pdf?origin=publication_detail

randomised controlled trial, double blind
citations: 4

———

"Intraoperative music provides an inexpensive nonpharmacological option to further reduce postoperative pain."

Intraoperative Music Reduces Perceived Pain After Total Knee Arthroplasty–A Blinded, Prospective, Randomized, Placebo-Controlled Clinical Trial
The Journal of Knee Surgery
https://www.thieme-connect.com/products/ejournals/html/10.1055/s-0030-1247831

randomised controlled trial, double blind
citations: 35

———

"The rhythmic music device was found to be efficacious in reducing high BP during 2 months of self-treatment by patients at home. Breathing pattern modification appears to be an important component in this reduction."

Treating hypertension with a device that slows and regularises breathing: a randomised, double-blind controlled study
Nature
https://www.nature.com/articles/1001148

randomised controlled trial, double blind
citations: 248

———

"two specific music programs significantly reduced burnout symptoms after 5 weeks."

"The effects were maintained over a long time period."

"This newly developed method of receptive music therapy was also evaluated for the treatment of depression and dysthymia, with significant outcomes."

Music Programs Designed to Remedy Burnout Symptoms Show Significant Effects after Five Weeks
Annals of the New York Academy of Sciences
https://nyaspubs.onlinelibrary.wiley.com/doi/pdf/10.1111/j.1749-6632.2009.04790.x

randomised controlled trial, double blind
citations: 40

———

"Our findings confirm that short-term therapeutic effects of music listening results in substantial reduction in physiological stress responses arising from anxiety in mechanically ventilated patients."

Effects of music intervention on physiological stress response and anxiety level of mechanically ventilated patients in China: a randomised controlled trial
Journal of Clinical Nursing
https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-2702.2009.02845.x

randomised controlled trial
citations: 177

———

"Music therapy significantly diminished patients' negative symptoms, increased their ability to converse with others, reduced their social isolation, and increased their level of interest in external events"

Rehabilitative Effect of Music Therapy for Residual Schizophrenia: A One-Month Randomised Controlled Trial in Shanghai
Psychiatric Rehabilitation in China
https://www.cambridge.org/core/journals/the-british-journal-of-psychiatry/article/rehabilitative-effect-of-music-therapy-for-residual-schizophrenia/B39038F33917B6CA95328499F9C687E6

randomised controlled trial
citations: 145

———

"music therapy treatment improved the following: global and social functioning in schizophrenia and/or serious mental disorders, gait and related activities in Parkinson's disease, depressive symptoms, and sleep quality."

Effectiveness of music therapy: a summary of systematic reviews based on randomized controlled trials of music interventions
Patient Preference and Adherence
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4036702/

randomised controlled trial
citations: 116
———

Effects of music relaxation video on pain and anxiety for women with gynaecological cancer receiving intracavitary brachytherapy: a randomised controlled trial
Journal of Research in Nursing
https://journals.sagepub.com/doi/10.1177/1744987114529298

randomised controlled trial
citations: 14
———

"listening to music during surgery under regional anesthesia has effects on cortisol levels (reflecting stress-reducing effects) and reduces sedative requirements to reach light sedation."

Effects of music listening on cortisol levels and propofol consumption during spinal anesthesia
Frontiers in Psychology
https://www.frontiersin.org/articles/10.3389/fpsyg.2011.00058/full

citations: 124
———

"Listening to music during bed rest after open-heart surgery has some effects on the relaxation system as regards s-oxytocin and subjective relaxations levels. This effect seems to have a causal relation from the psychological (music makes patients relaxed) to the physical (oxytocin release)."

Soothing music can increase oxytocin levels during bed rest after open-heart surgery: a randomised control trial
Nilsson Journal of Clinical Nursing
https://www.ncbi.nlm.nih.gov/m/pubmed/19583647/

randomised controlled trial
citations: 307
———

“the use of a specially designed music reduced anxiety and increased well-being in patients during coronary angiographic procedures.”

Music interventions in patients during coronary angiographic procedures: a randomized controlled study of the effect on patients' anxiety and well-being
European Journal of Cardiovascular Nursing
https://www.ncbi.nlm.nih.gov/m/pubmed/20685168/

randomised controlled trial
citations: 95

———

“Music is a simple, safe, and effective method of reducing potentially harmful physiologic and psychologic responses arising from pain in patients post-PCI undergoing a C-clamp procedure”

Effects of music on patients undergoing a C-clamp procedure after percutaneous coronary interventions: a randomized controlled trial
Heart Lung
https://www.ncbi.nlm.nih.gov/m/pubmed/18005804/

randomised controlled trial
citations: 86

———

Stress reduction and analgesia in patients exposed to calming music postoperatively: a randomized controlled trial
European Journal of Anaesthesiology
https://www.cambridge.org/core/journals/european-journal-of-anaesthesiology/article/stress-reduction-and-analgesia-in-patients-exposed-to-calming-music-postoperatively-a-randomized-controlled-trial/F4C1C17F3A173BE3270ECD27CB50933A

randomised controlled trial
citations: 304

———

”music therapy may improve symptoms of depression and increase HRV in demented patients”

Effects of Music Therapy on Psychological Symptoms and Heart Rate Variability in Patients with Dementia. A Pilot Study
Current Aging Science
https://www.ingentaconnect.com/content/ben/cas/2010/00000003/00000003/art00009

citations: 57

———

"musical performance would lead to a greater effect of emotion-related modulation in cardiac autonomic nerve activity than musical perception"

modul8 infinity hypothesize that music with visuals modulate the brain and heart equivalent to performing music

Emotion-related Changes in Heart Rate and Its Variability during Performance and Perception of Music
Annals of the New York Academy of Sciences
https://nyaspubs.onlinelibrary.wiley.com/doi/abs/10.1111/j.1749-6632.2009.04788.x

citations: 87

———

"listening to the music has the influences on autonomic control. Using HRV analysis can help evaluating the effects of music therapy"

Using heart rate variability analysis to assess the effect of music therapy on anxiety reduction of patients
Computers in Cardiology
https://ieeexplore.ieee.org/abstract/document/1291194

citations: 27

———

"These findings promote the potential for HRV to reflect complex emotional stimuli and suggest that residual emotional reactions continue to occur in the vegetative state. HRV descriptors and data-mining appear applicable in brain function research in the absence of consciousness."

Heart Rate Variability, Emotions, and Music
Journal of Psychophysiology
https://econtent.hogrefe.com/doi/abs/10.1027/0269-8803/a000021?journalCode=jop&

citations: 22

———

"We suggest that excitatory heavy metal music acutely decreases global HRV as compared with classical music"

Auditory stimulation with music influences the geometric indices of heart rate variability in men
International Archives of Medicine
https://intarchmed.biomedcentral.com/articles/10.1186/1755-7682-7-27

citations: 28

———

Music and heart rate variability. Study of the effect of music on heart rate variability in healthy adolescents
Europe PMC
https://europepmc.org/abstract/med/10412282

citations: 27
———

Art and science: how musical training shapes the brain
Frontiers in Psychology
https://www.frontiersin.org/articles/10.3389/fpsyg.2013.00713/full?utm_source=newsletter&utm_medium=email&utm_campaign=Psychology-w45-2013

citations: 97
———

A Little Goes a Long Way: How the Adult Brain Is Shaped by Musical Training in Childhood
Journal of Neuroscience
http://www.jneurosci.org/content/32/34/11507

citations: 207
———

The effects of music on the cardiovascular system and cardiovascular health
Heart
https://heart.bmj.com/content/96/23/1868

citations: 163
———

"some combination of endorphin release and self-other merging lead to the social bonding effects of music"

Music and social bonding: "self-other" merging and neurohormonal mechanisms
Frontiers in Psychology
https://www.frontiersin.org/articles/10.3389/fpsyg.2014.01096/full

citations: 272
———

Increase in salivary oxytocin and decrease in salivary cortisol after listening to relaxing slow-tempo and exciting fast-tempo music
Plos One
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0189075

citations: 26
———

Music and Learning-Induced Cortical Plasticity
Annals of the New York Academy of Sciences
https://nyaspubs.onlinelibrary.wiley.com/doi/full/10.1196/annals.1284.054

citations: 207

Music structure determines heart rate variability of singers
Frontiers in Psychology
https://www.frontiersin.org/articles/10.3389/fpsyg.2013.00334/full

citations: 160

The neurochemistry and social flow of singing: bonding and oxytocin
Frontiers in Human Neuroscience
https://www.frontiersin.org/articles/10.3389/fnhum.2015.00518/full#B6

citations: 97

Effects of choir singing or listening on secretory immunoglobulin A, cortisol, and emotional state
Journal of Behavioural Medicine
https://link.springer.com/article/10.1007/s10865-004-0006-9

citations: 569

Does singing facilitate social bonding
Music and Medicine
https://s3.amazonaws.com/academia.edu.documents/43377505/Music__Medicine__2014__Volume_6_Issue_220160304-7044-13ng0hf.pdf?response-content-disposition=inline%3B%20filename%3DMusic_and_Medicine_2014_Volume_6_Issue_2.pdf&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAIWOWYYGZ2Y53UL3A%2F20190608%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20190608T031604Z&X-Amz-Expires=3600&X-Amz-SignedHeaders=host&X-Amz-Signature=801b272888cd02323929bc6a3f2cd3ccc4b3177e88b25bedf84bfa7eaf6d5f98

citations: 125

Tapping polyrhythms in music activates language areas
Neuroscience Letters
https://www.sciencedirect.com/science/article/abs/pii/S0304394011003077

citations: 57

Music and social bonding: "self-other" merging and neurohormonal mechanisms
Frontiers in Psychology
https://www.frontiersin.org/articles/10.3389/fpsyg.2014.01096/full

citations: 272

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"The use of the Music Care application within this setting proved good feasibility"

A patient-controlled, smartphone-based music intervention to reduce pain—A multi-center observational study of patients with chronic pain
European Journal of Integrative Medicine
https://www.sciencedirect.com/science/article/abs/pii/S1876382016300026?via%3Dihub

citations: 13

———

"music is effective in facilitating both implicit learning and recall of the advertised product, showing that, under non-attentive conditions, there appears to be a certain mechanism of unconscious elaboration of the musical signal"

Music and Memory in Advertising: Music as a Device of Implicit Learning and Recall
Music, Sound and the Moving Image
https://online.liverpooluniversitypress.co.uk/doi/abs/10.3828/msmi.1.1.7

citations: 38

———

Listening to polyphonic music recruits domain-general attention and working memory circuits
Cognitive, Affective & Behavioural Neuroscience
https://link.springer.com/article/10.3758/CABN.2.2.121

citations: 296

———

Effects of Background Music on Phonological Short-Term Memory
Quarterly Journal of Experimental Psychology
https://journals.sagepub.com/doi/abs/10.1080/14640748908402355

citations: 537

———

Music-based memory enhancement in Alzheimer's Disease: Promise and limitations
Neuropsychologia
https://www.sciencedirect.com/science/article/pii/S0028393212003910

citations: 118

———

Music-dependent memory: The roles of tempo change and mood mediation
APA Psychiatry
https://psycnet.apa.org/record/1996-06524-003

citations: 162

Remembrance of Things Past: Music, Autobiographical Memory and Emotion
Advances in Consumer Research
http://www.acrwebsite.org/search/view-conference-proceedings.aspx?Id=7363

citations: 295

Music, memory and emotion
Journal of Biology
https://jbiol.biomedcentral.com/articles/10.1186/jbiol82

citations: 226

An overview of research on music and learning
New England Conservatory
https://www.researchgate.net/publication/245362946_An_overview_of_research_on_music_and_learning

citations: 86

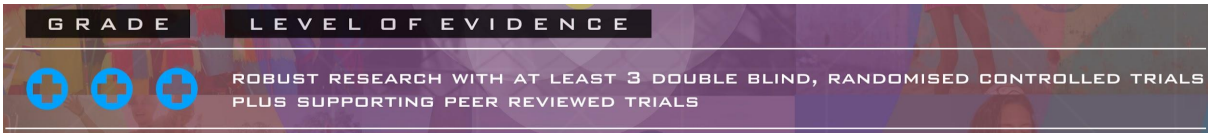
“an improvement in co-ordination was observed”

The Effect of Background Music on Learning
Psychology of Music
https://journals.sagepub.com/doi/abs/10.1177/0305735699272005

citations: 115

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viii. music and sub conscious therapeutic affirmations reduce pain and improve recovery



"Music or music in combination with therapeutic suggestions during surgery and postoperative recovery can be used as an adjunct in multimodal regimes to improve patient recovery process"

The effect of music and music in combination with therapeutic suggestions on postoperative recovery
Linköping University, Department of Medicine and Care, Nursing Science
http://www.diva-portal.org/smash/record.jsf?pid=diva2%3A248080&dswid=8121

randomised controlled trial, double blind
citations: 23

———

"study results are similar to that of some previous studies and has clearly proven that music and music with therapeutic suggestions have a definite role in the improvement on the postoperative events as this group required less rescue analgesics and antiemetics."

Does Intraoperative Music Therapy or Positive Therapeutic Suggestions during General Anaesthesia affect the postoperative outcome? A double blind randomised controlled trial.
Indian Journal of Anaesthesia

randomised controlled trial, double blind
citations: 17

———

"intra-operative music and music in combination with therapeutic suggestions may have some beneficial effects on postoperative recovery after hysterectomy"

Improved recovery after music and therapeutic suggestions during general anaesthesia: a double-blind randomised controlled trial
Acta Anaesthesioligica Scandanavica
https://onlinelibrary.wiley.com/doi/abs/10.1034/j.1399-6576.2001.045007812.x

randomised controlled trial, double blind
citations: 243

ix. Baroreflex modulates Heart Rate Variability via 111hz Mayer Waves, modulating oxytocin, and in turn the sympathetic and parasympathetic nervous system



"Our data suggest that OT terminals in the solitary vagal complex modulate reflex control of the heart, acting to facilitate vagal outflow and the slowdown of the heart."

Baroreflex control of heart rate by oxytocin in the solitary-vagal complex
American Journal of Physiology - Regulative, Integrative & Comparative Physiology
https://journals.physiology.org/doi/full/10.1152/ajpregu.00806.2000

citations: 140

———

"Oxytocin is modulated differently after aortic baroreceptor deafferentation"

Effect of aortic baroreceptor deafferentation on plasma vasopressin and oxytocin in the conscious rat
Brain Research
https://www.sciencedirect.com/science/article/abs/pii/S000689939600813X

citations: 5

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"Several studies have shown that OTergic drive from PVN to brainstem increases baroreflex sensitivity and improves autonomic control of the circulation"

"reduced PVN oxytocin protein levels were positively correlated with decreased baroreflex sensitivity"

Ovarian Hormone Deprivation Reduces Oxytocin Expression in Paraventricular Nucleus Preautonomic Neurons and Correlates with Baroreflex Impairment in Rats
Frontiers in Integrative Physiology
https://www.frontiersin.org/articles/10.3389/fphys.2016.00461/full

citations: 13

———

Neurochemical Circuits Subserving Fluid Balance and Baroreflex: A Role for Serotonin, Oxytocin, and Gonadal Steroids
Europe PMC
https://europepmc.org/article/NBK/NBK200961

citations: 19

Oxytocin augments baroreflex bradycardia in conscious rats
Peptides
https://www.sciencedirect.com/science/article/abs/pii/0196978194900493

citations: 22
———

Baroreceptor influences on oxytocin and vasopressin secretion.
Hypertension
https://www.ahajournals.org/doi/abs/10.1161/01.HYP.13.2.110

citations: 66
———

"OT's role in BP and baroreflex/autonomic balance. The data provide the first experimental evidence for a role of endogenous OT in the maintenance of tonic BP"

"Alterations in baroreflex function were noted in mice lacking OT"

"There is much evidence to show that brain peptides regulate BP and autonomic balance"

"For the highly complementary peptides vasopressin and OT, there are effects on BP, HR, and baroreflex activity"

Oxytocinergic regulation of cardiovascular function: studies in oxytocin-deficient mice
https://journals.physiology.org/doi/full/10.1152/ajpheart.00774.2002

citations: 77
———

Differential effects of vasopressin (AVP) and oxytocin (OXT) on the baroreceptor reflex in conscious rats
Clinical and Experimental Hypertension. Part A: Theory and Practice
https://www.tandfonline.com/doi/abs/10.3109/10641968409046106

citations: 9
———

The cardiovascular effects of oxytocin in conscious male rats
Pubmed
https://pubmed.ncbi.nlm.nih.gov/4029259/

citations: 67
———

"These findings indicate that oxytocin in the dorsal motor nucleus of the vagus may act as a regulator of vagally-mediated gastric and cardiovascular functions"

Dorsal medullary oxytocin, vasopressin, oxytocin antagonist, and TRH effects on gastric acid secretion and heart rate
Peptides
https://pubmed.ncbi.nlm.nih.gov/3938842/

citations: 126
———

Heart Rate Variability Biofeedback as a Method for Assessing Baroreflex Function: A Preliminary Study of Resonance in the Cardiovascular System
Applied Psychophysiology and Biofeedback
https://link.springer.com/article/10.1023/A:1014587304314

citations: 274
———

"Here we address the question as to whether the low frequency variability (~0.1 Hz) of the heart period may depend on a baroreflex-mediated response to blood pressure oscillations, induced by the α-sympathetic drive on the peripheral resistance."

"We conclude that in supine rest conditions, the oscillation of RR at low frequency is almost entirely accounted for by a baroreflex mechanism, since it is not produced in the absence of a 0.1 Hz pressure oscillation."

"The results provide physiological support for the use of non-invasive estimates of the closed-loop baroreflex gain from cross-spectral analysis of blood pressure and heart period variability in the 0.1 Hz range."

Baroreflex and oscillation of heart period at 0.1 Hz studied by α-blockade and cross-spectral analysis in healthy humans
The Journal of Physiology
https://physoc.onlinelibrary.wiley.com/doi/full/10.1111/j.1469-7793.2001.0235j.x

citations: 169
———

"LF and LFa were unrelated to myocardial 6-[18F]fluorodopamine-derived radioactivity."

"LF power, with or without respiratory adjustment, reflects baroreflex modulation and not cardiac sympathetic tone."

Supine low-frequency power of heart rate variability reflects baroreflex function, not cardiac sympathetic innervation
Cleveland Clinic Journal of Medicine

citations: 203

Heart Rate Variability, Baroreflex Function and Heart Rate Turbulence: Possible Origin and Implications
Hellenic Journal of Cardiology
https://hellenicjcardiol.org/archive/full_text/2007/5/2007_5_278.pdf

citations: 65
———

Effects of respiratory time ratio on heart rate variability and spontaneous baroreflex sensitivity
Journal of Applied Physiology
https://journals.physiology.org/doi/full/10.1152/japplphysiol.00163.2013

citations: 26
———

"Heart rate variability biofeedback had strong long-term influences on resting baroreflex gain and pulmonary function. It should be examined as a method for treating cardiovascular and pulmonary diseases.

Also, this study demonstrates neuroplasticity of the baroreflex."

Heart Rate Variability Biofeedback Increases Baroreflex Gain and Peak Expiratory Flow
Psychosomatic Medicine - Journal of Biobehavioral Medicine
https://journals.lww.com/psychosomaticmedicine/Abstract/2003/09000/Heart_Rate_Variability_Biofeedback_Increases.12.aspx

citations: 507
———

Arterial baroreflex influence on heart rate variability: A mathematical model-based analysis
Medical and Biological Engineering and Computing
https://link.springer.com/article/10.1007/BF02344775

citations: 41
———

"It is often assumed that the power in the low-(around 0.10 Hz) and high-frequency (around 0.25 Hz) bands obtained by power spectral analysis of cardiovascular variables reflects vagal and sympathetic tone respectively."

Physiology and Pathophysiology of Heart Rate and Blood Pressure Variability in Humans: Is Power Spectral Analysis Largely An Index of Baroreflex Gain?
Portland Press: Clinical Science
https://portlandpress.com/clinsci/article-abstract/88/1/103/76359

citations: 348
———

Frequency-dependent baroreflex modulation of blood pressure and heart rate variability in conscious mice
American Journal of Physiology: Heart & Circulatory Physiology
https://journals.physiology.org/doi/full/10.1152/ajpheart.01224.2004

citations: 60

———

How Does Heart Rate Variability Biofeedback Work? Resonance, the Baroreflex, and Other Mechanisms
Biofeedback
https://meridian.allenpress.com/biofeedback/article-abstract/41/1/26/113345/How-Does-Heart-Rate-Variability-Biofeedback-Work?redirectedFrom=fulltext

citations: 79

———

"Mayer waves are oscillations of arterial pressure occurring spontaneously in conscious subjects at a frequency lower than respiration (~0.1 Hz in humans)."

The enigma of Mayer waves: Facts and models
Oxford Academic: Cardiovascular Research
https://academic.oup.com/cardiovascres/article/70/1/12/408540?login=true

citations: 569

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"Physiological coherence is used broadly and includes all of the specific approaches for quantifying the various types of coherence measures, such as cross-coherence (frequency entrainment between respiration, BP, and heart rhythms), or synchronization among systems (e.g., synchronization between various EEG rhythms and the cardiac cycle), auto-coherence (stability of a single waveform such as respiration or HRV patterns), and system resonance."

"A coherent heart rhythm is defined as a relatively harmonic (sine-wave-like) signal with a very narrow, high-amplitude peak in the LF region of the HRV power spectrum with no major peaks in the VLF or HF regions. Coherence is assessed by identifying the maximum peak in the 0.04–0.26 Hz range of the HRV power spectrum, calculating the integral in a window 0.030 Hz wide, centered on the highest peak in that region, and then calculating the total power of the entire spectrum. The coherence ratio is formulated as: (Peak Power/[Total Power – Peak Power])"

"With practice, people can learn to breathe at their cardiovascular system's resonance frequency. This aligns the three oscillators (baroreflex, HR, and BP) at that frequency and moves the peak frequency from the HF range (≈0.2 Hz) to the LF range (≈0.1 Hz). Breathing at the resonance frequency more than doubles the energy in the LF band (0.04–0.15 Hz). This corresponds to the Institute of HeartMath's heart rhythm coherence, which is associated with a "narrow, high-amplitude, easily visualized peak" from 0.09 to 0.14 Hz"

"The resonance frequency model uses the mechanism of slow breathing at an individual's resonance frequency, while the psychological coherence model adds and emphasizes the importance of the pattern and stability of vagal afferent traffic as important components of emotional self-regulation. Both approaches have been associated with increased overall HRV over time."

"There is agreement that the concept of "coherence" in the heart rate signal is observed as a peak in the HRV power spectrum around 0.1 Hz, which reflects resonance frequency of breathing and resonance in the baroreflex. The authors' "psychophysiological coherence model" advocates biofeedback training that actually creates a maximum of HRV variability at this virtual single frequency."

"It is known that the power spectrum distribution of normal fluctuations in a healthy heart follows an 1/f pattern, characteristic of fractal signals. So, this leads us to juxtapose Coherence and simplicity vs fractals and complexity."

A healthy heart is not a metronome: an integrative review of the heart's anatomy and heart rate variability
Frontiers in Psychology
https://www.frontiersin.org/articles/10.3389/fpsyg.2014.01040/full

citations: 807

———

"The most common methods evaluate fractal-like properties of the heart rate time series. For example, the heart rate frequency (f) spectrum exhibits an inverse power law relationship"

Heart rate variability – a historical perspective
Frontiers in Physiology: Clinical & Translational Physiology
https://www.frontiersin.org/articles/10.3389/fphys.2011.00086/full

citations: 646

Hypothetical Control of Heart Rate Variability
Frontiers in Physiology: Fractal & Network Physiology
https://www.frontiersin.org/articles/10.3389/fphys.2019.01078/full

citations: 4

Fractals in the nervous system: conceptual implications for theoretical neuroscience
Frontiers in Physiology: Fractal & Network Physiology
https://www.frontiersin.org/articles/10.3389/fphys.2010.00015/full

citations: 213

Fractal physiology and the fractional calculus: a perspective
Frontiers in Physiology: Fractal & Network Physiology
https://www.frontiersin.org/articles/10.3389/fphys.2010.00012/full

citations: 201

Does sample rate introduce an artifact in spectral analysis of continuous processes?
Frontiers in Physiology: Fractal & Network Physiology

citations: 12

"Together with the broader literature on OT and HRV, findings suggest that acute administration of OT may facilitate a fundamental psychophysiological feature of social behaviour, increasing capacity for social engagement.

Findings also suggest that HRV changes may provide a novel biomarker of response to OT nasal spray that can be incorporated into research on response to treatment."

Oxytocin Increases Heart Rate Variability in Humans at Rest: Implications for Social Approach-Related Motivation and Capacity for Social Engagement
PLOS ONE
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0044014

citations: 123

Intranasally administered oxytocin decreases heart rate and increases heart rate variability in dogs
Journal of Veterinary Behavior
http://kisanna.web.elte.hu/doc/tdk/j.jveb.2014.09.050.pdf

citations: 22
———

"oxytocin increased high frequency heart rate variability, a relatively pure measure of parasympathetic cardiac control, and decreased pre-ejection period, a well-validated marker of enhanced sympathetic cardiac control"

Oxytocin increases autonomic cardiac control: Moderation by loneliness
Biological Psychology
https://www.sciencedirect.com/science/article/abs/pii/S0301051110003030

citations: 174
———

"Autonomic nervous system (ANS) dysfunction (i.e., increased sympathetic and/or decreased parasympathetic activity) has been proposed to contribute to psychosis vulnerability."

Intranasal oxytocin increases heart-rate variability in men at clinical high risk for psychosis: a proof-of-concept study
Nature
https://www.nature.com/articles/s41398-020-00890-7

citations: 1
———

"Oxytocin significantly increased indexes of parasympathetic activity, including heart rate variability"

Benefits of oxytocin administration in obstructive sleep apnea
American Journal of Physiology: Lung, Cellular & Molecular Physiology
https://journals.physiology.org/doi/full/10.1152/ajplung.00206.2017

citations: 11
———

music modulates our DNA

x.music modulates gene expression, improving mood and reducing stress via rhythmic and melodic entrainment



"Relaxation through active engagement in Recreational Music Making may be more effective than quiet reading at altering gene expression and thus more clinically useful for stress amelioration"

Recreational Music-Making alters gene expression pathways in patients with coronary heart disease
National Centre for Biotechnology Informatics
https://www.ncbi.nlm.nih.gov/pubmed/23435350

citations: 21

"gene expression changes for interferon-gamma, interleukin-10, and improved mood, this RMM protocol has significant potential for utilization in the corporate wellness environment"

Recreational music-making modulates natural killer cell activity, cytokines, and mood states in corporate employees
Medical Science Monitor
https://www.ncbi.nlm.nih.gov/pubmed/17261984

citations: 68

Early music exposure modifies GluR2 protein expression in rat auditory cortex and anterior cingulate cortex
Neuroscience Letters
https://www.sciencedirect.com/science/article/abs/pii/S0304394007005435

citations: 29

“improvement in immunological profile and mood states in the older group and the low level of energy required for participation suggest this music-making protocol has potential as a health improvement strategy for older individuals”

Recreational music-making modulates immunological responses and mood states in older adults
Journal of Medical and Dental Sciences
https://www.ncbi.nlm.nih.gov/pubmed/20099470

citations: 45

———

xi. music modulates the NMDA gene receptor, improving learning and memory



"In the hippocampus, following activation of NMDA receptors and extracellular signal-regulated kinase (essential events in several forms of LTP) acetylation and phosphorylation of histone H3 are increased (Reul et al., 2009)"

"In neuroscience, LTP long-term potentiation is a persistent strengthening of synapses based on recent patterns of activity. These are patterns of synaptic activity that produce a long-lasting increase in signal transmission between two neurons (Wikipedia)"

Toward an Epigenetic View of Our Musical Mind
Frontiers in Genetics
https://www.frontiersin.org/articles/10.3389/fgene.2011.00111/full

citations: 15

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"the expression of N-Methyl-aspartate (NMDA) glutamate receptors has been implicated in increasing synaptic efficacy to bring about learning, with more effective learning taking place in animals with more NMDA receptors (for review, see Lau and Zukin, 2007)"

"Auditory enrichment early in life through music exposure has been linked to increased NMDA receptor expression in the rat auditory cortex, paralleled by improved performance on auditory discrimination and learning and memory tasks (Tang et al., 1999; Xu et al., 2009; see also Cui et al., 2009)"

"it is possible that early enrichment through musical training may cause an upregulation of proteins like NMDA receptors to promote synaptic efficacy and improve auditory learning abilities, in turn feeding back to promote musical development"

Art and science: how musical training shapes the brain
Frontiers in Psychology
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3797461/

citations: 97

———

"the effects of light on the mammalian circadian system of rodents are mediated by glutamatergic mechanisms and that the N-methyl-D-aspartate (NMDA) receptor plays an important role in this regulation (Colwell 2001)"

"Mozart music could involve modulation of the NMDA receptor (Cruz et al. 2011)"

"music may enrich some environments in comparison to silence"

Anxiolytic effect of Mozart music over short and long photoperiods as part of environmental enrichment in captive Rattus norvegicus (Rodentia: Muridae)
Scandinavian Journal of Laboratory Animal Science
http://www.sjlas.org/index.php/SJLAS/article/download/341/266

citations: 5

———

"Music can improve the spatial memory of rats and increase the expression of the NMDA receptor and the mRNA coding the NMDA receptor in hippocampus."

Effect of music on learning and memory and expression of NMDA receptor on rat hippocampus
School of Medicine,Shandong University
http://en.cnki.com.cn/Article_en/CJFDTOTAL-TSYX200705021.htm

citations:

———

"Music can enhance the expression of NMDA-receptor and its mRNA in the rat hippocampus in a time-dependent manner, and effect is more obvious postnatally."

Effect of music on expression of NMDA-receptor in rat hippocampus
Department of Anatomy, Medical School, Shandong University
http://en.cnki.com.cn/Article_en/CJFDTOTAL-JPXZ200304019.htm

citations: 1

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"rats exposed to music shortly after the onset of hearing, for example, demonstrate enhanced NMDA receptor (NMDAR) subunit NR2B protein expression in auditory cortex (Xu et al., 2009), which facilitates the experience-dependent synaptic plasticity that undergirds learning"

Biological impact of preschool music classes on processing speech in noise
Developmental Cognitive Neuroscience
https://www.sciencedirect.com/science/article/pii/S187892931300042X

citations: 87

———

"Bliss and Lømo (1973) were the first to discover that a few sec- onds of high-frequency stimulation enhanced the synaptic activity between two neurons and that this change persisted for weeks.

Thus, this phenomenon has been labeled long-term potentiation (LTP). It was later discovered that specific receptors, N-methyl-D- aspartate (NMDA), mediate the LTP response"

Exploring a Neuroplasticity Model of Music Therapy
Journal of Music Therapy
http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.893.2493&rep=rep1&type=pdf

citations: 73

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"Our findings suggest that early auditory enrichment with music influences NMDA-mediated neural plasticity, which results in enhanced auditory discrimination learning"

Early auditory enrichment with music enhances auditory discrimination learning and alters NR2B protein expression in rat auditory cortex
Behavioural Brain Research
https://www.sciencedirect.com/science/article/pii/S0166432808003690

citations: 54

———

"Rats exposed to music showed a significant increase in the expression of the NMDA receptor NR2B protein in their auditory cortex (Xu et al., 2009)"

Music Affects Rodents: A Systematic Review of Experimental Research
Frontiers in Behavioral Neuroscience
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6302112/

citations: 11

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xii. music modulates interleukin 6 (IL—6) gene, reducing inflammation and human stress conditions

"The resultant amelioration of stress-induced genomic expression supports the underlying premise that recreational music making warrants additional consideration as a rational choice within our armamentarium of stress reduction strategies"

"While reversing individualized genomic stress induction signatures rather than describing the modulation or mecha- nisms associated with individual gene expression is the focus of this investigation, it should be noted that statistically significant Phase II reversals common to both the resting control and recreational music making experimental groups include: IL6 (interleukin 6)"

Recreational music-making modulates the human stress response: a preliminary individualized gene expression strategy
Medical Science Monitor
https://www.ncbi.nlm.nih.gov/pubmed/15668624

citations: 99

xiii. music modulates endogenous DMT and the DMT gene, a systems biology hypothesis

"There has been some evidence that NMDA receptors may also play a role in mediating the effects of DMT. DMT partially blocked the discriminative stimulus effects of phencyclidine, which produces hallucinations through it actions at NMDA receptors (West et al., 2000). In addition, activation of sigma-1 receptor by DMT may lead to potentiation of NMDA receptors (Cozzi et al., 2009)."

Neuropharmacology of N,N-Dimethyltryptamine
Brain Research Bulletin
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5048497/

citations: 71

electromagnetism modulates our DNA

xiv. low frequency (sub 10 HZ) electromagnetic brain entrainment modulates depression (electromagnetic waves via Transcranial Magnetic Stimulation TMS)



Efficacy and Safety of Transcranial Magnetic Stimulation in the Acute Treatment of Major Depression: A Multisite Randomized Controlled Trial
Biological Psychiatry
https://www.sciencedirect.com/science/article/pii/S0006322307001461

randomised controlled trial
citations: 1304

———

Daily Left Prefrontal Transcranial Magnetic Stimulation Therapy for Major Depressive Disorder A Sham-Controlled Randomized Trial
JAMA Psychiatry
https://jamanetwork.com/journals/jamapsychiatry/article-abstract/210744

randomised controlled trial
citations: 761

———

A randomized clinical trial of repetitive transcranial magnetic stimulation in the treatment of major depression
Biological Psychiatry
https://www.sciencedirect.com/science/article/pii/S0006322399002437

randomised controlled trial
citations: 432

———

“Compared to sham, active rTMS at 10hz significantly improved auditory hallucinations”

Low frequency repetitive transcranial magnetic stimulation improves source monitoring deficit in hallucinating patients with schizophrenia
Schizophrenia Research
https://www.sciencedirect.com/science/article/pii/S0920996405004767

citations: 144

———

"A 2–4 week randomized, prospective trial comparing rTMS at 10hz to ECT produced comparable therapeutic effects in severely depressed patients"

Repetitive transcranial magnetic stimulation versus electroconvulsive therapy for major depression: preliminary results of a randomized trial
Biological Psychiatry
https://www.sciencedirect.com/science/article/pii/S0006322302015081

citations: 278

"These results affirm that the resonant features of alpha frequency EEG play an important role in the pathophysiology of schizophrenia"

Therapeutic Effects of Individualized Alpha Frequency Transcranial Magnetic Stimulation (αTMS) on the Negative Symptoms of Schizophrenia
Schizophrenia Bulletin
https://academic.oup.com/schizophreniabulletin/article/32/3/556/1907392

citations: 161

"These preliminary results suggest that low-frequency rTMS may be beneficial mainly in major depression and to some extent in schizophrenia"

Preliminary evidence for a beneficial effect of low-frequency, repetitive transcranial magnetic stimulation in patients with major depression and schizophrenia
Depression and Anxiety
https://onlinelibrary.wiley.com/doi/abs/10.1002/(SICI)1520-6394(1998)7:2%3C65::AID-DA2%3E3.0.CO;2-4

citations: 177

"The results we report are encouraging and support an important role for rTMS at 10hz in the treatment of severe major depressive disorder"

Repetitive transcranial magnetic stimulation is as effective as electroconvulsive therapy in the treatment of nondelusional major depressive disorder: an open study
Biological Psychiatry
https://www.sciencedirect.com/science/article/pii/S0006322399002541

citations: 435

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xv. light modulates DNA (laser and LED)



Psychological benefits 2 and 4 weeks after a single treatment with near infrared light to the forehead: a pilot study of 10 patients with major depression and anxiety
Behavioural and Brain Functions
https://behavioralandbrainfunctions.biomedcentral.com/articles/10.1186/1744-9081-5-46

citations: 205

Significant Improvements in Cognitive Performance Post-Transcranial, red/near infrared Light Emitting Diode Treatments in Chronic, Mild Traumatic Brain Injury: Open Protocol Study
Journal of Neurotrauma
https://www.liebertpub.com/doi/abs/10.1089/neu.2013.3244

citations: 134

Mitochondrial signal transduction in accelerated wound and retinal healing by near-infrared light therapy
Mitohcondrion
https://www.sciencedirect.com/science/article/pii/S156772490400145X

citations: 447

Effects of low-power light therapy on wound healing: LASER x LED*
Anais Brasileiros de Dermatologia
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4148276/

citations: 175

Low-level laser (light) therapy (LLLT) on muscle tissue: performance, fatigue and repair benefited by the power of light
Photonics & lasers in medicine
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3635110/

citations: 173

Biphasic Dose Response in Low Level Light Therapy
Dose Response
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2790317/

citations: 966

Interplay between up-regulation of cytochrome-c-oxidase and hemoglobin oxygenation induced by near-infrared laser
Nature
https://www.nature.com/articles/srep30540

citations: 92

Argon Laser Treatment of Radiation Proctitis
Jama Surgery
https://jamanetwork.com/journals/jamasurgery/article-abstract/594011

citations: 42

Etiopathogenetic basis for using magnetolaser therapy in the complex treatment of male infertility
Urologiia
https://www.ncbi.nlm.nih.gov/pubmed/12811920/

citations: 4

---

xvi. electromagnetic fields EMFs modulates DNA



Turning on stem cell cardiogenesis with extremely low frequency magnetic fields
The FASEB Journal
http://www.fasebj.org/doi/abs/10.1096/fj.04-2695fje?url_ver=Z39.88-2003&rfr_id=ori%3Arid%3Acrossref.org&rfr_dat=cr_pub%3Dpubmed&

citations: 108

Elf-pulsed magnetic fields modulate opioid peptide gene expression in myocardial cells
Cadiovascular Research
https://academic.oup.com/cardiovascres/article/45/4/1054/299145

citations: 72

Anti-senescence efficacy of radio-electric asymmetric conveyer technology
GeroScience
https://link.springer.com/article/10.1007/s11357-013-9537-8
citations: 31

Expression of ACTN2, alpha-actin and TNNT2 in rat bone marrow-derived mesenchymal stem cells induced by low frequency pulsed electromagnetic fields
Sichuan Da Xue Xue Bao Yi Xue Ban
https://www.ncbi.nlm.nih.gov/pubmed/23230735

citations: 2

Non-thermal effects of power-line magnetic fields (50 Hz) on gene expression levels of pluripotent embryonic stem cells-the role of tumour suppressor p53
Mutation Research
https://www.ncbi.nlm.nih.gov/pubmed/14706519

citations: 78

Electromagnetic fields affect transcript levels of apoptosis-related genes in embryonic stem cell-derived neural progenitor cells
FASEB Journal
https://www.ncbi.nlm.nih.gov/pubmed/16116041

citations: 206

Electromagnetic field change the expression of osteogenesis genes in murine bone marrow mesenchymal stem cells
Journal of Huazhong University of Science and Technology [Medical Sciences]
https://www.ncbi.nlm.nih.gov/pubmed/18480985

citations: 29

Pulsed electromagnetic fields accelerate proliferation and osteogenic gene expression in human bone marrow mesenchymal stem cells during osteogenic differentiation
Bioelectromagnetics
https://www.ncbi.nlm.nih.gov/pubmed/19866474

citations: 96

The light of life: evidence that the sun modulates human lifespan
Medical Hypotheses
https://www.ncbi.nlm.nih.gov/pubmed/17951015/

citations: 56

---

xvii. sound modulates DNA



Life Rhythm as a Symphony of Oscillatory Patterns: Electromagnetic Energy and Sound Vibration Modulates Gene Expression for Biological Signaling and Healing
Glob Adv Health Med
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4010966/

citations: 47

Targeted microbubbles for ultrasound mediated gene transfection and apoptosis induction in ovarian cancer cells
Ultrasonics Sonochemistry
https://www.sciencedirect.com/science/article/pii/S1350417712001381

citations: 66

Therapeutic Potential of Transcranial Focused Ultrasound for Rett Syndrome
International Medical Journal of Experimental and Clinical Research
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5087669/

citations: 4

Ultrasound-mediated DNA transfer for bacteria
Nucleic Acids Research
https://academic.oup.com/nar/article/35/19/e129/2402701

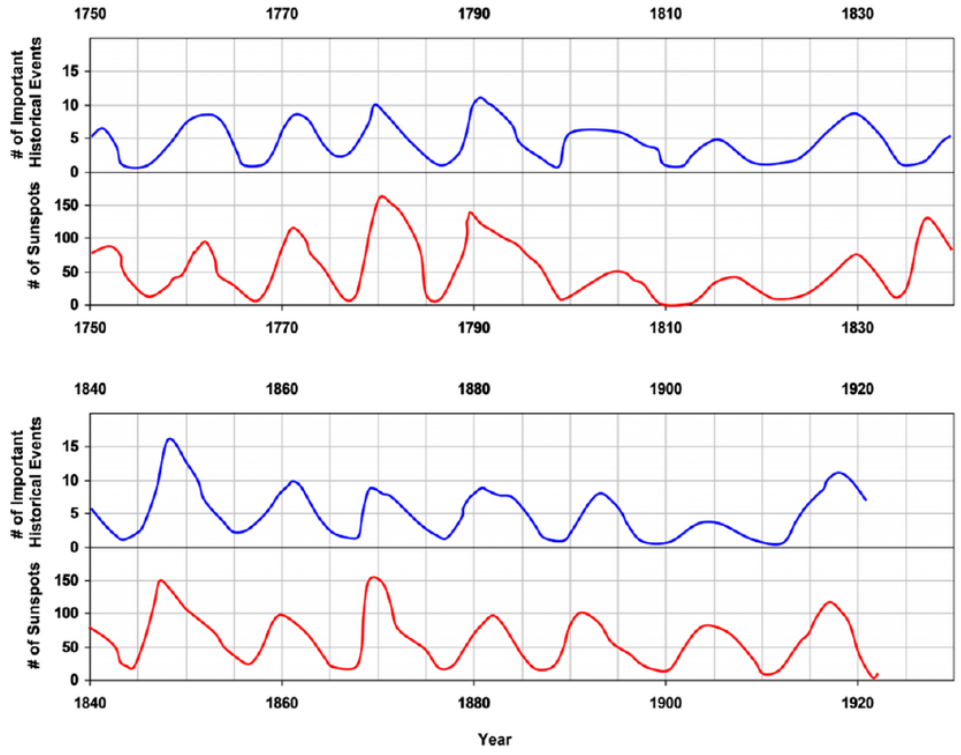
citations: 77

---

xviii. schumann resonance, solar wind and geomagnetic activity modulate heart rate variability, mental health and human culture



"Evidence of Significant Events Corresponding with Solar Cycles"



The Global Coherence Initiative - Opportunities for Scientific Research and Health Promotion
African Journal for Physical Activity and Health
https://www.researchgate.net/figure/Evidence-of-Significant-Events-Corresponding-with-Solar-Cycles_fig3_286869493

citations: 3

———

"energetic environmental phenomena affect psychophysical processes that can affect people in different ways depending on their sensitivity, health status and capacity for self-regulation."

"Increase in cosmic rays, solar radio flux, and Schumann resonance power was all associated with increased HRV and parasympathetic activity."

"daily autonomic nervous system activity responds to changes in geomagnetic and solar activity"

Long-Term Study of Heart Rate Variability Responses to Changes in the Solar and Geomagnetic Environment
Scientific Reports
https://www.heartmath.org/research/research-library/coherence/long-term-study-of-hrv-responses-to-changes-in-solar-geomagnetic-enviroment/

citations: 41

———

"daily autonomic nervous system activity not only responds to changes in solar and geomagnetic activity, but is synchronized with the time-varying magnetic fields associated with geomagnetic field-line resonances and Schumann resonances"

Synchronization of Human Autonomic Nervous System Rhythms with Geomagnetic Activity in Human Subjects
International Journal of Environmental Research and Public Health
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5551208/

citations: 50

———

"The results of this study are consistent with other studies showing that daily autonomic nervous system activity responds to changes in geomagnetic activity"

"this study also confirms the surprising degree of synchronized activity between the slow wave dynamics in heart rhythms and changes in the Earth's time-varying magnetic field in a frequency range that includes both Schumann resonances and geomagnetic field line resonances, which have similar frequencies as the rhythms produced by human brains and hearts"

Identification of a Group's Physiological Synchronization with Earth's Magnetic Field
Environmental Research & Public Health
https://www.mdpi.com/1660-4601/14/9/998/htm

citations: 17

———

"The results of the conducted study indicate that the applied Heart Lock-In meditation technique has a positive impact on the synchronization between the human heart rate variability and the Earth's magnetic field. Thus, such medical techniques are shown to be valuable not only psychologically (in sustaining positive emotions), but also physiologically since high synchronization between heart rate variability and geomagnetic activity has been associated with better health conditions."

The Influence of Heart Coherence on Synchronization Between Human Heart Rate Variability and Geomagnetic Activity
Journal of Complexity in Health Sciences
https://www.heartmath.org/assets/uploads/2019/01/influence-of-heart-coherence-on-synchronization-human-hrv-and-geomagnetic-activity.pdf

citations: 5

Research on the health effects of electromagnetic radiation - Parliament of Australia
Parliament of Australia
https://www.aph.gov.au/Parliamentary_Business/Committees/Senate/Environment_and_Communications/Completed_inquiries/1999-02/emr/report/c02

citations:

Schumann Resonance Frequencies Found within Quantitative Electroencephalographic Activity: Implications for Earth-Brain Interactions
Laurentian University
https://www.researchgate.net/publication/279743125_Schumann_Resonance_Frequencies_Found_within_Quantitative_Electroencephalographic_Activity_Implications_for_Earth-Brain_Interactions

citations: 34

Does Schumann resonance affect our blood pressure?
Biomedicine and Pharmacotherapy
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2656447/

citations: 50

"In 1954 and 1960 Koenig and his colleagues described the remarkable similarities of spectral power density profiles and patterns between the earth-ionosphere resonance and human brain activity"

"Our results confirm the measurements reported by earlier researchers that demonstrated unexpected similarities in the spectral patterns and strengths of electromagnetic fields generated by the human brain and the earth-ionospheric cavity"

Similar Spectral Power Densities Within the Schumann Resonance and a Large Population of Quantitative Electroencephalographic Profiles: Supportive Evidence for Koenig and Pobachenko
Plos One
http://journals.plos.org/plosone/article?id=10.1371/journal.pone.0146595

citations: 21

———

"Natural and technology-based variables affecting the Schumann parameters might be reflected in human brain activity, including modifications of cognition and dream-related memory consolidation"

"The results are consistent with the congruence of the frequency, magnetic field intensity, voltage gradient, and phase shifts that are shared by the human brain and the earth-ionospheric spherical wave guide"

Human Quantitative Electroencephalographic and Schumann Resonance Exhibit Real-Time Coherence of Spectral Power Densities: Implications for Interactive Information Processing
JSIP
https://www.scirp.org/journal/PaperInformation.aspx?PaperID=56609

citations: 15

———

Physical Factors of the Historical Process by A. L. Tchijevsky
Cycles
https://cyclesresearchinstitute.org/pdf/cycles-history/chizhevsky1.pdf

citations: 107

———

Wars and increased solar-geomagnetic activity: aggression or change in intraspecies dominance?
Perception and Motor Skills
https://www.ncbi.nlm.nih.gov/pubmed/10485122/

citations: 44

———

Effects of Geomagnetic Activity and Atmospheric Power Variations on Quantitative Measures of Brain Activity Replication of Azerbaijani Studies
Advances in Space Research
https://www.sciencedirect.com/science/article/abs/pii/S0273117709007558

citations: 49
———

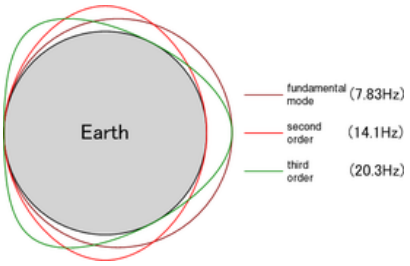
“The principal device of the Timewave Zero theory is a fractal function (constructed using numerical values derived from the King Wen Sequence of I Ching hexagrams) which maps time onto 'novelty'.”

“the "revealed" axiom that all phenomena are at root constellated by a wave form which is the hierarchical summation of its constituent parts, morphogenetic patterns related to those in DNA. ... We argue that the theory of the hyperspatial nature of superconductive bonds, and the experiment we devised to test that theory, yielded ... a modular wave-hierarchy theory of the nature of time that we have been able to construe, using a particular mathematical treatment of the I Ching, into a general theory of systems, which illuminates the nature of time and organism and provides an idea model which explains the interconnection of physical and psychological phenomena from the submolecular to the macrocosmic level.”

Timewave Zero - I Ching fractal map of time
Terrance McKenna, Dennis McKenna and Peter Meyer
http://www.fractal-timewave.com/

citations:
———

xix. Schumann Resonances are harmonics of the speed of light



Schumann Resonances discovery and formula (in original German)
W.O. Schumann
https://www.degruyter.com/downloadpdf/j/zna.1952.7.issue-2/zna-1952-0202/zna-1952-0202.pdf

citations:

In an ideal cavity, the resonant frequency of the n-th mode f_n is determined by the Earth radius a and the speed of light c.[15]

$$f_n = \frac{c}{2\pi a}\sqrt{n(n+1)}$$

ELF Electromagnetic Waves from Lightning: The Schumann Resonances
Atmosphere
https://res.mdpi.com/atmosphere/atmosphere-07-00116/article_deploy/atmosphere-07-00116.pdf?filename=&attachment=1

citations:

brain entrainment & music modulate mental health in at risk communities

xx. brain entrainment modulates pain, anxiety and heart rate variability in at risk communities



"The use of binaural beat technology in the theta brainwave frequency embedded into music decreases physical and psychological indications of stress. Binaural beat technology embedded with beta and delta frequencies may improve cognitive functioning and sleep quality, respectively."

The Effect of Binaural Beat Technology on the Cardiovascular Stress Response in Military Service Members With Post Deployment Stress
Journal of Nursing Scholarship
https://sigmapubs.onlinelibrary.wiley.com/doi/abs/10.1111/jnu.12304

randomised controlled trial, double blind
citations: 11

———

"When placed under an acute stressor after using music embedded with binaural beat , participants showed a decrease in sympathetic response and an increased in parasympathetic response. Those who used music alone had the opposite effect. In other words, those who did not use the technology exhibited more cardiovascular signs of chronic stress exposure."

The Sound Mind Warrior Study: Using Sound Technology to Combat Stress in Military Service Members
Sigma Global Nursing
https://sigma.nursingrepository.org/handle/10755/616141

randomised controlled trial, double blind
citations:

———

Group Therapy for severely traumatized refugees with a focus on sleep disorders
Hemi-Sync Journal, The Monroe Institute
https://www.monroeinstitute.org/article/2724

citations: 1

———

*"**Fifty veterans with PTSD** received 10 days of sham-controlled iTBS (1,800 pulses/day), followed by 10 unblinded sessions."*

*"**iTBS appears to be a promising new treatment for PTSD**."*

*"intermittent **theta** burst **stimulation** fires a frequency of 50hz at 5 cycles per minute, **5hz**, for a therapeutic duration of 3 minutes"*

Theta-Burst Transcranial Magnetic Stimulation for Posttraumatic Stress Disorder

AM J Psychiatry | Randomized Controlled Trial

xxi. music modulates mental health with homeless and refugees



"These results seem promising in the light of domestic violence research, which has found that a greater amount of personal resources is a crucial aspect of abused women's recovery process. Reduction of anxiety and improvement of sleep quality can be considered as increased personal resources, and seem feasible through the use of music therapy"

Effect of Music Therapy on the Anxiety Levels and Sleep Patterns of Abused Women in Shelters
Journal of Music Therapy
https://academic.oup.com/jmt/article-abstract/42/2/140/937831

citations: 138
———

Hip Hop Therapy: An Exploratory Study of a Rap Music Intervention with At-Risk and Delinquent Youth
Journal of Poetry Therapy
https://link.springer.com/article/10.1023/A:1019795911358

citations: 193
———

Needs of Children Experiencing Homelessness who are Living in Shelters: A Qualitative Investigation of Perceptions of Care Workers to Inform Music Therapy Clinical Practice
Voices
https://voices.no/index.php/voices/article/view/2298

citations: 6
———

A Music/Nonmusic Intervention with Homeless Children
Journal of Music Therapy
https://academic.oup.com/jmt/article-abstract/30/4/236/870085

citations: 26
———

Singing for Healing and Hope: Music Therapy Methods that Use the Voice with Individuals Who Are Homeless and Mentally Ill
Music Therapy Perspectives
https://academic.oup.com/mtp/article-abstract/29/1/14/1571106

citations: 21
———

Exploring the Meaning of a Performance in Music Therapy for Children and Their Families Experiencing Homelessness and Family Violence
Music Therapy Perspectives
https://academic.oup.com/mtp/article-abstract/35/1/36/2631812

citations: 24

Music: Making the Connection with Persons who are Homeless
Journal of Psychosocial Nursing and Mental Health Services
https://www.healio.com/psychiatry/journals/jpn/1993-7-31-7/%7Be5581e7c-5143-4f52-95e7-8c39d1b943a6%7D/music-making-the-connection-with-persons-who-are-homeless

citations: 8

Evaluation of a school-based creative arts therapy program for adolescents from refugee backgrounds
The Arts in Psychotherapy
https://www.sciencedirect.com/science/article/abs/pii/S0197455615000817

citations: 55

Neo-colonialism In Music Therapy: A Critical Interpretive Synthesis of the Literature Concerning Music Therapy Practice With Refugees
Voices
https://voices.no/index.php/voices/article/view/2299

citations: 20

“The beat will make you be courage”: The role of a secondary school music program in supporting young refugees and newly arrived immigrants in Australia
Research Studies in Music Education
https://journals.sagepub.com/doi/abs/10.1177/1321103X12466138

citations: 91

A Pilot Analysis of the Psychological Themes Found During the CARING at Columbia—Music Therapy Program with Refugee Adolescents from North Korea
Journal of Music Therapy
https://academic.oup.com/jmt/article-abstract/47/4/380/981501

citations: 44

Traumatised refugees: An integrated dance and verbal therapy approach
The Arts in Psychotherapy
https://www.sciencedirect.com/science/article/abs/pii/S0197455609000768

citations: 85

Music Therapy with Traumatized Refugees in a Clinical Setting
Voices
https://voices.no/index.php/voices/article/view/1715

citations: 59

Holding a Steady Beat: The Effects of a Music Therapy Program on Stabilising Behaviours of Newly Arrived Refugee Students
British Journal of Music Therapy
https://journals.sagepub.com/doi/abs/10.1177/135945750501900205

citations: 26

From healing rituals to music therapy: bridging the cultural divide between therapist and young Sudanese refugees
Australia b School of Music, The University of Queensland, Brisbane
http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.463.7780&rep=rep1&type=pdf

citations: 84

The effect of music therapy services on classroom behaviours of newly arrived refugee students in Australia—a pilot study
Emotional and Behavioural Difficulties
https://www.tandfonline.com/doi/abs/10.1080/13632750601022170

citations: 89

Action Research and Music Therapy: Group Music Therapy with Young Refugees in a School Community
Voices
https://voices.no/index.php/voices/article/view/1709/1469

citations: 53

Sounds of Trauma: An Introduction to Methodology in Music Therapy With Traumatized Refugees in Clinical and Outpatient Settings
APA Psychiatry
https://psycnet.apa.org/record/2004-19247-017
citations: 31

Introduction: Music and Migration
Journal of Ethnic & Migration Studies
https://www.tandfonline.com/doi/abs/10.1080/13691830500487266

citations: 179
———

Reporting on Music Therapy in Kenya
Nordisk Tidsskrift for Musikkterapi
https://www.tandfonline.com/doi/abs/10.1080/08098130009477987?journalCode=rnjm19

citations: 23
———



"A disc jockey plays music from the top of a truck. Festivals, fashion shows, and even a Miss Bidibidi pageant have been held in the settlement"

"The world's second largest refugee camp is slowly but surely transforming into a permanent city."

"He envisions a world-class city emerging from the forest. "I say the sky's the limit."

How Bidibidi, a refugee camp in Uganda, is becoming a city
National Geographic
https://www.nationalgeographic.com/magazine/article/how-bidibidi-uganda-refugee-camp-became-city

citations:
———

"The next evening, the apprehensive Atilio morphs into "Lil Ton", a local musician who captivates audiences with his soulful voice."

"The theatre is Yuga's compound, now swept and decorated with flowers he worked for weeks at a restaurant to be able to buy. Wani is Lil Ton's "mentor". He tells him over and over again that he must earn money. And music might be the key to the magic future he is looking for."

"We want a peace. We want a peace. Peace in South Sudan," sings "Fabulous", Atilio's brother with whom, off stage, they share a name. Together, the three young men form the group Super Talent, who made it to the semi-finals of the last "Bidi Bidi's Got Talent" show."

Inside the world's largest refugee camp
The Guardian
https://www.theguardian.com/world/2018/may/19/inside-world-largest-refugee-camp-uganda-south-sudan-civil-war

citations:
———



"DJ Snake, went live from Bangladesh yesterday in a plight to raise netizen's attention to the over one million Rohingya refugees from Myanmar"

"The French DJ and producer is currently there walking among the Rohingya refugees in their camps and posted a video encouraging his fans to help raise resources for the stateless minority that would help improve living conditions by providing food, education, medicine, clothing, shelter and more."

"Together with activist network Love Army, founded by Jérôme Jarre who believes the everyone with an online audience has a voice, yesterday they went live and met with refugees, talked to them about their experiences and learned about their needs."

WATCH DJ SNAKE AS HE VISITS BANGLADESH TO RAISE AWARENESS FOR THE ROHINGYA REFUGEE CRISIS
Mixmag
https://mixmag.asia/read/watch-dj-snake-as-he-visits-bangladesh-to-raise-awareness-for-the-rohingya-refugee-crisis-local

citations:
———

DJ Snake & Love Army for Rohingya - $2,258,710 raised
Gofundme
https://www.gofundme.com/f/love-army-for-rohingya

citations:
———

“An informal programme of hip-hop workshops in the region's Palestinian refugee camps offers a valuable lesson in how cultural development initiatives should be run.”

“You can use big terms if you want - empowerment, conflict prevention - but for me it's enough that we come here and give people the chance to do something that makes them feel good.”

“he has been taking hip-hop workshops to refugee camps on and off for the last two years.”

Hip-hop workshops for refugees in Jordan
The National News
https://amp.thenationalnews.com/arts-culture/music/hip-hop-workshops-for-refugees-in-jordan-1.558121

citations:
———

“She explained her mother had met the Iraqi DJ just as he came out of the boat. “I wasn’t in the camp at the time but later, every volunteer was talking about how amazing this man was. He came the whole way from Turkey listening to his music on his huge headphones and, while he was at Platanos, he showed the songs to some volunteers.”

“Some seven days later, on the night of New Year’s Eve, Gabriela made her way to the registration camp. “While we were doing some work there, the DJ from Iraq stopped my mom because he remembered her. I was in the middle of Moria and this man put his headphones on my ears.”

“I started listening to an absolutely amazing remix of “Hello” by Adele, that he made. I couldn’t stop smiling and I almost started crying. Now, I think a lot about how I would like to hear that again.”

45 Days in a Greek Refugee Camp
UNICEF
https://blogs.unicef.org/blog/45-days-in-a-greek-refugee-camp/

citations:
———

"he devotes his time to informing and educating the community by radio."

"The messages have had a positive impact on the daily life of the people that is in here."

"I am a link between the UN organizations and the refugees."

This refugee DJ is a voice for change
Medium
https://medium.com/world-food-programme-insight/this-refugee-dj-is-a-voice-of-change-68a4f5339dda

citations:
———

"a rich musical culture flourishes against all odds"

"The Sound of Refugees", the series focuses on a camp situated in the Islamic Republic of Mauritania, just miles away from war-torn Mali and hosts 50,000 refugees. Despite having lost their homes, Tuareg and Arab musicians inhabiting the camp are determined not to lose hope. Even in the shade, temperatures reach 45 degrees, but undeterred by the merciless heat, the community play on."

"I am where the desert is/I ask my soul to hold on living" – Tuareg refugee song"

Meet the stranded refugees making music in the Malian heat
Dazed
https://www.dazeddigital.com/art-photography/article/36833/1/jean-marc-caimi-the-stranded-refugees-making-music-in-the-malian-heat

citations:
———

"In a Congolese refugee camp, a DJ launched his own radio state for the camp, taking old electronic parts and fashioning them into an improvised transmitter."

Refugee camps are surprising hot spots of innovation
PRI
https://www.pri.org/stories/2014-10-22/refugee-camps-are-surprising-hot-spots-innovation

citations:
———

Art project helps transform lives of Syrian refugee children
BBC
https://www.bbc.com/news/av/world-middle-east-23441902

citations:
———

"Red Cross staff use music, dance and house-to-house visits to share critical health information"

Using music and dance to improve health in a Tanzania refugee camp
Community Engagement Hub
https://communityengagementhub.org/resource/using-music-and-dance-to-improve-health-in-a-tanzania-refugee-camp/

citations:
———

"Music and dance also helps to enhance a message of hope, while providing useful education."

Music as a Strategy of Youth Resilience in Dadaab Refugee Camp Kenya
Child Hub
https://childhub.org/en/child-protection-online-library/music-strategy-youth-resilience-dadaab-refugee-camp-kenya?language=sq

citations:
———

"In his song Sorry, rapper STIB pleads for forgiveness from the girlfriend he was forced to leave behind in Burundi, while another act explains how God's will guided him through the Congolese jungle to safety after his village was burned and family separated."

Young refugees of Kakuma turn to music to forget their troubles
The Irish Times
https://www.irishtimes.com/news/world/africa/young-refugees-of-kakuma-turn-to-music-to-forget-their-troubles-1.3090923?mode=amp

citations:
———

"Tanzania Red Cross Society's Health Information Team (HIT), a group of nearly 170 Burundian refugees who use music, dance, and door-to-door social mobilization to share critical health information."

Music, Dancing and Mobilizing for Health
American Red Cross
https://www.redcross.org/about-us/news-and-events/news/Music-Dancing-and-Mobilizing-for-Health.html

citations:
———



“Congolese choreographer Fabrice Don de Dieu Bwabulamutima believes in the healing power of dance.”

“He travels with his drama and dance company to refugee camps across the Democratic Republic of the Congo, teaching people who have experienced war and violence how to overcome trauma, rebuild their self-confidence and learn to live with other people again through the medium of dance.”

“Dance touches everybody, regardless of your position of power, your age or your race,” says the 40-year-old instructor who has put his own dance career on hold.
His company, Kongo Drama, has been conducting a four-month dance, drama and music programme entitled ‘Refugees on the Move’ in Inke camp, in Nord-Ubangi province, which hosts more than 16,000 refugees from the Central African Republic.”

Congo dance project helps refugees take steps towards rebuilding lives
UNHCR
https://www.unhcr.org/en-au/news/stories/2018/3/5aa666c14/congo-dance-project-helps-refugees-steps-towards-rebuilding-lives.html

citations:
———

Hope and Recognition. A Music Project among Youth in a Palestinian Refugee Camp
Voices
https://voices.no/index.php/voices/article/view/1853/1616

citations:
———

“Music, dance and drumming are omnipresent in the countries of Africa’s Great Lakes region. In the Nyarugusu refugee camp in western Tanzania, volunteers with the Tanzania Red Cross Society are using music and dance to share critical health information.”

Music and dance to improve health in a Burundian refugee camp
European Union
https://ec.europa.eu/echo/field-blogs/videos/music-and-dance-improve-health-burundian-refugee-camp_en

citations:
———

“The goal is to create a living museum, where the various cultures living within the camp can come together to share music, dance, art and handicrafts with each other and with the local community, as well as visitors to the camp”

“it soon resembled an African version of the universal "Got Talent" series or a combined dance and music copy of "American Idol." Watching frenzied dancers kick up dust in an impromptu contest between Congolese and Burundian troupes, the camp residents cheered and clapped. "It is wonderful to see them using this forum as positive energy to celebrate their differences," said Botelho.”

“Almost everyone was determined to take part. Even a group of elderly Congolese women got up and danced to the songs of their youth, much to the delight of the younger set.”

UNHCR and UNESCO help create "living museum" in Mozambique camp
MDG Fund
http://www.mdgfund.org/node/1227

citations:
———

“Music education has been proven to be an incredibly effective medium for achieving long-term social change for both the children involved and their families worldwide.”

“El Sistema Greece’s ultimate goal is to promote the social inclusion of children refugees in Greek / European society through music that becomes a platform for dialogue and unity among different communities.”

“El Sistema Greece provides daily musical activities led by Greek and foreign teachers who volunteer in refugee camps, professional musicians and internationally renowned artists.”

Music as an opportunity for social inclusion of refugees: El Sistema Greece
Get Close To Opera
https://www.getclosetoopera.eu/music-as-an-opportunity-for-social-inclusion-of-refugees-el-sistema-greece/

citations:
———

"Every other Saturday, LHI would put on a festival of events and activities, headlined by a Yazidi talent show. Every time, there were at least two or three musical performances, mainstays of a tradition that brought together culture and community to fill the void created by daily refugee life."

"It's amazing to see the difference between those who do and don't have access to those kinds of classes," Katelyn says. "They feel like they accomplished something...it's almost therapeutic [for them] to even have a schedule."

"For many refugees, the classes are far more than just a creative outlet. Most asylum seekers are fleeing from some sort of war or conflict. Many of them have lost family members and loved ones. As victims of ethnic genocide, many of the Yazidis in Serres experienced unimaginable trauma. And unfortunately, as Katelyn tells me, the traumatic events do not always stop with the arrival at a refugee camp."

Music & Refugees: Improving Lives In Uncertain Times
Hyperbot
https://www.hypebot.com/hypebot/2019/07/music-refugees-improving-lives-in-uncertain-times.html

citations:
———

"music and dance provides an important opportunity for the sharing of knowledge networks. The elders sing not only to provide entertainment, but also to entrench cultural values in their societies through music and dance."

" On the other hand, the youth hopes and ambitions are carried through their songs and dances."

"Their determination to keep track of hip hop, rap and reggae music, as well as their mode of dressing demonstrates that youth have a common identity with youths well beyond the refugee camps. Music and dance also helps to enhance a message of hope, while providing useful education."

Music as a Strategy of Youth Resilience in Dadaab Refugee Camp Kenya
Research on Humanities and Social Sciences
http://www.ku.ac.ke/schools/humanities/images/stories/2016/14853-17497-1-PB_2.pdf

citations:
———

"A British composer is taking music to refugee camps in Lebanon, so that children can try to deal with what has happened to them by taking part in performances."

Helping Syria's child refugees with music
BBC
https://www.bbc.com/news/av/world-middle-east-35449400

citations:
———

“this library also offers opportunities for children in the camp to learn drama, music, media production and dance.”

Aida Refugee Camp Library
Seraj Library
https://www.serajlibraries.org/aida-refugee-camp.html

citations:
———

“Aida music school | established in 2019, to be the first music school in the camp that embrace children and youth provide them with tools, practice, and knowledge. The school work with participants on many several levels of performance, analyzing, creation, and the relation between music and science, assuming our responsibilities to improve our society as a creative environment for knowledge and Progressive.”

Aida Music School
Aida Center
https://www.aidacenter.org/project/key-of-return/

citations:
———

“In the face of adversity, comes creativity; the community of Dzaleka are using dance as a vehicle for change, redrawing the parameters of what it means to be a refugee.”

The Dance Crew’s of Dzaleka Refugee Camp
Trippin
https://trippin.world/feature/the-dance-crews-of-dzaleka-refugee-camp

citations:
———

“Declan Byrne chats to three refugees making music in Australia. Despite the struggles of identity, they are spreading messages of positivity”

“Once settled in Australia, refugees can add immense value to our society, including telling their stories through music.”

“Music is so powerful," says P-UniQue, "I realised that I have the ability to voice my opinions and what I’m feeling and what’s affecting me and the people around me. People will bob their heads, but there is a message behind it. They’ll go home bobbing to my music but it’s drilled within their heads what I’m trying to say.”

These refugees making music in Australia are using their platforms to change minds
Triple J, ABC
https://www.abc.net.au/triplej/news/musicnews/refugees-making-music-australia-dyspora-p-unique-arig/11200432

citations:
———

"Elijah Gentle, a Melbourne-based advocate for displaced people, believes that music is an ideal vehicle for raising awareness and has established a project that documents the experiences of refugees in Eastern Europe through their own songs."

"The name of this project is No Borders Music, and in it Gentle has recorded the music-making he encountered in refugee camps that he visited in Bosnia, Serbia, Macedonia, Greece and Lebanon. His aim it to combine these recordings with instrumentation prepared back in Australia, and thereby create a cross-cultural synthesis that embraces hip-hop, traditional Arabic music, Sufi, and East-West fusions."

"Along with an audio album that will be released later this year, a documentary filmed by a Danish film maker is being made. A GoFundMe campaign to raise funds to cover costs is underway"

Refugees' Stories Through Song: No Borders Music
Music Australia
https://musicaustralia.org.au/2019/04/refugees-stories-through-song-no-borders-music/

citations:
———

"We compose lyrics that seek to inspire young refugees to dream amid setbacks that hover around their lives"

"My desire is to let the entire world discover that refugees are endowed with talents that can be harnessed to transform humanity. I have performed in refugee camps and other formal settings and the audience has been very receptive to my songs"

"So far, more than 700 young people living in Kakuma refugee camp have undergone auditions for various genres of performing arts including music, modelling, photography and dancing."

Kenya's second largest refugee camp becomes melting pot for artistic talent and integration
Xinhuanet
http://www.xinhuanet.com/english/2018-12/08/c_137659915.htm

citations:
———

"Through dance, I can express experiences that I would not be able to verbalize in any other way. The possibility of using dance to say something about oneself, one's own feelings and views is not to be found in traditional dance, which was what I learnt first. Rather, it is contemporary dance that makes that possible."

"But in the workshops I feel it is important to start with tradition in order to show the participants that dance has something to do with them. It is part of their culture and can also become a means through which they can express themselves."

"Dancing together brings them into the present, it creates a specific moment. That may be the starting point for actively shaping the future. Through reviving the culture the refugees feel they have lost, we lead them to a new way of dealing with it. It is as if we were opening up the possibility of a retrospective view, but one that focuses on what was beautiful and not on the suffering."

Dancing with Refugees - Expressing Feelings and Experiences
Goethe Institut
https://www.goethe.de/en/m/kul/tut/gen/tan/20710799.html

citations:

xxii. music modulates mental health with indigenous youth



Music Therapy to Manage Asthma Symptoms in Young Indigenous People in an Urban Setting
Aboriginal and Islander Health Work Journal
https://search.informit.com.au/documentSummary;dn=999889170996022;res=IELFSC

citations: 5

Attitudes towards Cross-Cultural Empathy in Music Therapy
Music Therapy Perspectives
https://academic.oup.com/mtp/article-abstract/24/2/108/1125006

citations: 31

The Roots of Music Therapy: Towards an Indigenous Research Paradigm
New York University
https://philpapers.org/rec/AIGTRO

citations: 97

Indigenous music therapy theory building through grounded theory research: The developing indigenous theory framework
The Arts in Psychotherapy
https://www.sciencedirect.com/science/article/abs/pii/S0197455608000609

citations: 39

Reflections on Music Therapy with Indigenous Families: Cultural Learning put into Practice
Voices
https://voices.no/index.php/voices/article/view/1799/1560

citations: 27

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xxiii. music modulates mental health with armed servicemen and women



A Virtual Hope Box Smartphone App as an Accessory to Therapy: Proof-of-Concept in a Clinical Sample of Veterans
Suicide and Life Threatening Behaviour
https://onlinelibrary.wiley.com/doi/abs/10.1111/sltb.12103

citations: 66

A Historical Review of Music Therapy and the Department of Veterans Affairs
Molloy College
https://digitalcommons.molloy.edu/cgi/viewcontent.cgi?referer=https://scholar.google.com.au/scholar?start=10&q=music+therapy+veterans+soldiers&hl=en&as_sdt=0,5&httpsredir=1&article=1011&context=etd

citations: 4

Soldier Attitudes about Technology-Based Approaches to Mental Health Care
Cyber Psychology and Behaviour
https://www.liebertpub.com/doi/abs/10.1089/cpb.2008.0071

citations: 136

Music for the Injured Soldier: A Contribution of American Women's Military Bands During World War II
Journal of Music Therapy
https://academic.oup.com/jmt/article-abstract/44/3/282/955009

citations: 35

Drumming through trauma: Music therapy with post-traumatic soldiers
The Arts in Psychology
https://www.sciencedirect.com/science/article/abs/pii/S0197455607000883

citations: 206

A pendulum between trauma and life: Group music therapy with post-traumatized soldiers
The Arts in Psychotherapy
https://www.sciencedirect.com/science/article/abs/pii/S0197455612000299

citations: 55

Music Therapy, War Trauma, and Peace: A Singaporean Perspective
Voices
https://voices.no/index.php/voices/article/view/1701

citations: 29

Re-entry: online virtual worlds as a healing space for veterans
The Engineering Reality of Virtual Reality
https://www.spiedigitallibrary.org/conference-proceedings-of-spie/7238/72380C/Re-entry--online-virtual-worlds-as-a-healing-space/10.1117/12.806121.short?SSO=1

citations: 11

Distant Thunder: Drumming with Vietnam Veterans
Music Therapy Perspectives
https://academic.oup.com/mtp/article-abstract/13/2/110/1074989

citations: 28

Participatory choral music as a means of engagement in a veterans' mental health and addiction treatment setting
Arts and Health
https://www.tandfonline.com/doi/abs/10.1080/17533015.2014.999246

citations: 13

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xxiv. virtual reality modulates emotions and heals PTSD with war veterans

Virtual reality exposure-based therapy for the treatment of post-traumatic stress disorder: a review of its efficacy, the adequacy of the treatment protocol, and its acceptability
Neuropsychiatric Disease and Treatment
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4599639/

citations: 157

Baseline psychophysiological and cortisol reactivity as a predictor of PTSD treatment outcome in virtual reality exposure therapy
Behaviour Research and Therapy
http://www.sciencedirect.com/science/article/pii/S0005796716300663

citations: 74

---

xxv. near infra red laser modulates cells and heals traumatic brain injury with war veterans

Treatments for traumatic brain injury with emphasis on transcranial near-infrared laser phototherapy
Neuropsychiatric Disease and Treatment
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4550182/

citations: 97

xxvi. art therapy modulates emotions with war veterans

Art therapy for PTSD and TBI: A senior active duty military service member's therapeutic journey
The Arts in Psychotherapy
https://www.sciencedirect.com/science/article/pii/S0197455616300636

citations: 52

---

xxvii. art therapy modulates emotions with at risk youth

The art of wellness: A 14-week art therapy program for at-risk youth
The Arts in Psychotherapy
https://www.sciencedirect.com/science/article/abs/pii/S019745561500057X

citations: 42

Ethical implications of cross-cultural international art therapy
The Arts in Psychotherapy
https://www.sciencedirect.com/science/article/abs/pii/S0197455616302386

citations: 16

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quantum computing empowers bioinformatics

xxviii. Quantum advantage enables transformational bioinformatics with gene expression modelling via cloud based quantum computers.



"Quantum computers promise faster algorithms that can affect molecular biology and bioinformatics – for example, in data analysis, electronic structure simulations and protein modeling."

"The potential of even small quantum computers to outperform the best supercomputers on certain tasks may prove transformative to computational biology, promising to make impossible problems difficult, and difficult problems routine."

"There are potential benefits across the entire field, from the ability to process vast amounts of information and run machine learning algorithms far more efficiently, to algorithms for quantum simulation that are poised to improve computational calculations in drug discovery, to quantum algorithms for optimization that may advance fields from protein structure prediction to network analysis."

The prospects of quantum computing in computational molecular biology
WIREs Computational Molecular Science
https://sigmapubs.onlinelibrary.wiley.com/doi/abs/10.1111/jnu.12304

citations: 4

———

"Quantum computing equipped with features such as superposition, entanglement, parallelism possesses the potential capability to tackle many problems that are classical computing unable to solve.

Although quantum computing is yet in a nascent phase, it paves the way towards a new era of power computing which will enable to solve complicated problems including genome assembly, discovering protein- DNA binding interaction, disease detection and many more."

Quantum Computing-Applications in Bioinformatics
International Journal of Computer Applications
https://www.ijcaonline.org/archives/volume177/number12/baiskhiyar-2019-ijca-919527.pdf

citations:

———

"Our findings suggest that the quantum simulation of classically intractable molecules may be possible without the overhead of quantum error correction."

Scalable Quantum Simulation of Molecular Energies
APS Physical Review
https://journals.aps.org/prx/abstract/10.1103/PhysRevX.6.031007

citations: 565

A Quantum Pattern Recognition Method for Improving Pairwise Sequence Alignment
Nature
https://www.nature.com/articles/s41598-019-43697-3

citations: 2

"Lattice protein folding models are a cornerstone of computational biophysics.

This first implementation of a biophysical problem paves the way towards studying optimization problems in biophysics and statistical mechanics using quantum devices."

Finding low-energy conformations of lattice protein models by quantum annealing
Nature
https://www.nature.com/articles/srep00571

citations: 234

"Our work is the first step towards scalable nuclear structure computations on a quantum processor via the cloud, and it sheds light on how to map scientific computing applications onto nascent quantum devices."

Cloud Quantum Computing of an Atomic Nucleus
arXiv Quantum Physics
https://arxiv.org/abs/1801.03897#:~:text=Our%20work%20is%20the%20first,applications%20onto%20nascent%20quantum%20devices.

citations: 172

"Transcription factors regulate gene expression, but how these proteins recognize and specifically bind to their DNA targets is still debated. Machine learning models are effective means to reveal interaction mechanisms. Here we studied the ability of a quantum machine learning approach to classify and rank binding affinities."

"Despite technological limitations, we find a slight advantage in classification performance and nearly equal ranking performance using the quantum annealer for these fairly small training data sets."

"Thus, we propose that quantum annealing might be an effective method to implement machine learning for certain computational biology problems.

Quantum annealing versus classical machine learning applied to a simplified computational biology problem
Nature
https://www.nature.com/articles/s41534-018-0060-8/

citations: 50
———

Quantum Computing at the Frontiers of Biological Sciences
arXiv Quantum Physics
https://arxiv.org/abs/1911.07127

citations: 4
———

Quantum Computing, Grover's Search Algorithm and its Application in Bioinformatics
Cosmos
https://www.worldscientific.com/doi/abs/10.1142/S0219607706000171

citations:
———

YinYang Bipolar Relativity: A Unifying Theory of Nature, Agents and Causality with Applications in Quantum Computing, Cognitive Informatics and Life Sciences
Georgia Southern University
https://www.igi-global.com/book.aspx?ref=yinyang-bipolar-relativity&titleid=45961&f=hardcover-e-book&f=hardcover-e-book

citations: 81
———

oxytocin receptor gene (OXTR) modulates happiness

xxix. Oxytocin receptor gene (OXTR) modulates dopamine receptor gene (DRD2) and Serotonin receptor gene (SLC6A4), counter acting internet app addiction, substance abuse and compulsive behaviour.



"Multilevel models revealed that history of care had a greater influence on maternal baseline cortisol (but not cortisol trajectory) for mothers with more plasticity alleles of SLC6A3 (10R) and OXTR (G), relative to mothers with fewer or no plasticity alleles."

"OXTR moderates between maternal history of care and cortisol during infant stress."

"SLC6A3 moderates between maternal history of care and cortisol during infant stress."

Maternal DRD2, SLC6A3, and OXTR genotypes as potential moderators of the relation between maternal history of care and maternal cortisol secretion in the context of mother-infant separation
Biological Psychology

https://www.sciencedirect.com/science/article/abs/pii/S0301051117302338

citations: 5

"The neuropeptide oxytocin acts within the STh to reduce the rewarding effects of the psychostimulant methamphetamine, through a proposed interaction with dopamine."

"Preference to dopamine treatment was blocked by the activation of local oxytocin receptors."

"These data suggest that dopamine neurotransmission in the STh produces rewarding effects that can be reduced by activation of local oxytocin receptors."

Oxytocin modulates dopamine-mediated reward in the rat subthalamic nucleus
Hormones and Behavior

https://www.sciencedirect.com/science/article/abs/pii/S0018506X12002930

citations: 44

"The results indicated that the OXTR rs53576 is connected with the striatal dopamine transporter DAT availability in vivo and modulates the interactions between the oxytocinergic and dopaminergic systems."

Oxytocin receptor gene rs53576 polymorphism modulates oxytocin–dopamine interaction and neuroticism traits—A SPECT study
Psychoneuroendocrinology

https://www.sciencedirect.com/science/article/abs/pii/S0306453014002042

citations: 50
———

"Oxytocin release decreases the activity of SNc DA dopamine neurons via inhibitory circuits"

Biased Oxytocinergic Modulation of Midbrain Dopamine Systems
Neuron

https://www.sciencedirect.com/science/article/pii/S0896627317305019

citations: 96
———

"Our data provide evidence that the VTA is the human brain site where oxytocin attaches salience to socially relevant cues. This mechanism might play an important role in triggering motivation to react at the prospect of social reward or punishment."

Oxytocin Influences Processing of Socially Relevant Cues in the Ventral Tegmental Area of the Human Brain
Biological Psychiatry

https://www.sciencedirect.com/science/article/abs/pii/S0006322313000395

citations: 187
———

"Oxytocin, a neurohypophyseal neuropeptide, also partially antagonized cocaine-induced motor hyperactivity.

Moreover, oxytocin antagonized the increased utilization of dopamine, elicited by cocaine in the nucleus accumbens."

"The data suggest that oxytocin may influence the behavioural effects of cocaine by affecting dopaminergic neurotransmission in some regions of the brain."

The role of oxytocin-dopamine interactions in cocaine-induced locomotor hyperactivity
Neuropharmacology

https://www.sciencedirect.com/science/article/abs/pii/0028390890900959

citations: 128

"In addition to psychosocial improvement, oxytocin has recently been implicated in mediating mesolimbic dopamine pathways during drug addiction and withdrawal. Central oxytocin pathways may serve as a potential therapeutic target to improve mood and socio-affiliative behaviors in patients with profound social deficits and/or drug addiction."

Dopamine and Oxytocin Interactions Underlying Behaviors: Potential Contributions to Behavioral Disorders
CNS Neuroscience & Therapeutics

https://onlinelibrary.wiley.com/doi/full/10.1111/j.1755-5949.2010.00154.x

citations: 315

———

"OXT receptors in the CNS—mainly those located in limbic and basal forebrain structures—are responsible for mediating various effects of OXT in the opiate- and cocaine-addicted organism."

"Tolerance to ethanol (e.g. hypothermia-inducing effect of ethanol) also was inhibited by OXT."

"In relation to cocaine abuse, OXT dose-dependently decreased cocaine-induced hyperlocomotion and stereotyped grooming behavior."

Oxytocin and Addiction - A Review
Pyschoneuroendocrinology

https://www.sciencedirect.com/science/article/abs/pii/S030645309800064X

citations: 265

———

"Dopamine D2 receptor (D2R)–oxytocin receptor (OTR) interactions exist within heterocomplexes with facilitatory effects on D2R recognition and Gi/o coupling."

"Based on the current findings, the blockade of the anxiolytic effects of oxytocin by the simultaneous intra-CeA administration of raclopride can be explained by a lack of facilitatory protomer interactions in D2R-OTR heterocomplexes."

"Dysfunction and/or disruption of such interactions in the central amygdala may lead to anxiety development. Restoration of such interactions may represent a new strategy for development of novel anxiolytic drugs."

Signaling in dopamine D2 receptor-oxytocin receptor heterocomplexes and its relevance for the anxiolytic effects of dopamine and oxytocin interactions in the amygdala of the rat
Molecular Basis of Disease

https://www.sciencedirect.com/science/article/pii/S0925443916301648

citations: 30

———

"Oxytocin and dopamine possess significant overlap in the modulation of life-essential behaviors. Here, Xiao et al. (2017) show that the activity of dopamine neurons of the ventral tegmental area and the substantia nigra is finely tuned by axonal release of oxytocin."

Oxytocin Mobilizes Midbrain Dopamine toward Sociality
Neuron

https://www.sciencedirect.com/science/article/pii/S0896627317306049

citations: 13

———

"Three oxytocin signaling genes have been frequently implicated in human social behavior: OXT (structural gene for oxytocin), OXTR (oxytocin receptor), and CD38 (oxytocin secretion)."

"Expression of the three selected oxytocin pathway genes was enriched in subcortical and olfactory regions and there was high co-expression with several dopaminergic and muscarinic acetylcholine genes, reflecting an anatomical basis for critical gene pathway interactions."

"The oxytocin signaling system may interact with dopaminergic and muscarinic acetylcholine signaling to modulate cognitive state processes involved in complex human behaviors."

Oxytocin pathway gene networks in the human brain
Nature

https://www.nature.com/articles/s41467-019-08503-8

citations: 76

———

"The data indicate that peripheral oxytocin treatment is capable of modifying dopaminergic neurotransmission in mouse basal forebrain regions."

Peripheral oxytocin treatment modulates central dopamine transmission in the mouse limbic structures
Neurochemistry International

https://www.sciencedirect.com/science/article/abs/pii/0197018686901385

citations: 30

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xxx. 9 alleles combined of OXTR Single-Nucleotide Polymorphisms (SNPs) rs1042778, rs2254298, rs53576 modulate human evolution, from basic human needs of safety, belonging and esteem, to self actualisation, self transcendence and ascension.



"Psychological resources—optimism, mastery, and self-esteem—buffer the deleterious effects of stress and are predictors of neurophysiological and psychological health-related outcomes. These resources have been shown to be highly heritable, yet the genetic basis for this heritability remains unknown."

"Here, we report a link between the oxytocin receptor (OXTR) SNP rs53576 and psychological resources, such that carriers of the "A" allele have lower levels of optimism, mastery, and self-esteem, relative to G/G homozygotes."

"OXTR was also associated with depressive symptomatology. Mediation analysis indicates that the effects of OXTR on depressive symptoms may be largely mediated by the influence of OXTR on psychological resources."

Oxytocin receptor gene (OXTR) is related to psychological resources
PNAS - Proceedings of the National Academy of Sciences

https://www.pnas.org/content/108/37/15118.short

citations: 277

———

"This work presents a mathematical model that establishes an interesting connection between ***nucleotide frequencies*** *in human single-stranded* ***DNA*** *and the famous* ***Fibonacci****'s numbers.*

"Under these two hypotheses, it is possible to ***predict*** *the human* ***nucleotide frequencies*** *with* ***accuracy****."*

"This result may be used as evidence to the Fibonacci string model that was proposed to the sequence growth of DNA repetitive sequences."

Nucleotide Frequencies in Human Genome and Fibonacci Numbers
Bulletin of Mathematical Biology

https://link.springer.com/article/10.1007/s11538-007-9261-6

citations: 188

———

"The current findings show that OXTR rs53576 interacts with 2D:4D to shape one's cognitive empathy aptitude."

The association between 2D:4D ratio and cognitive empathy is contingent on a common polymorphism in the oxytocin receptor gene (OXTR rs53576)
Psychoneuroendocrinology

https://www.sciencedirect.com/science/article/abs/pii/S030645301500147X

citations: 67

———

"Oxytocin receptor gene polymorphism modulates effects of social support in the context of stress."

"Social support increases HRV only in carriers of at least one G allele of rs53576."

"Social support influences HRV specifically during direct social interaction."

Oxytocin receptor gene polymorphism modulates the effects of social support on heart rate variability
Biological Psychology

https://www.sciencedirect.com/science/article/abs/pii/S0301051116300357

citations: 23

———

"homozygous carriers of the G-allele of the single nucleotide polymorphism rs53576 have been reported to display more trust, empathy, and prosocial behavior and were less sensitive toward stress and maltreatment during childhood when compared to A-allele carriers."

"In A-allele carriers of rs53576, childhood trauma was differentially associated with empathy for psychological pain, which was not apparent in GG carriers."

"the present study demonstrates an association of childhood maltreatment and empathy for psychological pain which was mediated by rs53576 genotype, suggesting that the A-allele may convey genetic plasticity to environmental factors"

The OXTR Single-Nucleotide Polymorphism rs53576 Moderates the Impact of Childhood Maltreatment on Empathy for Social Pain in Female Participants: Evidence for Differential Susceptibility
Frontiers in Psychiatry - Behavioural and Psychiatric Genetics

https://www.frontiersin.org/articles/10.3389/fpsyt.2018.00359/full

citations: 16

———

The association of single-nucleotide polymorphisms in the oxytocin receptor and G protein–coupled receptor kinase 6 (GRK6) genes with oxytocin dosing requirements and labor outcomes
American Journal of Obstetrics and Gynacology

https://www.sciencedirect.com/science/article/abs/pii/S0002937817306324

citations: 10

———

"Our data suggest that both oxytocin (OXTR gene polymorphism) and fetal testosterone (2D:4D) may significantly affect emotional (anger) and cognitive (hostility) aggression in humans, given the leveling the role of culture."

Oxytocin receptor gene polymorphism (rs53576) and digit ratio associates with aggression: comparison in seven ethnic groups
Journal of Psychological Anthropology

https://jphysiolanthropol.biomedcentral.com/articles/10.1186/s40101-020-00232-y

citations: 1

———

"Differential stress responses between the genotype groups were observed depending on the presence or absence of social support."

"Only individuals with one or two copies of the G allele of rs53576 showed lower cortisol responses to stress after social support, compared with individuals with the same genotype receiving no social support."

"These results indicate that genetic variation of the oxytocin system modulates the effectiveness of positive social interaction as a protective buffer against a stressful experience."

Common oxytocin receptor gene (OXTR) polymorphism and social support interact to reduce stress in humans
PNAS

https://www.pnas.org/content/108/50/19937.short

citations: 279

———

Oxytocin Receptor Polymorphisms are Differentially Associated with Social Abilities across Neurodevelopmental Disorders
Nature

https://www.nature.com/articles/s41598-017-10821-0

citations: 24

———

*"The results revealed a significant effect in the interaction between OXTR rs53576 genotype and maternal postpartum depression on externalising problems in children with AA genotype (**β** = 0.136, 95% CI 0.032 to 0.240), but not in those with GG/GA genotype."*

"This indicates that an interaction of vulnerable genotypes (i.e., A allele of OXTR rs53576) with an environmental burden (i.e. maternal postpartum depression) may be one of the potential elements that predisposes the infant to developing behavioural problems early in life."

"Hence, special attention needs to be paid to children exposed to environmental risks such as maternal postpartum depression, to facilitate the provision of appropriate care."

Interaction effect of oxytocin receptor (OXTR) rs53576 genotype and maternal postpartum depression on child behavioural problems
Nature

https://www.nature.com/articles/s41598-019-44175-6

citations: 7

———

A common variant in OXTR rs53576 impacts topological patterns of brain functional networks
European Child and Adolescent Psychiatry

https://link.springer.com/article/10.1007/s00787-019-01414-5

citations: 3

———

A common oxytocin receptor gene (OXTR)polymorphism modulates intranasal oxytocin effects on the neural response to social cooperation in humans
Genes, Brain & Behaviour

https://onlinelibrary.wiley.com/doi/pdf/10.1111/gbb.12234

citations: 84

———

"It is hypothesized on the base of spatial distribution of the evolutionary novel A alleles of the both OXTR gene loci, that the spread of alleles of rs22542298 and rs53376 SNPs may be associated to some extant with manipulation of parental investment in humans."

Polymorphisms of two loci at the oxytocin receptor gene in populations of Africa, Asia and South Europe
BMC Genetics

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4704389/

citations: 26

———

"In conclusion, genetic variation in the rs53576 influences general sociality, which further implies that it is worthy to systematically examine whether the rs53576 is a valid genetic marker for socially related psychiatric disorders."

Association of Oxytocin Receptor Gene (OXTR) rs53576 Polymorphism with Sociality: A Meta-Analysis
PLOS ONE

https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0131820

citations: 138

———

"The findings highlight the influences of the OXTR gene on adaptive interpersonal interactions, especially when individuals face changing social situations."

OXTR rs53576 polymorphism impacts interpersonal adaptability: Dispositional forgiveness as a mediator
Psychoneuroendochrinology

https://www.sciencedirect.com/science/article/abs/pii/S0306453018308941

citations: 4

———

"Overall, genetic variations of OXTR rs53576 may be an important candidate in understanding the development of social functioning within the context of emotional childhood abuse."

Examining the link between emotional childhood abuse and social relationships in midlife: The moderating role of the oxytocin receptor gene
Child Abuse & Neglect

https://www.sciencedirect.com/science/article/abs/pii/S014521341930328X

citations: 5

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rs1042778
Self Decode

https://selfdecode.com/snp/rs1042778/

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rs2254298
Self Decode

https://selfdecode.com/snp/rs2254298/

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rs53576
Self Decode

https://selfdecode.com/snp/rs53576/

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rs2254298
SNPedia

https://www.snpedia.com/index.php/Rs2254298

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rs53576
SNPedia

https://www.snpedia.com/index.php/Rs53576

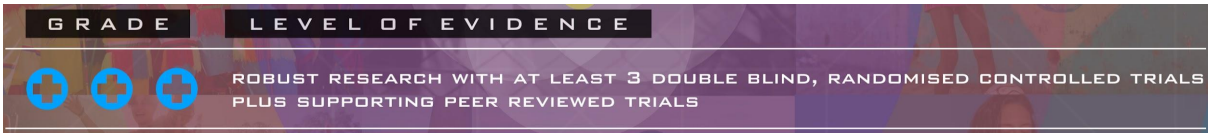
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rs1042778
SNPedia

https://www.snpedia.com/index.php/Rs1042778

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xxxi. Oxytocin creates connection between human beings, dissolving perceived separation including in context war and racism via empathy



"Overall these findings provide the first demonstration that OT can facilitate amygdala-dependent, socially reinforced learning and emotional empathy in men.'

Oxytocin Enhances Amygdala-Dependent, Socially Reinforced Learning and Emotional Empathy in Humans
The Journal of Neuroscience

https://www.jneurosci.org/content/30/14/4999.short

randomised controlled trial, double blind
citations: 832

———

Does intranasal oxytocin promote prosocial behavior to an excluded fellow player? A randomized-controlled trial with Cyberball
Psychoneuroendochrinology

https://www.sciencedirect.com/science/article/abs/pii/S0306453013000024

randomised controlled trial, double blind
citations: 115

———

Genetic modulation of oxytocin sensitivity: a pharmacogenetic approach
Nature

https://www.nature.com/articles/tp2015163

randomised controlled trial, double blind
citations: 67

———

"Israeli Jewish participants were asked to report their empathy to the pain of in-group (Jewish), neutral out-group (European), and adversary out-group (Palestinian) members.

"Oxytocin remarkably increased empathy to the pain of Palestinians, attenuating the effect of in-group empathy bias observed under the placebo condition."

"This effect, we argue, is driven by the general role of oxytocin in increasing the salience of social agents which, in turn, may interfere with processes pertaining to derogation of out-group members during intractable conflicts."

Giving peace a chance: Oxytocin increases empathy to pain in the context of the Israeli–Palestinian conflict
Psychoneuroendochrinology

https://www.sciencedirect.com/science/article/abs/pii/S030645301300334X

randomised controlled trial, double blind
citations: 105

———

"The present study demonstrates for the first time that a single dose of OT given intranasally improves affective empathy and approach motivation in women with BPD."

Effects of intranasal oxytocin administration on empathy and approach motivation in women with borderline personality disorder: a randomized controlled trial
Nature

https://www.nature.com/articles/s41398-019-0658-4

randomised controlled trial
citations: 10

———

"participants with high acute clinician-rated PTSD symptom severity did show beneficial effects of oxytocin. Although replication is warranted, these findings suggest that oxytocin administration is a promising preventive intervention for PTSD for individuals with high acute PTSD symptoms."

Intranasal Oxytocin to Prevent Posttraumatic Stress Disorder Symptoms: A Randomized Controlled Trial in Emergency Department Patients
Biological Psychiatry

https://www.sciencedirect.com/science/article/abs/pii/S0006322316330670#!

randomised controlled trial
citations: 86

———

"Results showed within group improvements in fear recognition, perspective taking, and a reduction in negative symptoms in the oxytocin group. These preliminary findings indicate oxytocin treatment may help improve certain components of functioning in schizophrenia."

A pilot six-week randomized controlled trial of oxytocin on social cognition and social skills in schizophrenia
Schizophrenia Research

https://www.sciencedirect.com/science/article/abs/pii/S0920996414001674

randomised controlled trial
citations: 82

———

Oxytocin Conditions Intergroup Relations Through Upregulated In-Group Empathy, Cooperation, Conformity, and Defense
Biological Psychiatry

https://www.sciencedirect.com/science/article/abs/pii/S0006322315002590

citations: 224

———

"Overall, our findings provide further support for the role of OXT in facilitating emotional empathy and demonstrate that effects are independent of culture and sex and involve modulatory effects on the amygdala and its interactions with other key empathy regions."

Oxytocin Enhancement of Emotional Empathy: Generalization Across Cultures and Effects on Amygdala Activity
Frontiers in Neuroscience

https://www.frontiersin.org/articles/10.3389/fnins.2018.00512/full

citations: 33

———

"OXT selectively enhanced affective empathy for fear"

Oxytocin increases attention to the eyes and selectively enhances self-reported affective empathy for fear
Neuropsychologia

https://www.sciencedirect.com/science/article/abs/pii/S0028393217303949#!

citations: 23

———

"We have shown that OT raised generosity in the UG by 80% over placebo"

Oxytocin Increases Generosity in Humans
PLOS ONE

https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0001128

citations: 1036

---

oxytocin modulates trust, growing businesses, economies and stock markets

xxxii. corporate social responsibility (csr) modulates customer happiness, employee happiness and share value via increases in trust, measured via oxytocin

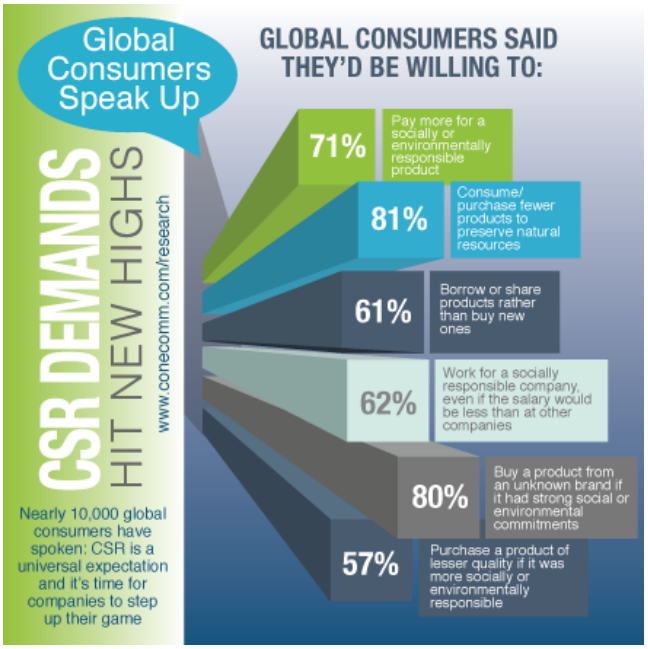


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"91% of global consumers expect companies to do more than make a profit, but also operate responsibly to address social and environmental issues"

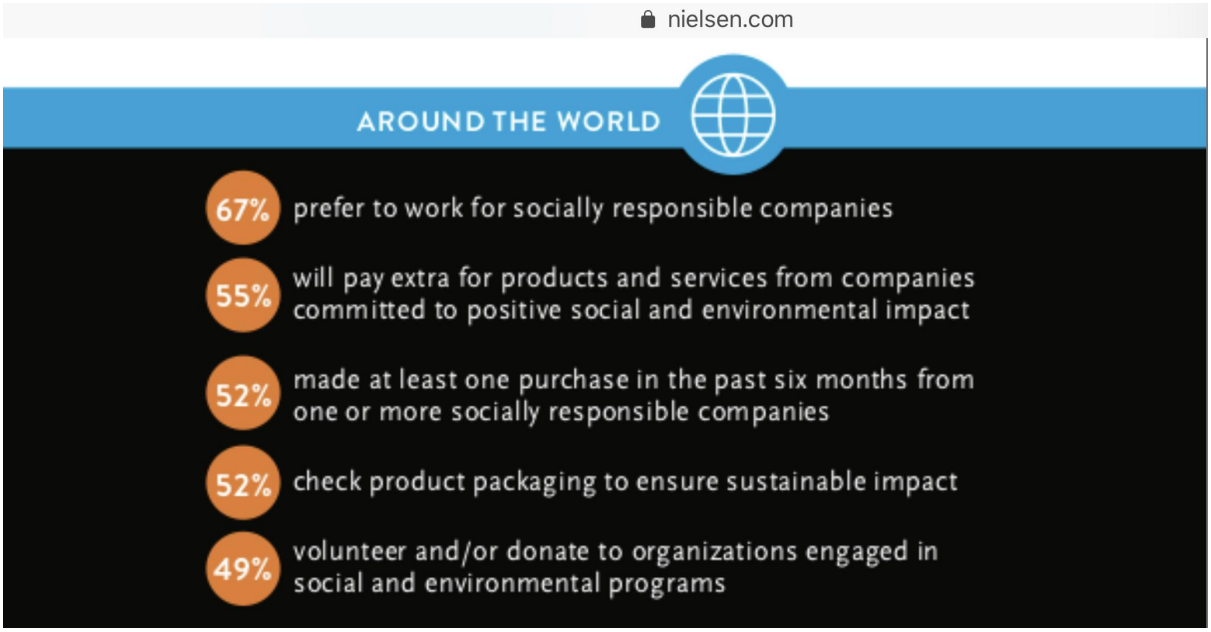
"84% say they seek out responsible products whenever possible"

"87% will purchase a product because the company advocated for a cause they care about"



2015 Cone Communications/Ebiquity Global CSR Study
Cone
http://www.conecomm.com/research-blog/2015-cone-communications-ebiquity-global-csr-study

citations: 5

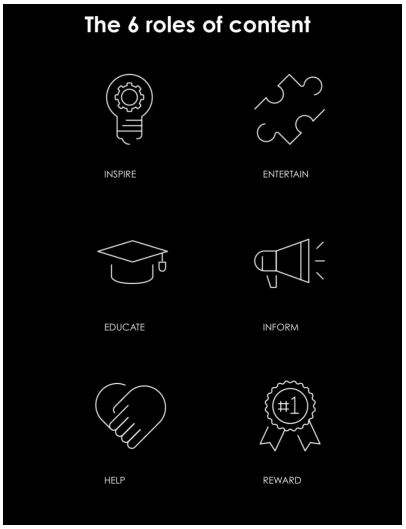


Doing Well by Doing Good: Increasingly, Consumers care about corporate social responsibility, but does it convert to consumption?
Nielson
https://www.nielsen.com/content/dam/nielsenglobal/apac/docs/reports/2014/Nielsen-Global-Corporate-Social-Responsibility-Report-June-2014.pdf

citations: 6

———

"58% of brand content is not meaningful"



Meaningful Brands
Havas Vivendi
https://www.meaningful-brands.com/en

citations: 15

"Many employees who are not engaged want a reason to be inspired."

"Approaching inspired engagement as a way to happiness, as a business strategy, yields better results than wellbeing on its own"

"Highly engaged organizations share common philosophies and practices"

The Right Culture: Not Just About Employee Satisfaction
Gallop
https://www.gallup.com/workplace/236366/right-culture-not-employee-satisfaction.aspx

citations: 9

———

"JB Hi-Fi recently commissioned a report on the program from Workplace Giving Australia, and discovered the positive impact of workplace giving on workplace culture, employee engagement and ROI"

"91 per cent of staff felt proud to be a JB Hi-Fi employee"

"76 per cent thought that workplace giving made JB Hi-Fi a better company to work for"

"65 per cent of workplace giving employees felt more actively engaged in their work"

"The company's Helping Hands workplace giving initiative, established in 2008, provides 12,000 employees the opportunity to donate to nine charity partners through regular payroll donations.

All donations are matched by the company, the program involves 80 per cent of staff, and has generated more than $15 million so far for its charity partners."

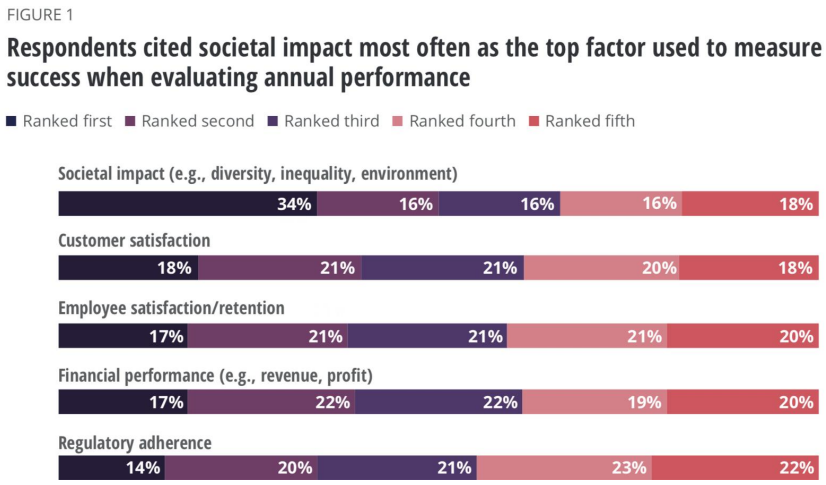
JB Hi-Fi Saving to the Tune of $8M With Workplace Giving Program
Workplace Giving Australia
https://probonoaustralia.com.au/news/2018/07/jb-hi-fi-saving-tune-8m-workplace-giving-program/

citations: 10

———

"65% of CEOs rated "inclusive growth" as a top three strategic concern, more than three times greater than the proportion citing "shareholder value""

"In 2019, the number one measurement of success cited by CEOs, is societal impact, including income, inequality, diversity and the environment"



Leading the social enterprise: Reinvent with a human focus 2019
Deloitte Global Human Capital Trends
https://www2.deloitte.com/content/dam/Deloitte/au/Documents/human-capital/deloitte-au-hc-human-capital-trends-2019-150419.pdf

citations: 17

The road ahead: The KPMG Survey of Corporate Responsibility Reporting 2017
KPMG
https://home.kpmg/content/dam/kpmg/xx/pdf/2017/10/kpmg-survey-of-corporate-responsibility-reporting-2017.pdf

citations: 23

"CSR Assets can support returns related to:

** Share price and market value*
** Sales and revenue*
** Reputation and brand*
** Human resources*
** Risk and license to operate*

CSR could avoid revenue losses of up to 7% of a company's market value.

It can also reduce the employee turnover rate by a staggering 50%."

Project ROI Report: Defining the Competitive and Financial Advantages of Corporate Responsibility and Sustainability
Project ROI
https://www.issuelab.org/resources/22448/22448.pdf

citations: 7
———

Perceived CSR and Corporate Reputation: The Mediating Role of Employee Trust
Vikalpa: The Journal for Decision Makers
https://journals.sagepub.com/doi/full/10.1177/02560909187948

citations: 593
———

Do Stock Market Investors Reward Companies with Reputations for Social Performance?
Corporate Reputation Review
https://link.springer.com/article/10.1057/palgrave.crr.1540048

citations: 111
———

CSR and Market Valuation: International Evidence
AFFI
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1833581

citations: 26
———

Corporate Social Responsibility and Stock Prices: The Environmental Awareness of Shareholders
Corporate Sustainability
http://corporate-sustainability.org/wp-content/uploads/arcs-2012-Flammer.pdf

citations: 42
———

Corporate Social Performance and Stock Returns: UK Evidence from Disaggregate Measures
Financial Management
https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1755-053X.2006.tb00149.x

citations: 732

The Effect of CSR on Stock Performance: New Evidence for the USA and Europe
Center of Economic Research at ETH Zurich
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1102528

citations: 54

Do social responsibility disclosures show improvements on stock price?
The Journal of Developing Areas
https://muse.jhu.edu/article/621336/summary

citations: 12

The influence of the implementation of CSR on business strategy: An empirical approach based on Cameroonian enterprises
The University of Yaounde II, Yaounde, Cameroon
https://www.sciencedirect.com/science/article/pii/S2214462516300135

citations: 4

The expansion of non-financial reporting: an exploratory study
Journal of Accounting & Business Research
https://www.tandfonline.com/doi/full/10.1080/00014788.2018.1470141

citations: 7

The Link Between Competitive Advantage and Corporate Social Responsibility
Harvard Business Review
https://www.comfama.com/contenidos/servicios/Gerenciasocial/html/Cursos/Columbia/Lecturas/Strategy-Society.pdf

citations: 10,194

Converting Social Problems into Business Opportunities: The New Meaning of Corporate Social Responsibility
California Business Review
https://search.proquest.com/openview/8dde78b449d8452c447619ddb82d15f9/1?pq-origsite=gscholar&cbl=42246

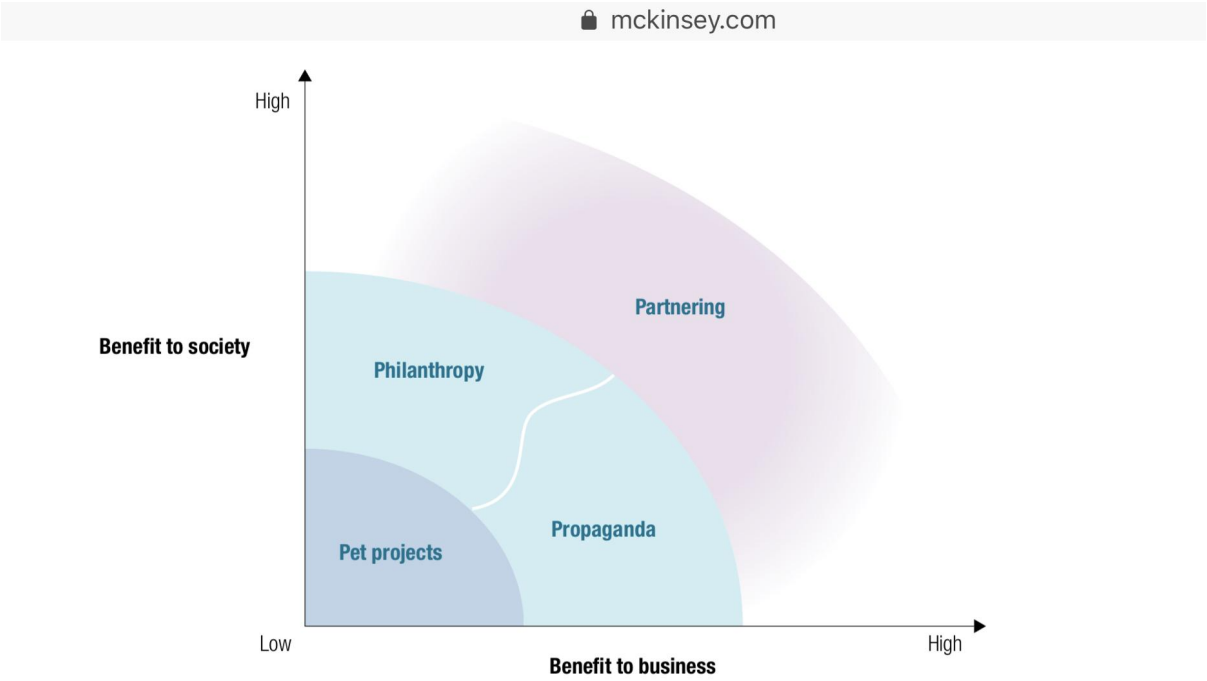
citations: 1126

Corporate Social Responsibility, Customer Satisfaction, and Market Value
Sage Journals
https://journals.sagepub.com/doi/abs/10.1509/jmkg.70.4.001

citations: 2715

Valuing social responsibility programs
McKinsey
https://www.mckinsey.com/business-functions/strategy-and-corporate-finance/our-insights/valuing-social-responsibility-programs

citations: 9



Making the most of corporate social responsibility
McKinsey
https://www.mckinsey.com/featured-insights/leadership/making-the-most-of-corporate-social-responsibility

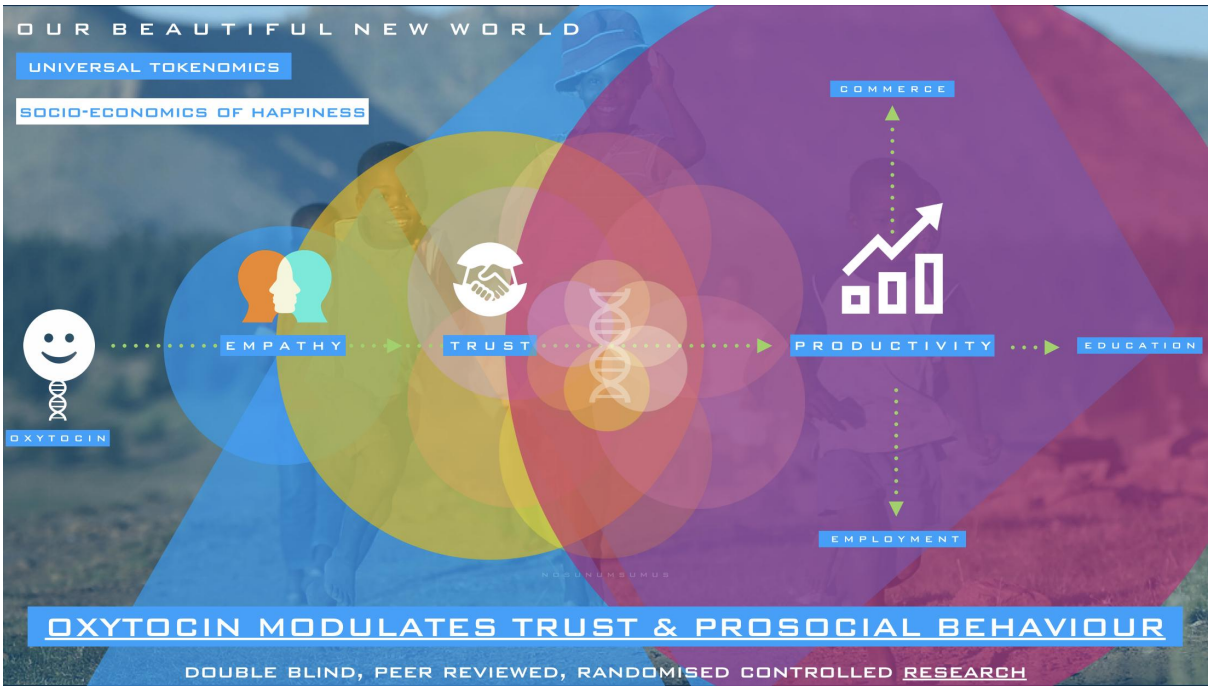
citations: 64

Beyond corporate social responsibility: Integrated external engagement
McKinsey
https://www.mckinsey.com/business-functions/strategy-and-corporate-finance/our-insights/beyond-corporate-social-responsibility-integrated-external-engagement

citations: 69

---

trust is a primary driver of personal, societal and economic growth [1, 2, 3]



1. “Empathy creates trust” 2. “trust empowers personal, societal and economic growth”

Oxytocin increases trust in humans
Nature
https://www.nature.com/articles/nature03701

citations: 4264

---

3. “Oxytocin plays an essential role in prosocial behaviours of empathy, trust and kindness”

How Our Brain Decides When to Trust
Multiple publications via Neuroeconomic Studies
http://www.neuroeconomicstudies.org/publications

citations: 157

---

xxxiii. the macroeconomics of happiness, trust and oxytocin modulate stock market growth



Spending Money on Others Promotes Happiness
Science
https://science.sciencemag.org/content/319/5870/1687

citations: 1621

The Macroeconomics of Happiness
Review of Economics and Statistics
https://www.mitpressjournals.org/doi/abs/10.1162/003465303772815745

citations: 1913

World happiness report [2012]
University of British Columbia Library
https://open.library.ubc.ca/media/download/pdf/52383/1.0053622/3

citations: 1893

Income and Happiness: Towards a Unified Theory
The Economic Journal
https://academic.oup.com/ej/article-abstract/111/473/465/5139977

citations: 3262

Happiness, Economy and Institutions
The Economic Journal
https://onlinelibrary.wiley.com/doi/abs/10.1111/1468-0297.00570

citations: 1957

What Can Economists Learn from Happiness Research?
Journal of Economic Literature
https://www.aeaweb.org/articles?id=10.1257/002205102320161320

citations:

Happiness and Economic Performance
The Economic Journal
https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-0297.1997.tb00085.x

citations: 4254

Happiness - A Revolution in Economics
MIT
https://pdfs.semanticscholar.org/f1e1/fcfea840733b67bec7487195b3693a7b79ff.pdf

citations: 1137

Happiness and the Human Development Index: The Paradox of Australia
NBER
https://www.nber.org/papers/w11416

citations: 214

Happiness and the Human Development Index: Australia Is Not a Paradox
Australian Economic Review
https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-8462.2006.00408.x

citations: 130

Happiness and Growth the World Over: Time Series Evidence on the Happiness-Income Paradox
IZA
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1369806

citations: 250

Development as Happiness: The Subjective Perception of Happiness and UNDP's Analysis of Poverty, Wealth and Development
Journal of Happiness Studies
https://link.springer.com/article/10.1007/s10902-007-9063-4

citations: 125

Wealth and Happiness Revisited – Growing National Income Does Go with Greater Happiness Social Indicator Research
https://link.springer.com/article/10.1023/A:1024790530822

citations: 626

United Nations Development Programme Human Development Index (HDI)
UNDP
http://hdr.undp.org/en/content/human-development-index-hdi

citations: 2

OECD Better Life Index
OECD
http://www.oecdbetterlifeindex.org/#/11111111111

citations: 1056

World Happiness Report
United Nations Sustainable Development Solutions Network
https://worldhappiness.report

citations: 1143

Happy Planet Index - Sustainable Wellbeing
New Economics Foundation
http://happyplanetindex.org

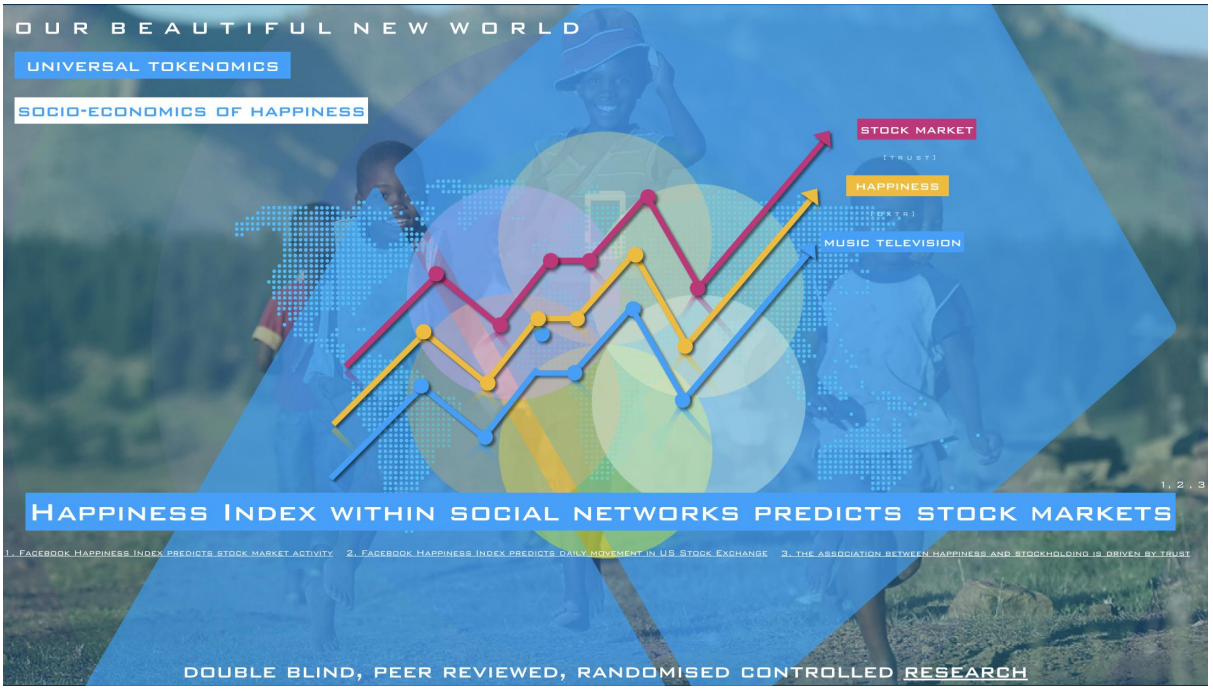
citations: 225

Global Happiness and Well-Being Policy Report
UN Sustainable Development Solutions Network
http://www.happinesscouncil.org

citations: 2

---

the association between happiness and stock holding is driven by trust [1, 2, 3]



“Facebook happiness predicts stock market activity”

1. Facebook’s daily sentiment and international stock markets
Journal of Economic Behavior & Organization
https://www.dropbox.com/s/kchixfrqw2kfdt6/Facebook%E2%80%99s%20daily%20sentiment%20and%20international%20stock%20markets.pdf?dl=0

Citations: 119

———

“Facebook Happiness Index predicts daily movement in US Stock Exchange”

2. Can Facebook Predict Stock Market Activity?
Goethe University Frankfurt
https://www.ecb.europa.eu/events/pdf/conferences/140407/Karabulut_CanFacebookPredictStockMarketActivitiy.pdf

Citations: 121

———

“the association between happiness and stockholding is driven by trust”
3. Happiness and Stock-Market Participation: Empirical Evidence from China
Journal of Happiness Studies
https://www.researchgate.net/publication/284942377_Happiness_and_Stock-Market_Participation_Empirical_Evidence_from_China

Citations: 22

xxxiv. gamification and behavioural economics modulate usage of digital health interventions



Improving Health Care by Gamifying It
Harvard Business Review
https://hbr.org/2019/05/improving-health-care-by-gamifying-it

citations: 5

Using Wearable Devices and Smartphones to Track Physical Activity: Initial Activation, Sustained Use, and Step Counts Across Sociodemographic Characteristics in a National Sample
Annals of Internal Medicine
https://annals.org/aim/article-abstract/2654953/using-wearable-devices-smartphones-track-physical-activity-initial-activation-sustained

citations: 31

"Gamification was commonly used by popular health and fitness mobile applications, but none used the specified behavioral economic principles to design rewards or points. Mobile applications could potentially improve their use if their design better leveraged principles from behavioral economic"

"Use of behavioral economic principles focused on designing rewards or points using loss aversion (allocated upfront and could be lost), variable reinforcement (not allocated constantly), and probability inflation (using drawings or lottery designs)."

"The top 50 rankes free health and fitness apps were downloaded from the App Store by Apple. Gamification was used by 64% of mobile applications. Most applications that included gamification (97%) targeted behaviors related to physical activity and weight loss. Applications focused on other areas such as reproductive health, meditation, and sleep used gamification less often (11%). Game elements used most commonly included goal setting (78%), social influences (78%), and challenges (63%), while less common elements included points (6%) and levels (3%). No applications incorporated behavioral economics principles specified in the study."

Gamification Use and Design in Popular Health and Fitness Mobile Applications
American Journal of Health Promotion
https://journals.sagepub.com/doi/abs/10.1177/0890117118790394?journalCode=ahpa

citations: 26

"Opt-out defaults, where clinically appropriate, could be a useful approach for increasing the generalizability of low-risk trials testing behavioral interventions in clinical settings."

A Randomized Controlled Trial of Opt-In Versus Opt-Out Enrollment Into a Diabetes Behavioral Intervention
American Journal of Health Promotion
https://journals.sagepub.com/doi/abs/10.1177/0890117116671673?journalCode=ahpa

randomised controlled trial
citations: 12

———

"Results of this study suggest that an opt-out approach to enrollment can significantly increase participation in trials of health system interventions."

Participation Rates With Opt-out Enrollment in a Remote Monitoring Intervention for Patients With Myocardial Infarction
JAMA Cardiology
https://jamanetwork.com/journals/jamacardiology/fullarticle/2549969

citations: 9

———

"Loss-framed financial incentives with personalized goal setting significantly increased physical activity among ischemic heart disease patients using wearable devices during the 16-week intervention, and effects were sustained during the 8-week follow-up"

Loss-Framed Financial Incentives and Personalized Goal-Setting to Increase Physical Activity Among Ischemic Heart Disease Patients Using Wearable Devices: The ACTIVE REWARD Randomized Trial
Journal of the American Heart Association
https://www.ahajournals.org/doi/10.1161/JAHA.118.009173

randomised controlled trial
citations: 34

———

Effect of a Game-Based Intervention Designed to Enhance Social Incentives to Increase Physical Activity Among Families. The BE FIT Randomized Clinical Trial
JAMA Internal Medicine
https://jamanetwork.com/journals/jamainternalmedicine/fullarticle/2655242

randomised controlled trial
citations: 68

———

Commitment Devices - Using Initiatives to Change Behavior
JAMA
https://jamanetwork.com/journals/jama/article-abstract/1866163

citations: 97
———

Framing Financial Incentives to Increase Physical Activity Among Overweight and Obese Adults: A Randomized, Controlled Trial
Annals of Internal Medicine
https://annals.org/aim/article-abstract/2491916/framing-financial-incentives-increase-physical-activity-among-overweight-obese-adults

randomised controlled trial
citations: 147
———

The Fresh Start Effect: Temporal Landmarks Motivate Aspirational Behavior
Management Science
https://pubsonline.informs.org/doi/10.1287/mnsc.2014.1901

citations: 171
———

"Using digital health devices to track behavior with a partner led to significant weight loss through 36 weeks"

Social Incentives and Gamification to Promote Weight Loss: The LOSE IT Randomized, Controlled Trial
Journal of General Internal Medicine
https://link.springer.com/article/10.1007%2Fs11606-018-4552-1

randomised controlled trial
citations: 20
———